

**FINITE ELEMENT ANALYSIS OF CONTACT MECHANICS
OF RIGID PUNCHES SLIDING OVER A GRADED
ORTHOTROPIC MEDIUM**

**ORTOTROPİK FONKSİYONEL DERECELENDİRİLMİŞ
MALZEMELERİN RİJİT ZİMBALAR ALTINDA TEMAS
MEKANİĞİNİN SONLU ELEMANLAR METHODU İLE
İNCELENMESİ**

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This work named “**Finite Element Analysis of Contact Mechanics of Rigid Punches Sliding Over a Graded Orthotropic Medium**” by **KORHAN BABACAN YILMAZ** has been approved as a **MASTER THESIS IN MECHANICAL ENGINEERING DEPARTMENT** by the below mentioned Examining Committee Members.

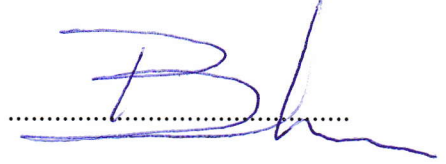
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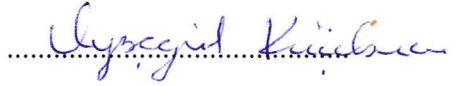
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ABSTRACT

FINITE ELEMENT ANALYSIS OF CONTACT MECHANICS OF RIGID PUNCHES SLIDING OVER A GRADED ORTHOTROPIC MEDIUM

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In this master thesis, the finite element analysis of contact mechanics of graded orthotropic half-plane loaded by a various punch profiles is conducted. The results that are obtained by the finite element solutions are compared with the analytical results that are in the literature. In general, it is hard to observe the solutions for the contact mechanics problems, because the complexity of the problem mainly depends on the problem parameters. For instance, the solution for the frictionless contact mechanics problems are easier than the solution for the frictional contact mechanics problems. The graded orthotropic half-plane, that used in the scope of this master thesis, is modeled as a linearly elastic material and a varying Young's modulus in the depth direction is used. The principal axes of orthotropy are assumed as parallel and perpendicular to the contact zone and generalized plane stress assumption is used for the solution of the

problem. The orthotropic graded half-plane is modeled by using quadratic finite elements and the material properties are defined at the center of each element for the finite element analysis. The presented numerical results that obtained by using the finite element method illustrates the effects of the coefficient of friction, the inhomogeneity parameter and the orthotropic elastic constants on the variation of the contact stresses and the in-plane stresses in order to find the most effective variation of the parameters for the minimum resulting contact and in-plane stress values. Furthermore, the comparison of the dimensionless load and error ratios that calculated by the finite element method and the analytical solution and the contact boundaries are given in the tables respectively.

Keywords: Contact mechanics, frictional contacts, functionally graded materials, finite element analysis, contact stresses, in-plane stresses

ÖZET

ORTOTROPIK FONKSİYONEL DERECELENDİRİLMİŞ MALZEMELERİN RIJIT ZİMBALAR ALTINDA TEMAS MEKANIĞININ SONLU ELEMANLAR METHODU İLE İNCELENMESİ

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Bu yüksek lisans tezinde; farklı rijit zımba profilleri kullanılarak ortotropik fonksiyonel derecelendirilmiş malzemelerin temas mekaniği problemleri iki boyutta sonlu elemanlar methodu ile incelenmiştir. Temas mekaniği problemleri genel olarak çözümü zor olan ve problem parametrelerine göre zorluğu artan problemlerdir. Örnek olarak; sürtünmenin olmadığı temas problemlerinin çözümü, sürtünme içeren temas problemlerinin çözümüne göre daha kolay olmaktadır. Bu yüksek lisans tezi kapsamında kullanılan ortotropik fonksiyonel derecelendirilmiş malzeme lineer elastik bir malzeme olarak modellenmiş ve derinlik yönünde

üstel olarak değişen elastisite modülü kullanılmıştır. Ortotropik özellikler için kullanılan asal eksenler, temas yüzeyine paralel ve dik olarak seçilmiş ve problemin çözümü sırasında genelleştirilmiş düzlem gerilme varsayımı kullanılmıştır. Ortotropik fonksiyonel derecelendirilmiş malzeme, sonlu elemanlar analizi sırasında kuadratik sonlu elemanlar ile modellenmiş ve malzeme özellikleri her bir sonlu elemanın merkezinde tanımlanmıştır. Çalışma sırasında, sürtünme katsayısının, homojen olmama parametresinin ve ortotropik elastik malzeme parameterlerinin, temas gerilmeleri ve düzlem gerilmeleri üzerindeki en etkili varyasyonları incelenmiş ve belirtilmiştir. Ayrıca, sonlu elemanlar methodu ve analitik çözüm yardımıyla hesaplanan boyutsuz yük değerlerinin karşılaştırılması ve hata oranları ile temas sınırları tablolar halinde verilmiştir.

Anahtar Kelimeler: Temas mekaniği, sürtünmeli temaslar, fonksiyonel derecelendirilmiş malzemeler, sonlu elemanlar analizi, temas gerilmeleri, düzlem gerilmeleri,

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Table 4.16. Comparison of the contact contact boundary, b/R , for a non-homogeneous orthotropic half-plane loaded by a semi-circular punch for various values of the effective Poisson's ratio, ν , and the stiffness ratio, δ^4 , with the parameters $\kappa=0.5$, $\beta(b)=0.5$, $b/R=0.001$73



NOMENCLATURE

SYMBOLS

E	effective stiffness
δ^4	stiffness ratio
κ	shear parameter
ν	effective Poisson's ratio
η	coefficient of friction
P	vertical force
Q	horizontal force
m	slope of the triangular punch
E^*	Young modulus
$(a + b) / R$	normalized contact radius
a / R	contact boundary (leading edge)
b / R	contact boundary (trailing edge)
βa	inhomogeneity parameter (flat punch)
βb	inhomogeneity parameter (triangular and semi-circular punches)
$\beta(a + b)$	inhomogeneity parameter (circular punch)
P^*	dimensionless load
P_0	maximum Hertzian contact pressur

1.INTRODUCTION

The purpose of this master thesis is to investigate the effects of the four independent material parameters; namely as the effective stiffness, E , the shear parameter, κ , the stiffness ratio, δ^4 , and the effective Poisson's ratio, ν , inhomogeneity parameter and the coefficient of friction, η , on the contact stress and the in-plane stress distributions for the problem of the graded orthotropic half-plane intended by a sliding flat, triangular, circular and semi-circular punch profiles are investigated in order to find the most effective variation of the parameters for the minimum resulting contact and in-plane stress values.

From past to present, the most salient method for transferring loads between the parts and the assemblies are to apply contacts to the bodies such as; brake discs, roller and cylindrical bearings and dove-tail connections. For instance, an example about a Hertzian contact problem for a roller bearing is illustrated in the Figure 1. Therefore, some methods have developed to solve the contact problems by analytically and numerically and with the help of the finite element modeling programs, the solutions became easier. In this master thesis, the finite element solutions are solved with using the most known finite element program ANSYS (2016). The mathematical formulations and the contact pressure distribution graphics are represented by using MathType (2014) and Sigma Plot (2014) programs. The pure penalty contact formulation is used in order to solve the finite element models. This contact formulation offers,

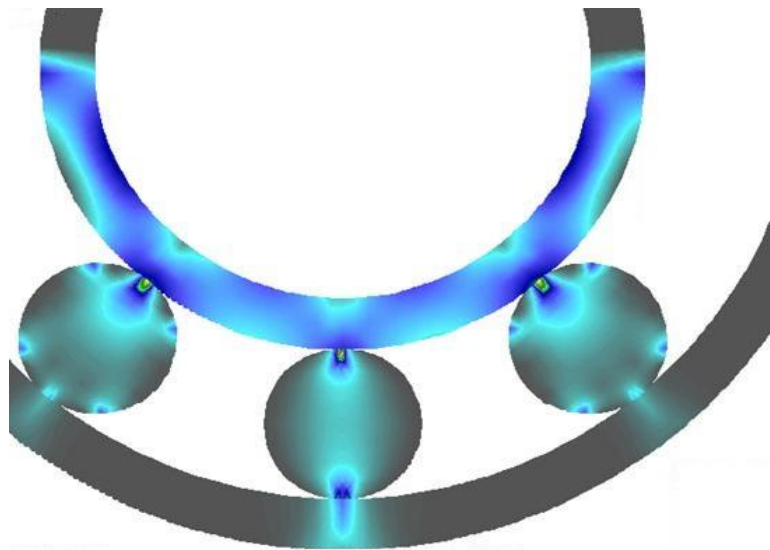


Figure 1.1. Example of the analysis of Hertz contact problem for a roller bearing

- Good convergence behavior,
- Contact detection at integration points,
- Wide-range of field of use.

Moreover, when any two curved bodies are in contact if they will initially touch each other at one or more points. The first analysis of this situation with using two curved bodies was conducted by Heinrich Hertz in 1881 and this analysis was set up by making the assumptions below,

- The contact surfaces of the curved bodies are continuous, frictionless, smooth and discordant,
- The contact area is much smaller than the contact bodies,
- Each contact surface can contemplate as an elastic half-plane

1.1.Literature Review

The application and the utilization of the composite materials was greatly increased for some engineering applications, particularly in aerospace, automotive and marine industry. With the increment on the usage of the application and the utilization of the composite materials, functionally graded materials (FGMs) have come into view. The concept of functionally graded material (FGM) was first contemplated in Japan in the mid-1980s during a space plane project for withstanding higher surface temperatures. Then it became popular in other countries, especially in Germany. The aim of proposing a functionally graded material (FGM) is to benefit from all the best material properties from the forming materials to alleviate unwanted problems; such as, fatigue, corrosion and fracture and to improve the tribological performance of the material or the mating parts in the assembly. As mentioned above, with the increment in the usage of the functionally graded materials (FGMs) some studies about the basic concepts, classification, preparation, history and the fabrication processes (FGMs) has been conducted by Upuda, Rao and Gangadharan [1] and by Somiya [2]. Furthermore, processing techniques for functionally graded materials (FGMs) are well investigated by Kieback, Neubrand and Riedel in [3]. Also, potential applications and the challenges in processing for functionally graded materials (FGMs) can be found in [4] by Bhatnagar and Srivatsan and some studies about the

improvements in wear resistance can be found in [5] by Yue and Li and by Zhang, Sun and Wang in [6]. Mishina, Inumaru and Kaitoku showed some examples about the improved tribological characteristics of functionally graded materials (FGMs) and their potential use in technological applications such as using the functionally graded materials (FGMs) as joint prostheses [7]. Furthermore, information on using the functionally graded materials (FGMs) for the cutting tool technology can be found in [8] by Nomura, Moriguchi and Tsuda.

The literature of the original work for the contact mechanics has started with the publication of the paper “On the contact of elastic solids” at 1882 by Heinrich Hertz after solving the contact problem of the two elastic bodies which have curved surfaces [9]. Between 1886 and 1889 Hertz has published two articles that are well-known as the backbone of the field of contact mechanics. The Hertzian contact stress has recourse to the localized stress, which mostly arise as two curved surfaces that are in contact and deform slightly under the imposed loads. This deformation is mainly depending on the elastic modulus of the materials that are in contact. In general, contact mechanics can be defined as the mechanics of interaction of deformable bodies that touch each other in a contact zone called “contact area” or “contact patch” [10]. Contact problems and contact mechanics are the main key for the mechanical problems and the solid mechanics, because applying contact is the most doctrine method for applying loads to a deformable body, the resulting stress concentration area is often the most critical area in the body. In many applications in engineering, the loads are mostly transferred between the parts in the assembly by contact regions with the presence of static and dynamic friction. Because of the stress concentrations and the sensitivity of the contact region, it is important to find the contact stresses to estimate the life span of the part or the assembly accurately. If the high stress concentrations can’t be measured correctly, the part or the assembly may fail due to wear, fatigue or fracture. On the other hand, unavoidable roughness of contacting surfaces generate complex local structures and this situation may lead to frictional heating and extreme local temperatures on the contact area. Barber and Ciavarella has summarized the field of contact mechanics in [11] in a very comprehensive manner. An extensive description about contact mechanics with examples can be found in [12], [13] and [14] respectively.

There are many publications for the contact mechanics problems for isotropic materials. The analytical solutions for the homogeneous isotropic contact mechanics problems are well

investigated by Mindlin in [15], Spence in [16] and Muskhelishvili in [17]. However, studies and research for the area of orthotropic materials and orthotropic contact mechanics problems are slightly limited if compared with the isotropic materials and isotropic contact mechanics problems. Especially the studies for thin films, fiber-reinforced composites, crystals and coatings, piezoelectric materials are much less common. Guler has analytically solved the problem for two-dimensional frictional contact intended by a rigid and parabolic punch with using the Jacobi polynomials in [18]. Kucuksucu, Guler and Avci has investigated the effects of the wedge-shaped punch intention on an orthotropic elastic half-plane in [19]. The explicit formulae and the fundamental solutions for the surface stresses are obtained by Zhou, Lee and Jang in [20]. They have shown that, the coefficient of friction has a more important role than the moving velocity in the contact problem. Zhou and Lee investigated the sliding frictional contact of the triangular and cylindrical punch profiles for the monoclinic piezoelectric materials in [21] and in [22] they have conducted the exact analysis of the frictional contact of the anisotropic piezoelectric materials indented by a flat and semi-parabolic punch profiles. Furthermore, again Zhou and Lee obtained the exact solutions of the 2D frictionless contact model for the orthotropic piezoelectric materials indented by a rigid sliding punch with a real fundamental solution approach in [23]. They have presented the effects of dimensionless sliding speed and the various mechanical-electrical and geometrical loadings on the surface in-plane stress and surface contact stress.

The solution for graded orthotropic contact problem is rather different from the homogeneous contact solutions because of the elastic gradation in the lateral direction, vertical direction or in both directions. Giannakopoulos and Pallot have provided an analytical closed form solution for the two dimensional contact problem for graded elastic material intended by a rigid cylinder [24]. Ke and Wang has solved the two-dimensional frictional contact mechanical problem with indiscriminately varying elastic properties in [25]. Kucuksucu, Guler and Avci has investigated the contact stresses on a two-dimensional sliding contact problem between a functionally graded orthotropic half-plane and a rigid punch and wedge-shaped punch with an arbitrary profile in [26].

There are two different techniques used by a majority while conducting finite element analysis with working on functionally graded materials for achieving spatial alterations in material

properties. While using the functionally graded finite element approach, the material properties calculate at each Gauss point of finite element and these material properties form the elemental stiffness matrix [27]. Conversely, the material properties point out at the centroid of each finite element in the homogeneous finite element approach [28]. Therefore, one can achieve a uniform material distribution over the given finite element model with using the homogeneous finite element approach. Moreover, with the help of the convenient size of the meshing and appropriate degree of mesh refinement, one can easily generate the same numerical results with using either method [29], [30].

Dag, Guler, Yildirim and Ozatag have achieved in solving the sliding frictional contact of an elastic graded half-plane by using a rigid punch both analytically and numerically in [31]. They performed a work to show the influence of the lateral non-homogeneity and the coefficient of friction on the contact stresses. The contact mechanics of thin films bonded to graded coatings are investigated by Guler, Gulver and Nart [32]. They used the finite element method while generating the numerical results and showed that mechanical and geometrical parameters and at the same time the loading type has a significant effect on the stress distribution and the on the thin film. The two-dimensional nonlinear partial slip contact problem for a graded half-plane has conducted by Vanini, Shahba and Kordani in [33]. They have carried out a finite element analysis to match the results that derived from the analytical solutions with the help of using single integral equations (SIEs) to find the effect of the friction coefficient on contact stresses, stress intensity factors and failure loads. Furthermore, Abhilash and Murthy have achieved to solve the response for an elastic punch that in contact with a functionally graded material with using an innovative finite element analysis for to handle the complex geometry of the material and the boundary conditions in [34]. They have considered different types of indenters on the functionally graded geometry; such as, flat and cylindrical indenters. Comparing the numerical results with the analytical results has a crucial importance while using the dual approach methodology. Hence, controlling the mesh convergence rate and the error percentage of the energy norm in the finite element models has a great importance. Demirhan and Kanber has conducted an article to investigate the slip and stick regions and contact stresses on the functionally graded coated elastic members in [35] with constantly checking the percent error in energy norm in finite element models and comparing mesh convergence rate with analytical results. In [36] the contact stresses for the linear, the elastic-perfect plastic and the elastic-

nonlinear hardening material properties are investigated by Sohoul, Goudarzi and Alashti. They have also calculated the contact stresses that derived from the rough surfaces analytically and compared them with the numerical results that obtained with using finite element method.

1.2.The Aim of the Thesis

Contact problems and contact mechanics are the main key for the mechanical problems and solid mechanics, because, in many applications in engineering (e.g. brake discs, roller and cylindrical bearings, dove-tail connections), loads are transferred between the mating parts or assemblies with using contacts. The aim of this master thesis, is to achieve to find the effects of the effective stiffness, E , the shear parameter, κ , the stiffness ratio, δ^4 , and the effective Poisson's ratio, ν , inhomogeneity parameter and the coefficient of friction, η , on the contact stress and the in-plane stress distributions for the problem of the graded orthotropic half-plane intended by a sliding flat, triangular, circular and semi-circular punch profiles for the benchmarks and to compare the finite element solutions with the analytical results that are in the literature.

1.3.Problem Definition and Method

There is a wide variety of application fields for the problem that defined in this master thesis such as, brake discs, roller and cylindrical bearings and dove-tail connections, cylinder linings, abradable seals in the gas turbine components. In general, it is hard to observe the solutions for the contact mechanics problems, because the complexity of the problem mainly depends on the problem parameters. For instance, the solution for the frictionless contact mechanics problems are easier than the solution for the frictional contact mechanics problems. The graded orthotropic half-plane, that used in the scope of this master thesis, is modeled as a linearly elastic material and a varying Young's modulus in the depth direction is used. The principal axes of orthotropy are assumed as parallel and perpendicular to the contact zone and generalized plane stress assumption is used for the solution of the problem. The orthotropic graded half-plane is modeled by using quadratic finite elements and the material properties are defined at the center of each element for the finite element analysis.

The finite element models for the finite element solutions are created in 2D plane and the plane stress assumptions are used while obtaining the finite element solutions. The effects of the four independent material parameters; namely as the effective stiffness, E , the shear parameter, κ , the stiffness ratio, δ^4 , and the effective Poisson's ratio, ν , inhomogeneity parameter and the coefficient of friction, η , on the contact stress and the in-plane stress distributions for the problem of the graded orthotropic half-plane intended by a sliding flat, triangular, circular and semi-circular punch profiles are investigated in order to find the most effective variation of the parameters for the minimum resulting contact and in-plane stress values. The preparation of the finite element solutions contains the below steps,

- Calculation of the independent material parameters for the specified finite element model from the equations that given in Section 2.
- Entering the values of the independent material parameters, inhomogeneity parameter and the coefficient of the friction to the finite element code
- Defining the boundary conditions of the finite element model
- Defining the finite element mesh properties of the finite element model
- Running the finite element code in the finite element program (ANSYS 2016)
- Gathering the contact and in-plane stress values for the specified finite element model
- Normalizing the contact and in-plane stress values for the specified finite element model

Furthermore, the finite element solutions for the graded orthotropic half-plane problem intended by a sliding flat and triangular punch profiles for using various values of the four independent material properties, $(E, \delta^4, \kappa, \nu)$, inhomogeneity parameters, βa and βb , and the coefficient of friction, η , are compared with the analytical results that derived from the article in [26]. Moreover, the finite element solutions for the circular and semi-circular punch profiles are also compared with the analytical results that are in [46]. The derivation and the explanation for the four independent material properties, $(E, \delta^4, \kappa, \nu)$, can be found in the Section 2. Additionally, the dimensionless load $P^* = P/(ER)$, and the contact boundaries, a/R and b/R , that calculated from the both finite element solutions and the analytical results are compared. This comparison helped to figure out the effects of the elastic gradation on material properties in a more evident way.

2.DERIVATION OF THE MODELING PARAMETERS

In this section, the four independent material parameters for the finite element solutions for the graded orthotropic half-plane intended by a sliding flat, triangular, circular and semi-circular punch are obtained. These four independent material parameters are used in the finite element models in order to obtain less complicated finite element solutions, because, for the case of plane orthotropic elasticity, the solution is quite complicated than the solution for the isotropic elasticity. As mentioned in the article by Krenk in [37] and the article by Cinar and Erdogan in [38], the four independent material parameters, which are the effective stiffness (E), the effective Poisson's ratio (ν), the stiffness ratio (δ) and the shear parameter (κ), are replaced by the four independent material parameters, E_{11} , E_{22} , G_{12} and ν_{12} . Furthermore, some comprehensive studies about the 3D and plane orthotropic elasticity can be found in the books by Saad in [39] and by Lekhnitskii in [40].

The stress state for the 3D orthotropic elasticity can be summarized as,

$$\sigma_{11} = \sigma_{11}(x, y, z) \quad (1a)$$

$$\sigma_{22} = \sigma_{22}(x, y, z) \quad (1b)$$

$$\sigma_{33} = \sigma_{33}(x, y, z) \quad (1c)$$

$$\sigma_{12} = \sigma_{12}(x, y, z) \quad (1d)$$

$$\sigma_{13} = \sigma_{13}(x, y, z) \quad (1e)$$

$$\sigma_{23} = \sigma_{23}(x, y, z) \quad (1f)$$

The Poisson's ratios are,

$$\nu_{12} = -\frac{\varepsilon_{22}}{\varepsilon_{11}} \quad (2a)$$

$$\nu_{23} = -\frac{\varepsilon_{33}}{\varepsilon_{22}} \quad (2b)$$

$$\nu_{21} = -\frac{\varepsilon_{11}}{\varepsilon_{22}} \quad (2c)$$

$$\nu_{31} = -\frac{\varepsilon_{11}}{\varepsilon_{33}} \quad (2d)$$

$$\nu_{32} = -\frac{\varepsilon_{22}}{\varepsilon_{33}} \quad (2e)$$

$$\nu_{13} = -\frac{\varepsilon_{33}}{\varepsilon_{11}} \quad (2f)$$

The elastic moduli are,

$$E_{11} = -\frac{\sigma_{11}}{\varepsilon_{11}} \quad (3a)$$

$$E_{22} = -\frac{\sigma_{22}}{\varepsilon_{22}} \quad (3b)$$

$$E_{33} = -\frac{\sigma_{33}}{\varepsilon_{33}} \quad (3c)$$

Using the simplified stress state, the corresponding strain field for the 3D orthotropic can be derived as,

$$\varepsilon_{11} = \frac{\sigma_{11}}{E_{11}} - \frac{\nu_{12}\sigma_{22}}{E_{11}} - \frac{\nu_{13}\sigma_{33}}{E_{11}} \quad (4a)$$

$$\varepsilon_{22} = -\frac{\nu_{21}\sigma_{11}}{E_{22}} + \frac{\sigma_{22}}{E_{22}} - \frac{\nu_{23}\sigma_{33}}{E_{22}} \quad (4b)$$

$$\varepsilon_{33} = -\frac{\nu_{31}\sigma_{11}}{E_{33}} - \frac{\nu_{32}\sigma_{22}}{E_{33}} + \frac{\sigma_{33}}{E_{33}} \quad (4c)$$

$$\varepsilon_{12} = \left(\frac{1 + \nu_{12}}{E_{11}} \right) \sigma_{12} \quad (4d)$$

$$\varepsilon_{23} = \left(\frac{1 + \nu_{23}}{E_{22}} \right) \sigma_{23} \quad (4e)$$

$$\varepsilon_{13} = \left(\frac{1 + \nu_{13}}{E_{33}} \right) \sigma_{13} \quad (4f)$$

where $\sigma_{nm}, E_{nm}, \nu_{nm}$ (for $n, m = x, y, z$) are the corresponding stress state, elastic moduli and the Poisson's ratio respectively.

The shear moduli, that are defined as the ratio of shear stress to engineering shear strain on the loading plane are,

$$G_{12} = \frac{\sigma_{12}}{\varepsilon_{12} + \varepsilon_{21}} = \frac{\sigma_{12}}{2\varepsilon_{12}} = \frac{\sigma_{12}}{y_{12}} \quad (5a)$$

$$G_{23} = \frac{\sigma_{23}}{\varepsilon_{23} + \varepsilon_{32}} = \frac{\sigma_{23}}{2\varepsilon_{23}} = \frac{\sigma_{23}}{y_{23}} \quad (5b)$$

$$G_{13} = \frac{\sigma_{13}}{\varepsilon_{13} + \varepsilon_{31}} = \frac{\sigma_{13}}{2\varepsilon_{13}} = \frac{\sigma_{13}}{y_{13}} \quad (5c)$$

where y_{nm} (for $n, m = x, y, z$) are the engineering shear strains of orthotropic elasticity. Note that, the engineering shears strains are two times of the normal shear strains.

$$y_{12} = \varepsilon_{12} + \varepsilon_{21} = 2\varepsilon_{12} \quad (6a)$$

$$y_{23} = \varepsilon_{23} + \varepsilon_{32} \quad (6b)$$

$$y_{13} = \varepsilon_{13} + \varepsilon_{31} \quad (6c)$$

The transformations from shear moduli to elastic moduli with using equations (2a), (2b) and (2f) are,

$$G_{12} = \frac{E_{11}}{2(1 + \nu_{12})} \quad (7a)$$

$$G_{23} = \frac{E_{22}}{2(1+\nu_{23})} \quad (7b)$$

$$G_{13} = \frac{E_{33}}{2(1+\nu_{13})} \quad (7c)$$

The derivation of the shear strains of the 3D orthotropic elasticity with using equations (4d), (4e), (4f) and (7) are,

$$\begin{aligned} \varepsilon_{12} &= \left(\frac{1+\nu_{12}}{E_{11}} \right) \sigma_{12} \rightarrow \frac{1+\nu_{12}}{E_{11}} = \frac{\varepsilon_{12}}{\sigma_{12}} \rightarrow G_{12} = \frac{E_{11}}{2(1+\nu_{12})} \\ &\rightarrow G_{12} = \frac{\sigma_{12}}{2\varepsilon_{12}} \rightarrow 2\varepsilon_{12} = y_{12} = \frac{\sigma_{12}}{G_{12}} \end{aligned} \quad (8a)$$

$$\begin{aligned} \varepsilon_{23} &= \left(\frac{1+\nu_{23}}{E_{22}} \right) \sigma_{23} \rightarrow \frac{1+\nu_{23}}{E_{22}} = \frac{\varepsilon_{23}}{\sigma_{23}} \rightarrow G_{23} = \frac{E_{22}}{2(1+\nu_{23})} \\ &\rightarrow G_{23} = \frac{\sigma_{23}}{2\varepsilon_{23}} \rightarrow 2\varepsilon_{23} = y_{23} = \frac{\sigma_{23}}{G_{23}} \end{aligned} \quad (8b)$$

$$\begin{aligned} \varepsilon_{13} &= \left(\frac{1+\nu_{13}}{E_{33}} \right) \sigma_{13} \rightarrow \frac{1+\nu_{13}}{E_{33}} = \frac{\varepsilon_{13}}{\sigma_{13}} \rightarrow G_{13} = \frac{E_{33}}{2(1+\nu_{13})} \\ &\rightarrow G_{13} = \frac{\sigma_{13}}{2\varepsilon_{13}} \rightarrow 2\varepsilon_{13} = y_{13} = \frac{\sigma_{13}}{G_{13}} \end{aligned} \quad (8c)$$

With the above definitions of stresses and strains, the 3D orthotropic elasticity can be summarized as,

$$[E] = [S][\sigma] \quad (9a)$$

$$\begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \varepsilon_{33} \\ 2\varepsilon_{12} \\ 2\varepsilon_{23} \\ 2\varepsilon_{13} \end{bmatrix} = \begin{bmatrix} \frac{1}{E_{11}} & -\frac{\nu_{12}}{E_{11}} & -\frac{\nu_{13}}{E_{11}} & 0 & 0 & 0 \\ -\frac{\nu_{21}}{E_{22}} & \frac{1}{E_{22}} & -\frac{\nu_{23}}{E_{22}} & 0 & 0 & 0 \\ -\frac{\nu_{31}}{E_{33}} & -\frac{\nu_{32}}{E_{33}} & \frac{1}{E_{33}} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{G_{12}} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{G_{23}} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{G_{13}} \end{bmatrix} \begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{12} \\ \sigma_{23} \\ \sigma_{13} \end{bmatrix} \quad (9b)$$

Because of the 3D orthotropic elasticity matrix is symmetrical, the following equations can be written,

$$\nu_{12}E_{22} = \nu_{21}E_{11} \quad (10a)$$

$$\nu_{23}E_{33} = \nu_{32}E_{22} \quad (10b)$$

$$\nu_{31}E_{11} = \nu_{13}E_{33} \quad (10c)$$

Furthermore, while working with a 2D reference plane, the z-components of both stress and strain field will vanish from the elasticity equations. In order to maintain a stress field independent of z, there must be no body forces or surface tractions on z direction.

The stress state for the 2D or plane orthotropic elasticity can be summarized as,

$$\sigma_{11} = \sigma_{11}(x, y) \quad (11a)$$

$$\sigma_{22} = \sigma_{22}(x, y) \quad (11b)$$

$$\sigma_{33} = 0 \quad (11c)$$

$$\sigma_{12} = \sigma_{12}(x, y) \quad (11d)$$

$$\sigma_{13} = 0 \quad (11e)$$

$$\sigma_{23} = 0 \quad (11f)$$

The Poisson' ratios are,

$$\nu_{12} = -\frac{\varepsilon_{22}}{\varepsilon_{11}} \quad (12a)$$

$$\nu_{23} = 0 \quad (12b)$$

$$\nu_{21} = -\frac{\varepsilon_{11}}{\varepsilon_{22}} \quad (12c)$$

$$\nu_{31} = 0 \quad (12d)$$

$$\nu_{32} = 0 \quad (12e)$$

$$\nu_{13} = 0 \quad (12f)$$

The elastic moduli are,

$$E_{11} = -\frac{\sigma_{11}}{\varepsilon_{11}} \quad (13a)$$

$$E_{22} = -\frac{\sigma_{22}}{\varepsilon_{22}} \quad (13b)$$

$$E_{33} = 0 \quad (13c)$$

Using the simplified plane stress state, the corresponding strain field for the 2D orthotropic elasticity is,

$$\varepsilon_{11} = \frac{\sigma_{11}}{E_{11}} - \frac{\nu_{12}\sigma_{22}}{E_{11}} \quad (14a)$$

$$\varepsilon_{22} = -\frac{\nu_{21}\sigma_{11}}{E_{22}} + \frac{\sigma_{22}}{E_{22}} \quad (14b)$$

$$\varepsilon_{33} = 0 \quad (14c)$$

$$\varepsilon_{12} = \left(\frac{1 + \nu_{12}}{E_{11}} \right) \sigma_{12} \quad (14d)$$

$$\varepsilon_{23} = 0 \quad (14e)$$

$$\varepsilon_{13} = 0 \quad (14f)$$

where $\sigma_{nm}, E_{nm}, \nu_{nm}$ (for $n, m = x, y, z$) are the corresponding stress state, elastic moduli and the Poisson ratio respectively.

The shear moduli, which are defined as the ratio of shear stress to engineering shear strain on the loading plane, can be written as,

$$G_{12} = \frac{\sigma_{12}}{\varepsilon_{12} + \varepsilon_{21}} = \frac{\sigma_{12}}{2\varepsilon_{12}} = \frac{\sigma_{12}}{y_{12}} \quad (15a)$$

$$G_{23} = 0 \quad (15b)$$

$$G_{13} = 0 \quad (15c)$$

where y_{nm} (for $n, m = x, y, z$) are the engineering shear strains of orthotropic elasticity. The engineering shears strains are two times of the normal shear strains.

$$y_{12} = \varepsilon_{12} + \varepsilon_{21} \quad (16a)$$

$$y_{23} = 0 \quad (16b)$$

$$y_{13} = 0 \quad (16c)$$

The transformations from shear moduli to elastic moduli with using equations (12a) and (12c) are,

$$G_{12} = \frac{E_{11}}{2(1 + \nu_{12})} \quad (17a)$$

$$G_{23} = 0 \quad (17b)$$

$$G_{13} = 0 \quad (17c)$$

The derivation of the shear strains of the 2D orthotropic elasticity with using equations (4d), (4e), (4f) and (7) is as follows,

$$\begin{aligned} \varepsilon_{12} &= \left(\frac{1+\nu_{12}}{E_{11}} \right) \sigma_{12} \rightarrow \frac{1+\nu_{12}}{E_{11}} = \frac{\varepsilon_{12}}{\sigma_{12}} \rightarrow G_{12} = \frac{E_{11}}{2(1+\nu_{12})} \\ \rightarrow G_{12} &= \frac{\sigma_{12}}{2\varepsilon_{12}} \rightarrow 2\varepsilon_{12} = y_{12} = \frac{\sigma_{12}}{G_{12}} \end{aligned} \quad (18a)$$

$$\varepsilon_{23} = 0 \quad (18b)$$

$$\varepsilon_{23} = 0 \quad (18c)$$

With the above definitions of stresses and strains, the 2D orthotropic elasticity can be summarized as,

$$[E] = [S][\sigma] \quad (19a)$$

$$\begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \varepsilon_{33} \\ 2\varepsilon_{12} \\ 2\varepsilon_{23} \\ 2\varepsilon_{13} \end{bmatrix} = \begin{bmatrix} \frac{1}{E_{11}} & -\frac{\nu_{12}}{E_{11}} & 0 & 0 & 0 & 0 \\ -\frac{\nu_{21}}{E_{22}} & \frac{1}{E_{22}} & 0 & 0 & 0 & 0 \\ -\frac{\nu_{31}}{E_{33}} & -\frac{\nu_{32}}{E_{33}} & \frac{1}{E_{33}} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{G_{12}} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{12} \\ \sigma_{23} \\ \sigma_{13} \end{bmatrix} \quad (19b)$$

For generalized plane stress conditions, there is no stress and strain components in z direction. Thus, the variables that are containing z-component can be remove from the matrix formulation. The modified 2D orthotropic elasticity matrix becomes as,

$$\begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ 2\varepsilon_{12} \end{bmatrix} = \begin{bmatrix} \frac{1}{E_{11}} & -\frac{\nu_{12}}{E_{11}} & 0 \\ -\frac{\nu_{21}}{E_{22}} & \frac{1}{E_{22}} & 0 \\ 0 & 0 & \frac{1}{G_{12}} \end{bmatrix} \begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \end{bmatrix} \quad (19c)$$

Because of the 2D orthotropic elasticity matrix is symmetric, the following equation can also be written,

$$\nu_{12}E_{22} = \nu_{21}E_{11} \quad (20)$$

While working with orthotropic materials in 2D with using the generalized plane stress conditions, instead of using the independent engineering constants directly, the following transformations can be used in order to obtain the following independent material parameters as,

effective stiffness,

$$E = \sqrt{E_{11}E_{22}} \quad (21)$$

the effective Poisson ratio,

$$\nu = \sqrt{\nu_{12}\nu_{21}} \quad (22)$$

the stiffness ratio,

$$\delta^4 = \frac{E_{11}}{E_{22}} = \frac{\nu_{12}}{\nu_{21}} \quad (23)$$

the shear parameter,

$$\kappa = \frac{1}{2} \sqrt{E_{11}E_{22}} \left(\frac{1}{G_{12}} - \frac{\nu_{12}}{E_{11}} - \frac{\nu_{21}}{E_{22}} \right) \quad (24)$$

With using the parameters that are defined above, the matrix components in equation (19c) can be written in a simpler form in order to simplify the calculations.

From the equations (21) and (23),

$$\delta^{-4} = \frac{E_{22}}{E_{11}} \quad (25a)$$

$$(\delta^{-4})^{1/2} = \left(\frac{E_{22}}{E_{11}} \right)^{1/2} \quad (25b)$$

$$\delta^{-2} = \sqrt{\frac{E_{22}}{E_{11}}} \quad (25c)$$

$$\frac{\delta^{-2}}{E} = \frac{\sqrt{E_{22}}}{\sqrt{E_{11}}} \frac{1}{\sqrt{E_{11}E_{22}}} = \frac{1}{E_{11}} \quad (25d)$$

From the equations (21) and (22),

$$\frac{\nu}{E} = \frac{\sqrt{\nu_{12}\nu_{21}}}{\sqrt{E_{11}E_{22}}} \quad (26)$$

Using equation (20), the equation (26) becomes,

$$\frac{\nu}{E} = \frac{\sqrt{\nu_{12}\nu_{21}}}{\sqrt{E_{11}E_{22}}} = \sqrt{\frac{\nu_{12}}{E_{11}}} \sqrt{\frac{\nu_{21}}{E_{11}}} = \frac{\nu_{12}}{E_{11}} \quad (27a)$$

Similarly,

$$\frac{\nu}{E} = \frac{\sqrt{\nu_{12}\nu_{21}}}{\sqrt{E_{11}E_{22}}} = \sqrt{\frac{\nu_{21}}{E_{22}}} \sqrt{\frac{\nu_{21}}{E_{22}}} = \frac{\nu_{21}}{E_{22}} \quad (27b)$$

Using equations (25c) and (21),

$$\frac{\delta^2}{E} = \frac{\sqrt{E_{11}}}{\sqrt{E_{22}}} \frac{1}{\sqrt{E_{11}E_{22}}} = \frac{1}{E_{22}} \quad (28)$$

From the equation (24),

$$\kappa = \frac{1}{2} \frac{E}{G_{12}} - \frac{1}{2} \sqrt{E_{11}E_{22}} \left(\frac{\nu_{12}E_{22} + \nu_{21}E_{11}}{E_{11}E_{22}} \right) \quad (29a)$$

$$\kappa = \frac{E}{2G_{12}} - \frac{1}{2} \left(\frac{2\nu_{12}E_{22}}{\sqrt{E_{11}E_{22}}} \right) \quad (29b)$$

$$\kappa = \frac{E}{2G_{12}} - \left(\frac{\sqrt{\nu_{12}E_{22}}}{\sqrt{E_{11}E_{22}}} \right) \sqrt{\nu_{12}E_{22}} \quad (29c)$$

$$\kappa = \frac{E}{2G_{12}} - \sqrt{\frac{\nu_{12}}{E_{11}}} \sqrt{\nu_{12}E_{22}} \quad (29d)$$

$$\kappa = \frac{E}{2G_{12}} - \sqrt{\frac{\nu_{12}\nu_{21}E_{11}}{E_{11}}} \quad (29e)$$

$$\kappa = \frac{E}{2G_{12}} - \sqrt{\nu_{12}\nu_{21}} \quad (29f)$$

$$\kappa = \frac{E}{2G_{12}} - \nu \quad (29g)$$

With using the above equations, the 2D orthotropic elastic matrix in equation (19c) can be written in a new form,

$$\begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ 2\varepsilon_{12} \end{bmatrix} = \frac{1}{E} \begin{bmatrix} \delta^{-2} & -\nu & 0 \\ -\nu & \delta^2 & 0 \\ 0 & 0 & 2(\kappa + \nu) \end{bmatrix} \begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{12} \end{bmatrix} \quad (30)$$

The term $\frac{2(\kappa + \nu)}{E}$ in equation (30) can be obtained as follows,

$$\frac{2(\kappa + \nu)}{E} = \frac{2}{\sqrt{E_{11}E_{22}}} \left[\frac{1}{2} \sqrt{E_{11}} \sqrt{E_{22}} \left(\frac{1}{G_{12}} - \frac{\nu_{12}}{E_{11}} - \frac{\nu_{21}}{E_{22}} \right) + \sqrt{\nu_{12}} \sqrt{\nu_{21}} \right] \quad (31a)$$

$$\frac{2(\kappa + \nu)}{E} = \left(\frac{1}{G_{12}} - \frac{\nu_{12}}{E_{11}} - \frac{\nu_{21}}{E_{22}} \right) + \left(\frac{2\sqrt{\nu_{12}\nu_{21}}}{\sqrt{E_{11}}\sqrt{E_{22}}} \right) \quad (31b)$$

$$\frac{2(\kappa + \nu)}{E} = \left(\frac{1}{G_{12}} - \frac{\nu_{12}}{E_{11}} - \frac{\nu_{21}}{E_{22}} \right) + \left(2\sqrt{\frac{\nu_{12}}{E_{11}}}\sqrt{\frac{\nu_{21}}{E_{22}}} \right) \quad (31c)$$

$$\frac{2(\kappa + \nu)}{E} = \frac{1}{G_{12}} \quad (31d)$$

The case of generalized plane strain condition is very similar to the case of generalized plane stress condition. As seen in the equations (14c), 14(e) and (14f) some of the strain components are zero.

From equation (4c),

$$\frac{-\nu_{31}}{E_{33}}\sigma_{11} - \frac{-\nu_{32}}{E_{33}}\sigma_{22} + \frac{\sigma_{33}}{E_{33}} = 0 \quad (32a)$$

$$\sigma_{33} = \nu_{31}\sigma_{11} + \nu_{32}\sigma_{22} \quad (32b)$$

If the equation (32b) substitutes in the equations (4a) and (4b),

$$\varepsilon_{11} = \frac{\sigma_{11}}{E_{11}} - \frac{\nu_{12}\sigma_{22}}{E_{11}} - \frac{\nu_{13}}{E_{11}}(\nu_{31}\sigma_{11} + \nu_{32}\sigma_{22}) \quad (33a)$$

$$\varepsilon_{11} = \frac{1}{E_{11}}(1 - \nu_{13}\nu_{31})\sigma_{11} - \frac{1}{E_{11}}(\nu_{12} + \nu_{13}\nu_{32})\sigma_{22} \quad (33b)$$

$$\varepsilon_{22} = -\frac{\nu_{21}\sigma_{11}}{E_{22}} + \frac{\sigma_{22}}{E_{22}} - \frac{\nu_{23}}{E_{22}}(\nu_{31}\sigma_{11} + \nu_{32}\sigma_{22}) \quad (33c)$$

$$\varepsilon_{22} = -\frac{1}{E_{22}}(\nu_{21} + \nu_{23}\nu_{31})\sigma_{11} + \frac{1}{E_{22}}(1 - \nu_{23}\nu_{32})\sigma_{22} \quad (33d)$$

With the above definitions, the 2D orthotropic elasticity matrix for generalized plane strain conditions can be written as,

$$\begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ 2\varepsilon_{12} \end{bmatrix} = \begin{bmatrix} \frac{(1-\nu_{13}\nu_{31})}{E} & -\frac{1}{E_{11}}(\nu_{12}+\nu_{13}\nu_{32}) & 0 \\ -\frac{1}{E_{22}}(\nu_{21}+\nu_{23}\nu_{31}) & \frac{(1-\nu_{23}\nu_{32})}{E} & 0 \\ 0 & 0 & \frac{1}{G_{12}} \end{bmatrix} \begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{12} \end{bmatrix} \quad (34)$$

Because of the elasticity matrix is again similar, equation (20) takes the form of

$$\frac{\nu_{12}+\nu_{13}\nu_{32}}{E_{11}} = \frac{\nu_{21}+\nu_{23}\nu_{31}}{E_{22}} \quad (35)$$

Instead of direct use of elastic constants, the following transformations can be used to simplify the mathematical operations. These transformations are,

effective stiffness,

$$E = \sqrt{\frac{E_{11}E_{22}}{(1-\nu_{13}\nu_{31})(1-\nu_{23}\nu_{32})}} \quad (36)$$

the effective Poisson ratio,

$$\nu = \sqrt{\frac{(\nu_{12}+\nu_{13}\nu_{32})(\nu_{21}+\nu_{23}\nu_{31})}{(1-\nu_{13}\nu_{31})(1-\nu_{23}\nu_{32})}} \quad (37)$$

the stiffness ratio,

$$\delta^4 = \frac{E_{11}}{E_{22}} \frac{1-\nu_{23}\nu_{32}}{1-\nu_{13}\nu_{31}} \quad (38)$$

the shear parameter,

$$\kappa = \frac{1}{2} \sqrt{\frac{E_{11}E_{22}}{(1-\nu_{13}\nu_{31})(1-\nu_{23}\nu_{32})}} \left(\frac{1}{G_{12}} - \frac{\nu_{12}+\nu_{13}\nu_{32}}{E_{11}} - \frac{\nu_{21}+\nu_{23}\nu_{31}}{E_{22}} \right) \quad (39)$$

In terms of these parameters, the 2D orthotropic elasticity matrix for generalized plane strain in equation (34) conditions can be modified with using below equations.

From the equations (36) and (38)

$$\delta^{-4} = \frac{E_{22}}{E_{11}} \frac{1 - \nu_{13}\nu_{31}}{1 - \nu_{23}\nu_{32}} \quad (40a)$$

$$(\delta^{-4})^{1/2} = \left(\frac{E_{22}}{E_{11}} \frac{1 - \nu_{13}\nu_{31}}{1 - \nu_{23}\nu_{32}} \right)^{1/2} \quad (40b)$$

$$\frac{(\delta^{-4})^{1/2}}{E} = \left[\frac{\left(\frac{E_{22}}{E_{11}} \frac{1 - \nu_{13}\nu_{31}}{1 - \nu_{23}\nu_{32}} \right)}{\left(\frac{E_{11}E_{22}}{(1 - \nu_{13}\nu_{31})(1 - \nu_{23}\nu_{32})} \right)} \right]^{1/2} = \frac{1}{E_{11}} (1 - \nu_{13}\nu_{31}) \quad (40c)$$

From the equations (35), (36) and (37)

$$\frac{\nu}{E} = \left[\frac{(\nu_{12} + \nu_{13}\nu_{32})(\nu_{21} + \nu_{23}\nu_{31})}{(1 - \nu_{13}\nu_{31})(1 - \nu_{23}\nu_{32})} \right]^{1/2} \left[\frac{(1 - \nu_{13}\nu_{31})(1 - \nu_{23}\nu_{32})}{E_{11}E_{22}} \right]^{1/2} \quad (41a)$$

$$\frac{\nu}{E} = \left[\frac{(\nu_{12} + \nu_{13}\nu_{32})}{E_{11}} \right]^{1/2} \left[\frac{(\nu_{21} + \nu_{23}\nu_{31})}{E_{22}} \right]^{1/2} \quad (41b)$$

$$\frac{\nu}{E} = \left[\frac{(\nu_{12} + \nu_{13}\nu_{32})}{E_{11}} \right]^{1/2} \left[\frac{(\nu_{12} + \nu_{13}\nu_{32})}{E_{11}} \right]^{1/2} \quad (41c)$$

$$\frac{\nu}{E} = \left[\frac{(\nu_{12} + \nu_{13}\nu_{32})}{E_{11}} \right] \quad (41d)$$

From the equations (36) and (40b),

$$\frac{\delta^2}{E} = \left[\frac{\frac{E_{11}}{E_{22}} \left(\frac{1 - \nu_{23}\nu_{32}}{1 - \nu_{13}\nu_{31}} \right)}{\frac{E_{11}E_{22}}{(1 - \nu_{13}\nu_{31})(1 - \nu_{23}\nu_{32})}} \right] \quad (42a)$$

$$\frac{\delta^2}{E} = \frac{1 - \nu_{23}\nu_{32}}{E_{22}} \quad (42b)$$

From the equation (39),

$$\begin{aligned} \kappa = & \frac{1}{2} \sqrt{\frac{E_{11}E_{22}}{(1 - \nu_{13}\nu_{31})(1 - \nu_{23}\nu_{32})}} \frac{1}{G_{12}} \\ & - \frac{1}{2} \sqrt{\frac{E_{11}E_{22}E_{11}E_{22}}{(1 - \nu_{13}\nu_{31})(1 - \nu_{23}\nu_{32})}} \left[\frac{\nu_{12} + \nu_{13}\nu_{32}}{E_{11}} + \frac{\nu_{21} + \nu_{23}\nu_{31}}{E_{22}} \right] \end{aligned} \quad (43a)$$

$$\kappa = \frac{E}{2G_{12}} - \nu \quad (43b)$$

The term in equation (30) is also can be written as same,

$$\frac{2(\kappa + \nu)}{E} = \frac{1}{G_{12}} \quad (44)$$

The 2D orthotropic elasticity matrix in equation (30) can also be simplified for the generalized plane strain conditions with using above transformations.

$$\begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ 2\varepsilon_{12} \end{bmatrix} = \frac{1}{E} \begin{bmatrix} \delta^{-2} & -\nu & 0 \\ -\nu & \delta^2 & 0 \\ 0 & 0 & 2(\kappa + \nu) \end{bmatrix} \begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{12} \end{bmatrix} \quad (45)$$

From using the equation (30),

$$\varepsilon_{11} = \frac{1}{E} \delta^{-2} \sigma_{11} - \frac{1}{E} \nu \sigma_{22} \quad (46a)$$

$$\varepsilon_{22} = -\frac{1}{E} \nu \sigma_{11} + \frac{1}{E} \delta^2 \sigma_{22} \quad (46b)$$

From the equations (46a) and (46b),

$$\sigma_{22} = \frac{E}{\delta^2} \varepsilon_{22} + \frac{\nu}{\delta^2} \sigma_{11} \quad (47a)$$

$$\varepsilon_{11} = \frac{1}{E} \delta^{-2} \sigma_{11} - \frac{1}{E} \nu \left[\frac{E}{\delta^2} \varepsilon_{22} + \frac{\nu}{\delta^2} \sigma_{11} \right] \quad (47b)$$

$$\varepsilon_{11} = \frac{1}{E} \delta^{-2} \sigma_{11} - \frac{1}{E} \frac{\nu E \varepsilon_{22}}{\delta^2} + \frac{1}{E} \frac{\nu^2}{\delta^2} \sigma_{11} \quad (47c)$$

$$\varepsilon_{11} E = \delta^{-2} \sigma_{11} - \frac{\nu E \varepsilon_{22}}{\delta^2} + \frac{\nu^2}{\delta^2} \sigma_{11} \quad (47d)$$

$$\sigma_{11} = \frac{E}{(1-\nu^2)} \left[\delta^2 \varepsilon_{11} + \nu \varepsilon_{22} \right] \quad (47e)$$

where E , δ^2 and ν can be taken from equations (21), (22) and (25c) respectively.

From the equations (47a) and (47e),

$$\sigma_{22} = \frac{E}{\delta^2} \varepsilon_{22} + \frac{\nu}{\delta^2} \left[\frac{E}{(1-\nu^2)} (\delta^2 \varepsilon_{11} + \nu \varepsilon_{22}) \right] \quad (48a)$$

$$\sigma_{22} = \frac{E}{\delta^2} \varepsilon_{22} + \left(\frac{\nu}{1-\nu^2} \right) E \varepsilon_{11} + \left(\frac{\nu^2}{1-\nu^2} \right) \frac{E}{\delta^2} \varepsilon_{22} \quad (48b)$$

$$\sigma_{22} = \frac{E}{\delta^2} \varepsilon_{22} \left(1 + \frac{\nu^2}{1-\nu^2} \right) + \left(\frac{\nu}{1-\nu^2} \right) E \varepsilon_{11} \quad (48c)$$

$$\sigma_{22} = \frac{E}{\delta^2} \varepsilon_{22} \left(\frac{1-\nu^2+\nu^2}{1-\nu^2} \right) + \left(\frac{\nu}{1-\nu^2} \right) E \varepsilon_{11} \quad (48d)$$

$$\sigma_{22} = \frac{E}{\delta^2} \varepsilon_{22} \left(\frac{1}{1-\nu^2} \right) + \left(\frac{\nu}{1-\nu^2} \right) E \varepsilon_{11} \quad (48e)$$

$$\sigma_{22} = \left(\frac{E}{1-\nu^2} \right) (\varepsilon_{22} \delta^{-2} + \nu \varepsilon_{11}) \quad (48f)$$

From the 2D orthotropic elasticity matrix in equation (30),

$$\sigma_{12} = \left(\frac{E}{\kappa + \nu} \right) \varepsilon_{12} \quad (49)$$

Then the 2D orthotropic elasticity matrix in equation (30) can be written as,

$$\begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{12} \end{bmatrix} = \frac{E}{1-\nu^2} \begin{bmatrix} \delta^2 & \nu & 0 \\ \nu & \delta^{-2} & 0 \\ 0 & 0 & \frac{1-\nu^2}{\kappa+\nu} \end{bmatrix} \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ 2\varepsilon_{12} \end{bmatrix} \quad (50)$$

3.FINITE ELEMENT METHOD

The finite element method is a numerical technique for solving the boundary value problems and partial differential equations approximately. Structural analyses, heat transfer, fluid flow and mass transfer problems are the examples of the solvable problems by using the finite element method. For the problems that have complicated geometries, loadings and boundary conditions, it is practically impossible to obtain the analytical solutions because the analytical solutions usually require to solve the complicated ordinary and partial differential equations. On the other hand, the finite element method divides a large and complicated problem into smaller and simpler parts called finite elements. After solving these finite elements separately, the general solution will be obtained by combining these finite elements into a one bigger solution. In general, the solution of the structural problems usually requires to find the displacements at each node and the stress components in each element that is subjected to the applied loads. The first references for the finite element method is conducted by Hrennikoff [41] in 1941 on the field of structural engineering and by McHenry [42] in 1943, who used a one dimensional line elements in order to solve the stress field in the continuous solids.

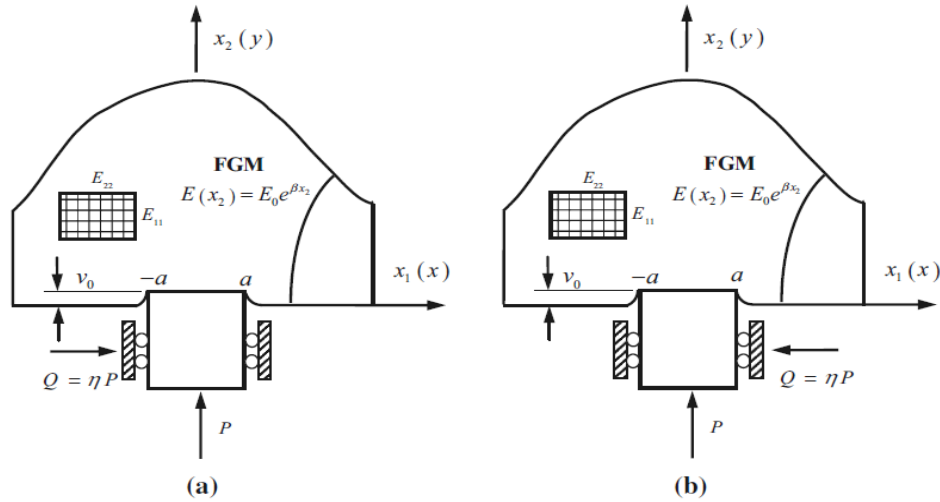


Figure 3.1. Functionally graded orthotropic half-plane in a sliding frictional contact intended by a flat punch.

The advantages of the finite elements over the whole geometry are given below [43].

- Easy representation of the final solution
- Accurate representation of the whole geometry

- Inclusion of dissimilar material properties
- Capture of local effects

The finite element analyses are solved by using the finite element analysis software called ANSYS (2016). In order to ensure the contact compatibility at the contact interface, ANSYS (2016) offers several different contact formulations. These contact formulations are given below.

- Pure Penalty
- Augmented Lagrange, Normal Lagrange
- Multi-Point Constraint Contact Formulation
- Beam Contact Formulation

Because of the contact and target bodies do not interpenetrate, it is needed to establish a sufficient relationship between the two surfaces to prevent them from passing through each other. Hence, it is an exigence to ensure the contact compatibility first. For the graded orthotropic half-plane problem that intended by a flat, triangular, circular and semi-circular punch profiles, pure penalty contact formulation is used.

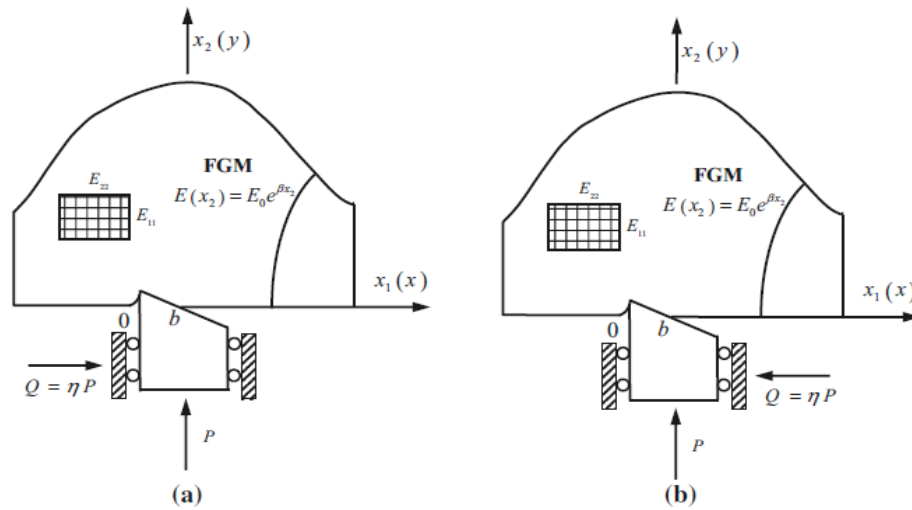


Figure 3.2. Functionally graded orthotropic half-plane in a sliding frictional contact intended by a triangular punch.

This contact formulation detects the contact with the use of integration points and it offers some good features; such as, good convergence behavior, contact detection at integration points and

it can be used with any type of contact behavior [44]. As will be shown in the results section, a very good matching between the analytical and numerical results are obtained with using the pure penalty contact formulation. The Figures for the graded orthotropic half-plane problem intended by a flat, triangular, circular and semi-circular punch profiles are illustrated in the Figure 3.1, Figure 3.2, Figure 3.3 and Figure 3.4 respectively.

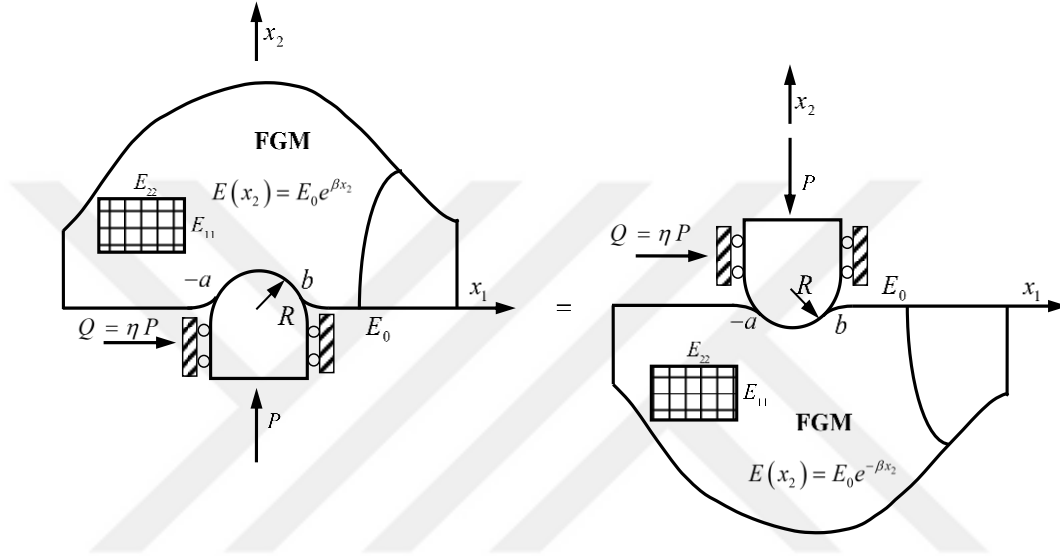


Figure 3.3. Functionally graded orthotropic half-plane in a sliding frictional contact intended by a circular punch.

The functionally graded half-plane is modeled by using 8-noded quadrilateral elements and the model of the element is shown in the Figure 3.5. In ANSYS (2016), *plane183*, *target169* and *conta172* contact line elements are used for defining the both punch profiles and the graded half-planes and they are illustrated in the Figure 3.6.

The plane element *plane183* is a higher order 2D, 8-noded or 6-noded element. It has quadratic displacement behavior and it is appropriate for modeling irregular meshes, such as produced by other CAD/CAM systems like, DS Solidworks, CATIA, ANSA etc., and it has two degrees of freedom at each node. The plane element *plane183* can be used for plane stress, plane strain and generalized plane strain conditions. Furthermore, it can also be used as an axisymmetric element. This element offers plasticity, hyperelasticity, creep, stress stiffening, large deflection and large strain capabilities. Assumptions and the restrictions for the *plane183* element are,

- The element must lie in a global x-y plane as illustrated in Figure 3.7.
- The area of the plane element must be greater than zero.
- A face with a removed midside node implies that the displacement varies linearly, rather than parabolically, along that face
- Use at least two elements to avoid hourglass mode
- Stress stiffening is always included in geometrically nonlinear analyses
- Enhanced strain formulation must be used in order to modeling the modal cyclic symmetry analyses.

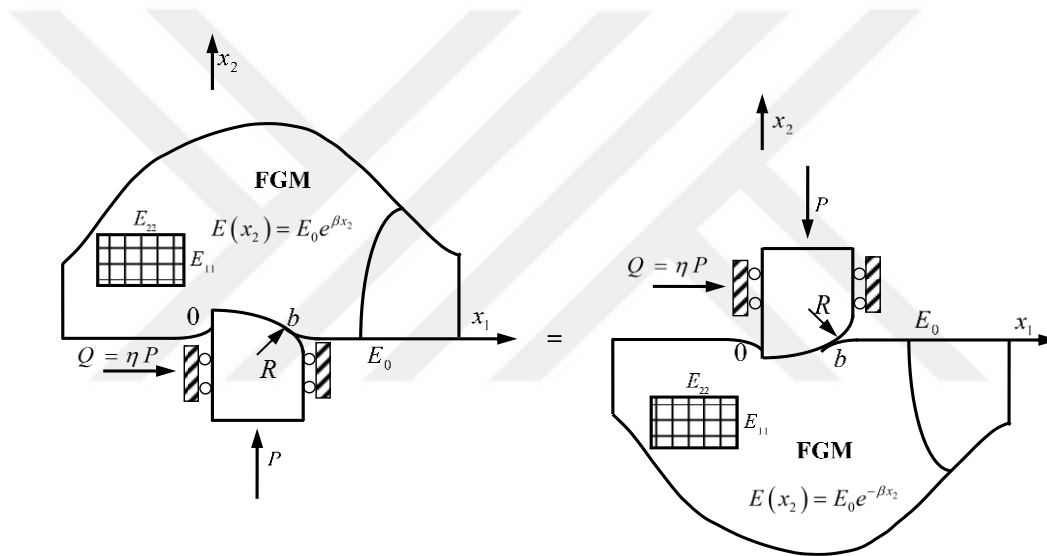


Figure 3.4. Functionally graded orthotropic half-plane in a sliding frictional contact intended by a semi-circular punch.

The *targe169* target element is used to illustrate various 2D target surfaces for the involving contact elements. The contact elements usually describe the boundary of a deformable body and are potentially in contact with the target surface, defined by *targe169*. Any translational or rotational displacement, temperature, voltage and magnetic potential can be imposed on the *targe169* element. Moreover, forces and moments can also be imposed on the target element *targe169*. Note that, for rigid targets, *targe169* can easily model complex target shapes. For flexible targets, these elements will overlay the solid elements describing the boundary of the deformable body. Note that, each target surface must be associated with only one contact surface. Here are some assumptions and restrictions for the *targe169* are given.

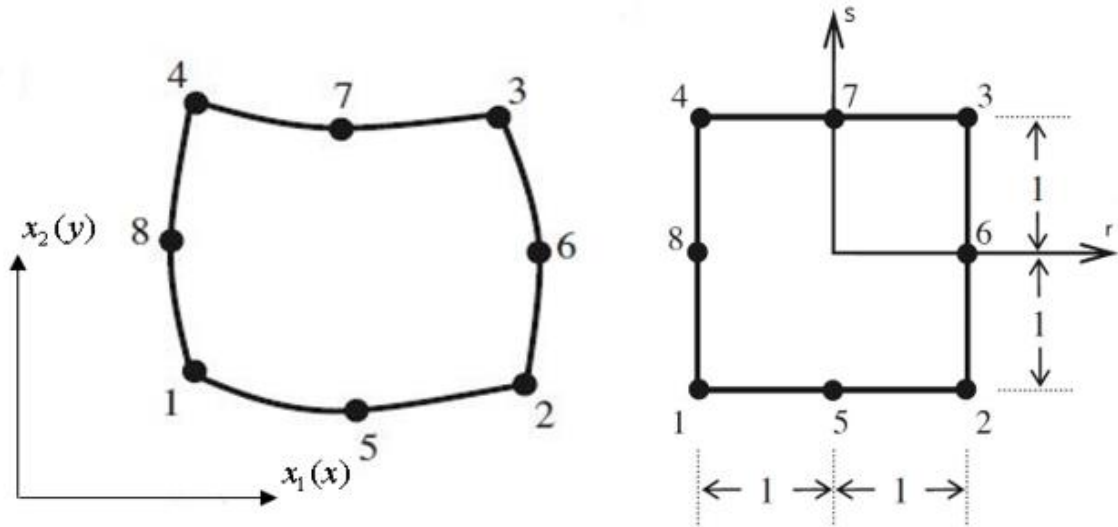


Figure 3.5. The quadrilateral finite element in the global and the local coordinate systems

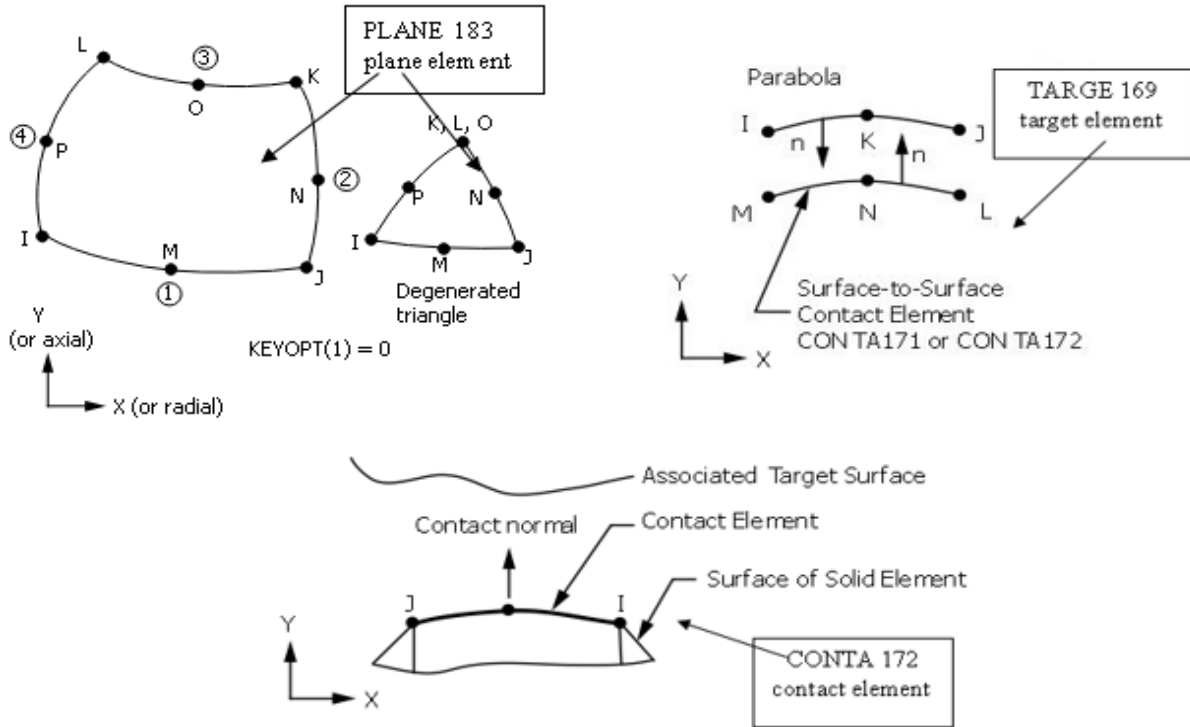


Figure 3.6. *Plane183*, *target169* and *conta172* contact elements

- The 2D segment element must be defined in an x-y plane.
- For circular arcs, the third node defines the actual center of the circle and must be defined accurately when the element is generated and must be moved consistently with the other nodes during the deformation process. If the third node is not moved consistently with

the other nodes, the arc shape will change with that node's movement. To ensure the correct behavior, apply all boundary conditions to a pilot node.

- For parabolic segments, the third point must lie at the middle of the parabola.
- The circle target segment is not supported in a general contact definition.

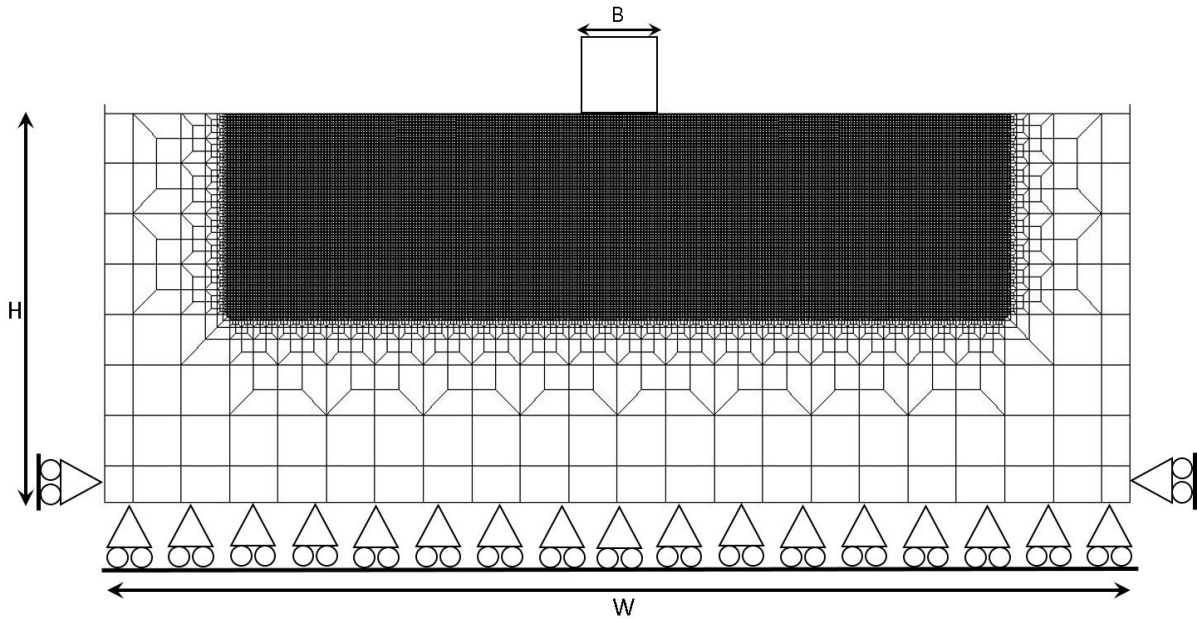


Figure 3.7. The finite element mesh and the punch profile used in the analysis of the flat punch problem.

The *conta172* contact element is used to represent contact and sliding between 2D target surfaces and a deformable surface, defined by this element. The *conta172* element can be used for modeling the D structural and coupled-field contact analyses and it can be used for pair-based contact definition. In a pair-based contact definition, the 2D contact elements (*conta172*) are associated with 2D target segment elements (*targe169*) with using a shared real constant set. The target surface can be either rigid or deformable. For rigid flexible contact, the rigid surface must be represented by a target surface. For flexible-flexible contact, one of the deformable surfaces must be overlaid by a target surface. Assumptions and the restrictions for the contact element *conta172* are given below.

- The 2-D contact element must be defined in an X-Y plane and the Y-axis must be the axis of symmetry for axisymmetric analyses.

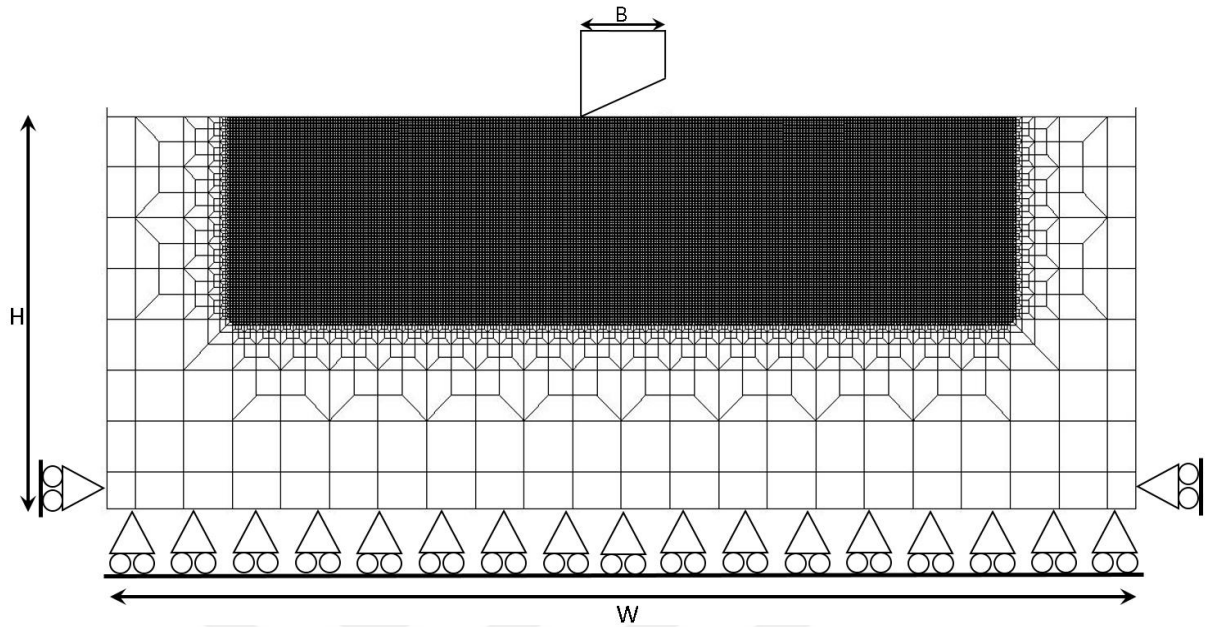


Figure 3.8. The finite element mesh and the punch profile used in the analysis of the triangular punch problem.

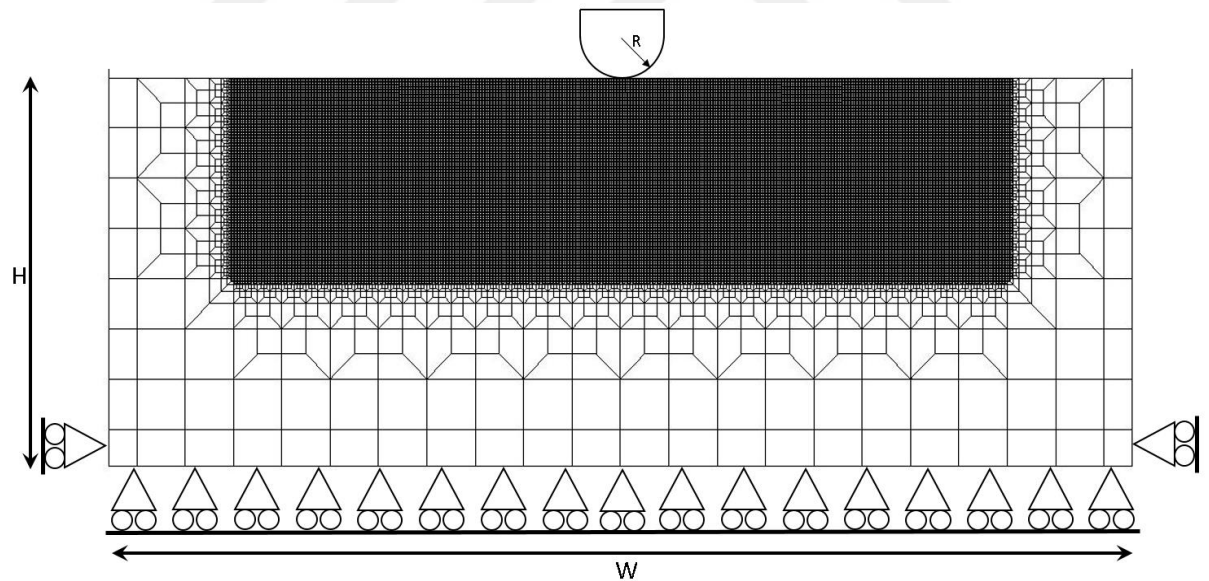


Figure 3.9. The finite element mesh and the punch profile used in the analysis of the circular punch problem.

- An axisymmetric structure should be modeled in the +X quadrants.
- The 2D contact element can be used by 3D target elements.
- *Conta172* can't be used with axisymmetric harmonic elements.

- The normal contact stiffness factor (FKN) must not be so large as to cause numerical instability.
- When nodal detection is used and the contact node is on the axis of symmetry in an axisymmetric analysis, the contact pressure on that node is not accurate since the area of the node is zero. The contact force is accurate in this situation.

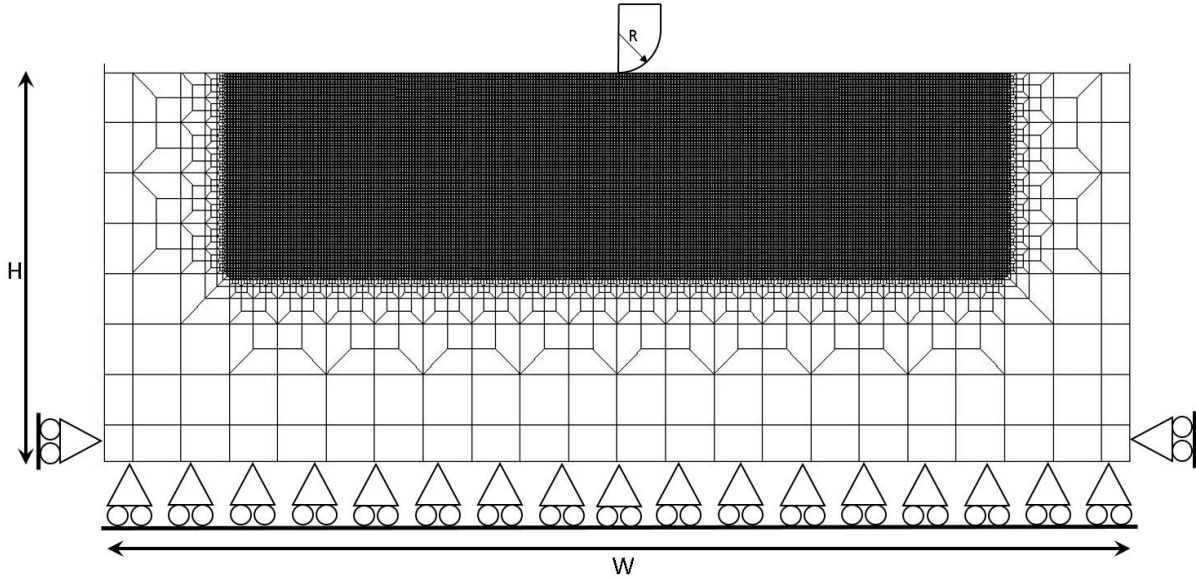


Figure 3.10. The finite element mesh and the punch profile used in the analysis of the semi-circular punch problem.

Note that, rigid target elements are used to represent the punch profiles. Solving the problem for the graded orthotropic contact mechanics is rather a bit complicated in ANSYS (2016). Thus, an extensive guide about how to use the commands for the program in [45] by ANSYS Inc. helped to solve the problem. The finite element mesh for the graded orthotropic half-plane loaded by a flat punch is shown in the Figure 3.7. The same mesh profiles are used in order to obtain the finite element solutions for the triangular, circular and semi-circular punch profiles respectively. The mesh for the finite element model comprises a total of 105857 quadrilateral elements and 318476 nodes. In order to not observe the sharp stress variations at the leading and the trailing edges of the contact zone, the mesh density is slightly increased from the edges of the half-plane to the contact zone. The rigid flat punch profile is connected to the graded orthotropic half-plane with using 400 contact line elements. On the other hand, the contact line elements for the triangular, circular and semi-circular punch profiles are 400, 900 and 900

respectively. Furthermore, the finite element mesh used in the analysis of the triangular, circular and semi-circular punch problems is illustrated in the Figures 3.8, 3.9 and 3.10. The dimensional parameters for the graded orthotropic half-plane problem intended by a flat and triangular punch profiles are the height and the width of the graded medium and the width of the punch profile and they are donated by, H , W and B respectively. For the circular and the semi-circular punch profiles, the radius of the punch is donated by R sign.

In order to preclude the negative effects of the problem boundaries, the dimensions for the graded medium are chosen as H/W is equal to $1/2$ and B/W is equal to $1/10$ for the flat and the triangular punch profiles and again H/W is chosen as $1/2$ for the circular and semi-circular punch profiles.

4.RESULTS AND DISCUSSIONS

4.1.Results And Discussions For The Graded Orthotropic Half-Plane Problem Loaded By A Rigid Flat Punch

In this section, the comparison between the finite element solutions and the analytical results for the problems that described in the Figs. 3.1, 3.2, 3.3 and 3.4 are given. As seen in these Figures, the graded orthotropic half-plane is loaded by a sliding flat, triangular, circular and semi-circular punch profiles. In order to simplify the 2D plane orthotropic elasticity equations, the four independent engineering constants, E_{11} , E_{22} , G_{12} and ν_{12} are replaced with the four independent material parameters; namely as the effective stiffness, E , the stiffness ratio, δ^4 , the shear parameter, κ , and the effective Poisson's ratio, ν , [30] and the derivation for the four independent material parameters for the plane stress and plane strain suppositions are given in Eqs. (21, 22, 23 and 29g).

The comparison of the finite element solutions for the graded orthotropic half-plane intended by a flat punch is conducted by using the analytical results that are given in the article by Kucuksucu, Guler and Avci in [19]. The collation of the contact stress distribution between the finite element solutions and the analytical results can be found in the Figures 4.1, 4.2, 4.3, 4.4 and 4.5 respectively. The non-dimensional material properties (E , δ^4 , κ , ν), the coefficient of friction, η , and the inhomogeneity parameter, βa , are chosen between the ranges ($-0.25 \leq \kappa \leq 5.0$, $0.25 \leq \delta^4 \leq 5.0$, $1/7 \leq \nu \leq 5/7$, $0.0 \leq \eta \leq 0.9$, $0.0 \leq \beta a \leq 1.0$) for the finite element solutions. Furthermore, for the convergence of the finite element solutions to the analytical results, the non-dimensional material properties also need to be in the following range ($-1 < \kappa < \infty$, $0.0 < \nu < 1.0$, $\kappa + \nu > 0.0$). Note that, the plane stress supposition is used for obtaining the independent material parameters for the analyses. The contact stress distribution is normalized by using $\sigma_0 = P/2a$ for the Figures 4.1, 4.2, 4.3, 4.4 and 4.5.

As seen in Figure 4.1, the contact stress distribution for the graded isotropic behavior is given in order make a comparison between the finite element solutions and the analytical results. The effect of the inhomogeneity parameter, βa , is investigated for the both frictionless and frictional contact, where $\eta = 0.0$ and $\eta = 0.5$ respectively. The stiffness ratio, δ^4 , and the shear

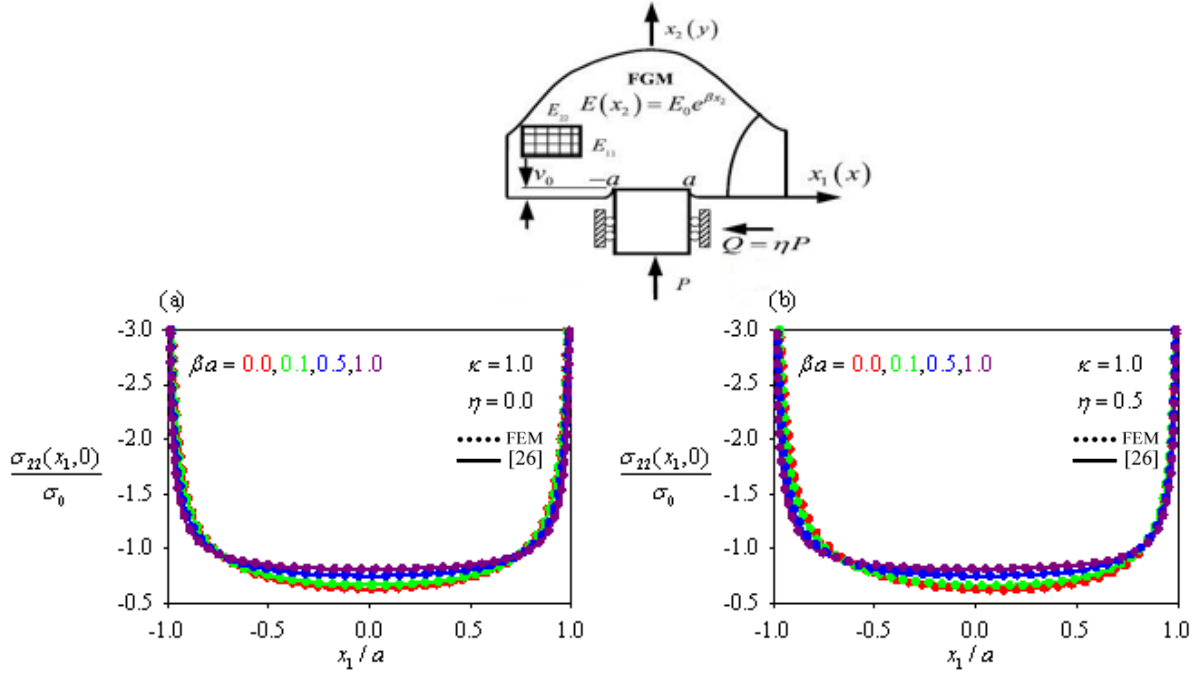


Figure 4.1. Contact stress, $\sigma_{22}(x_1, 0)$, distribution at the surface of the graded orthotropic half-plane loaded by a flat punch for various values of the inhomogeneity parameter, βa with the parameters $\kappa = 1.0, \nu = 3/7, \sigma_0 = P/2a$, (a) $\eta = 0.0, \delta^4 = 1.0$, (b) $\eta = 0.5, \delta^4 = 1.0$

parameter, κ , are chosen as, $\delta^4 = 1.0$ and $\kappa = 1.0$. Note that, the effective Poisson's ratio is fixed at $\nu = 3/7$. An impeccable match is obtained between the finite element solutions and the

In the Figure 4.2, the analogy of the effect of the inhomogeneity parameter, βa , and the shear parameter, κ , is investigated for the values of the coefficient of friction $\eta = 0.0$ and $\eta = 0.5$. Again an excellent convergence behavior is captured between the analytical and the finite element solutions. It can be easily seen from the Figure 4.2 that, the inhomogeneity parameter, βa , has a prominent effect on the contact stress distribution. For the frictionless contact, that is shown in Figures 4.2(a), 4.2(b) and 4.2(c), where $\eta = 0.0$, as the inhomogeneity parameter, βa , increases, the curves of the contact stresses, $\sigma_{22}(x_1, 0)$, becomes more straight at the center parts. Moreover, because of the coefficient of friction is equal to $\eta = 0.0$ in the frictionless contact, the contact stress distribution is symmetrical. For the Figures 4.2(d), 4.2(e) and 4.2(f) the contact stress distribution is not symmetrical because of the frictional contact, where $\eta = 0.5$. It can also be seen that as shear parameter, κ , changes, the contact stresses begin to prone to

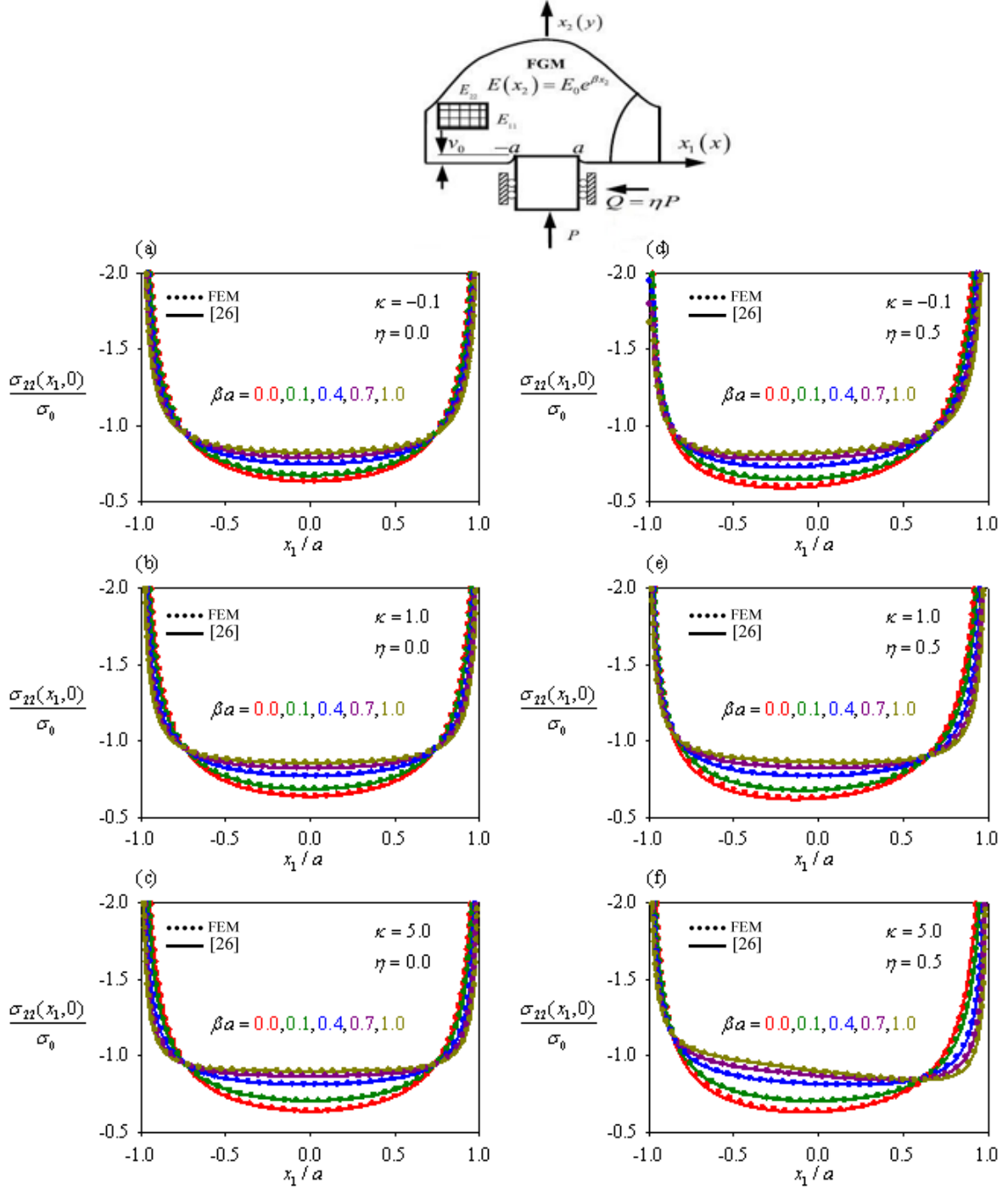


Figure 4.2. Contact stress, $\sigma_{22}(x_1, 0)$, distribution at the surface of the graded orthotropic half-plane loaded by a flat punch for various values of the inhomogeneity parameter, βa with the parameters, $\delta^4 = 0.2$, $\nu = 3/7$, $\sigma_0 = P/2a$, (a) $\eta = 0.0$, $\kappa = -0.1$, (b) $\eta = 0.0$, $\kappa = 1.0$, (c) $\eta = 0.0$, $\kappa = 5.0$, (d) $\kappa = -0.1$, (e) $\eta = 0.5$, $\kappa = 1.0$, (f) $\eta = 0.5$, $\kappa = 5.0$

either to the leading or to the trailing edge for both frictionless and frictional contact. Again, the shear parameter, κ , has a quiet large effect to the frictional contact, $\eta = 0.5$, than the frictionless contact, $\eta = 0.0$ and this can be seen flawlessly with crosschecking the Figures (a) to (f) in Figure 4.2. Note that, as expected, the contact stress distribution are singular at the leading and the trailing edges of the flat punch because of the sharp edges $(-a, a)$ of the punch. The Figure 4.3 shows a quiet apprehensible demonstration for the effect of the stiffness ratio, δ^4 , and the shear parameter, κ , on the contact stress distribution. The stiffness ratio, δ^4 , is chosen between $0.25 < \delta^4 < 2.0$ and the shear parameter, κ , is chosen between $-0.25 < \kappa < 2.0$ for the given finite element solutions. In the Figures 4.3(a), 4.3(b) and 4.3(c), the contact stress distribution is plotted for the frictionless contact case, where $\eta = 0.0$. It can be easily seen that, as the stiffness ratio, δ^4 , increases, the curves of the contact stresses becomes more straight at the center parts and the change on the shear parameter, κ , affects the contact stress distribution for the both frictionless and frictional contact. Furthermore, as the coefficient of friction changes from $\eta = 0.0$ to $\eta = 0.9$, the curves of the contact stresses begin to prone either to the leading or to the trailing edge of the contact. As an example, in the Figures 4.3(d), 4.3(e) and 4.3(f), for the frictional contact, the curves of the contact stresses prone to the leading edge for the shear parameter is $\kappa = -0.25$ and begin to prone to the trailing edge as the shear parameter, κ , increases to $\kappa = 2.0$. The effective Poisson's ratio, ν , and the inhomogeneity parameter, βa , is fixed at $\nu = 3/7$ and $\beta a = 1.0$ for the frictionless and frictional contact cases. In addition, because of the flat punch's leading and trailing edges are sharp, the contact stress distribution is singular at the $(-a, a)$ edges of the contact.

In the Figures 4.4(a) and 4.4(b), the effect of the shear parameter, κ , and the effect of the stiffness ratio, δ^4 , is investigated. The coefficient of friction, η , and the inhomogeneity parameter, βa , have the following constant values, $\eta = 0.5$ and $\beta a = 1.0$. For the Figure 4.4(a), one can easily observe that, as the shear parameter, κ , increases, the curves of the contact stresses prone to the trailing edge of the contact and as the shear parameter, κ , decreases, the curves of the contact stresses prone to the leading edge of the contact. In Figure 4.4(b), however, this alteration is much smaller than the Figure 4.4(a), because of the stiffness ratio, δ^4 , has a

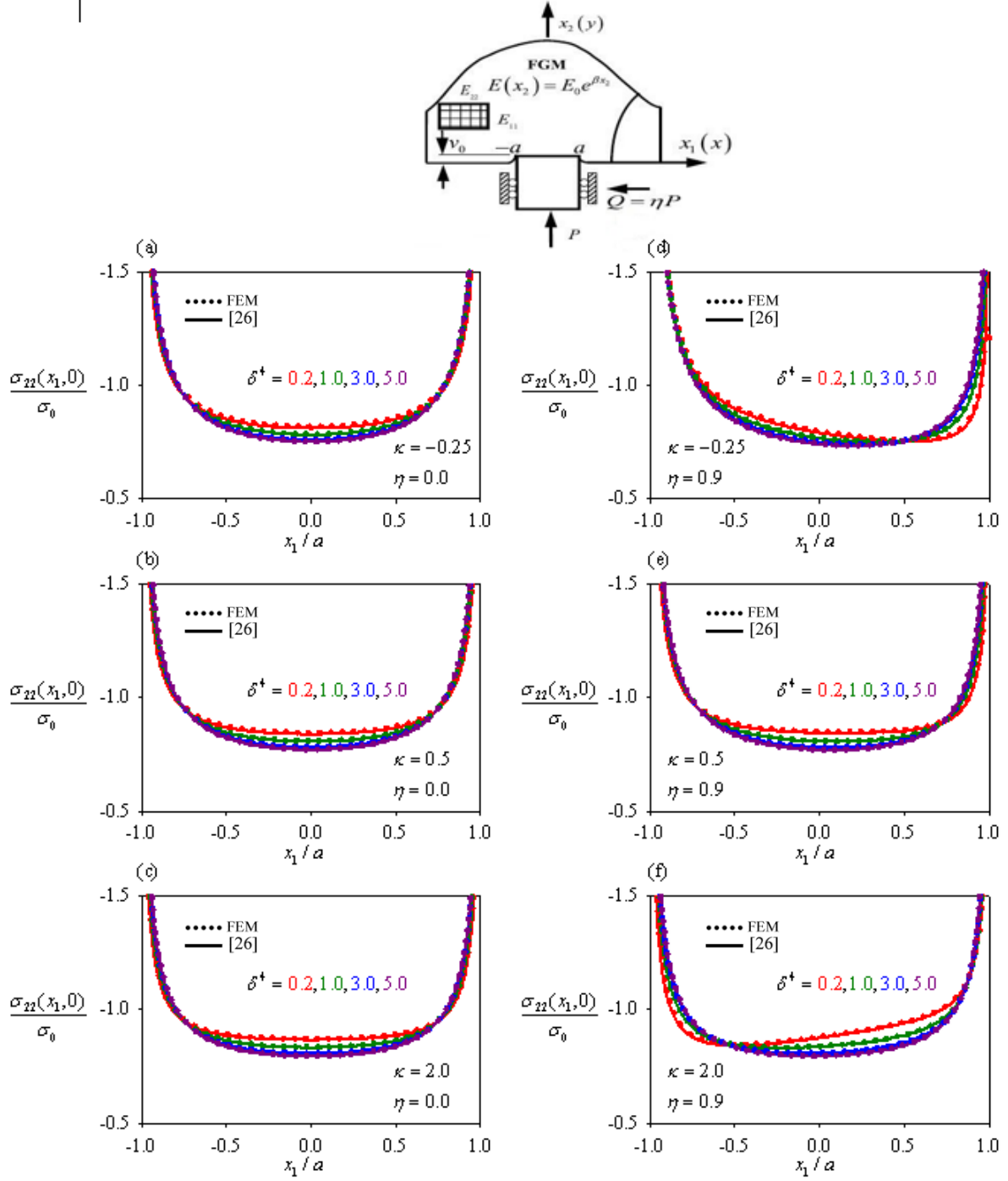


Figure 4.3. Contact stress, $\sigma_{22}(x_1, 0)$, distribution at the surface of the graded orthotropic half-plane loaded by a flat punch for various values of the stiffness ratio, δ^4 , with the parameters, $\beta a = 1.0, \nu = 3/7, \sigma_0 = P/2a$, (a) $\eta = 0.0, \kappa = -0.25$, (b) $\eta = 0.0, \kappa = 0.5$, (c) $\eta = 0.0, \kappa = 2.0$, (d) $\eta = 0.5, \kappa = -0.25$, (e) $\eta = 0.5, \kappa = 0.5$, (f) $\eta = 0.5, \kappa = 2.0$

very high value of $\delta^4 = 2.0$. The Figures 4.4(c) and 4.4(d) has the presentation of the effects of the effective Poisson's ratio, ν , and the stiffness ratio, δ^4 , on the contact stress distribution. As the effective Poisson's ratio, ν , increases, the curves of the contact stresses prone to the trailing edge from the leading edge for the stiffness ratio, δ^4 , is constant at $\delta^4 = 0.25$. Moreover, the effect of the effective Poisson's ratio, ν , is slightly smaller in the Figure 4.4(d) than the Figure 4.4(c) because of the higher value of the stiffness ratio, δ^4 . Note that, when the x_1 / a is equal to zero, the resulting value of the contact stresses is bigger for the Figures 4.4(b) and 4.4(d) than the Figures 4.4(a) and 4.4(c). Again, the effective Poisson's ratio, ν , is fixed at $\nu = 3/7$ for the Figures 4.4(a) and 4.4(b) and the stiffness ratio, κ is fixed at $\kappa = 0.5$ for the Figures 4.4(c) and 4.4(d) respectively.

The Figure 4.5 is mainly focused on the effect of the coefficient of friction, η , with using the various values of the shear parameter, κ , and the stiffness ratio, δ^4 . From the Figures 4.5(a) to 4.5(c), the shear parameter, κ , is increased from $\kappa = -0.25$ to $\kappa = 2.0$ while the stiffness ratio, δ^4 , is fixed at $\delta^4 = 0.25$. It can easily be seen that, the coefficient of friction, η , has a major effect on the contact stress distribution for the smaller values of the shear parameter, κ , and the stiffness ratio, δ^4 . For the Figure 4.5(a) and 4.5(d), as the coefficient of friction, η , increases, the curves of the contact stress distribution prone to the leading edge of the contact. However, as the shear parameter, κ , and the stiffness ratio, δ^4 , increases, it becomes quite difficult to observe the effect of the coefficient of friction, η . Furthermore, it can easily be seen that, the curves of the contact stress distribution prone to the leading edge of the contact for the lower values of the shear parameter, κ . As the shear parameter increases, the curves of the contact stress distribution prone to the trailing edge of the contact. Also, the curves of the contact stress distribution are nearly straight at the center parts for the value of the stiffness ratio is $\kappa = 0.5$. Moreover, the value of the contact stresses at the middle of the flat punch is larger for the higher values of the stiffness ratio, δ^4 and one can observe this by comparing the contact stress distributions on the Figures 4.5(b) and 4.5(e). Note that, the effective Poisson's ratio, ν , and the inhomogeneity parameter, βa , is not changing and have a constant value at, $\nu = 3/7$ and $\beta a = 1.0$ between the Figures 4.5(a) to 4.5(f). In addition to, the curves of the contact stress

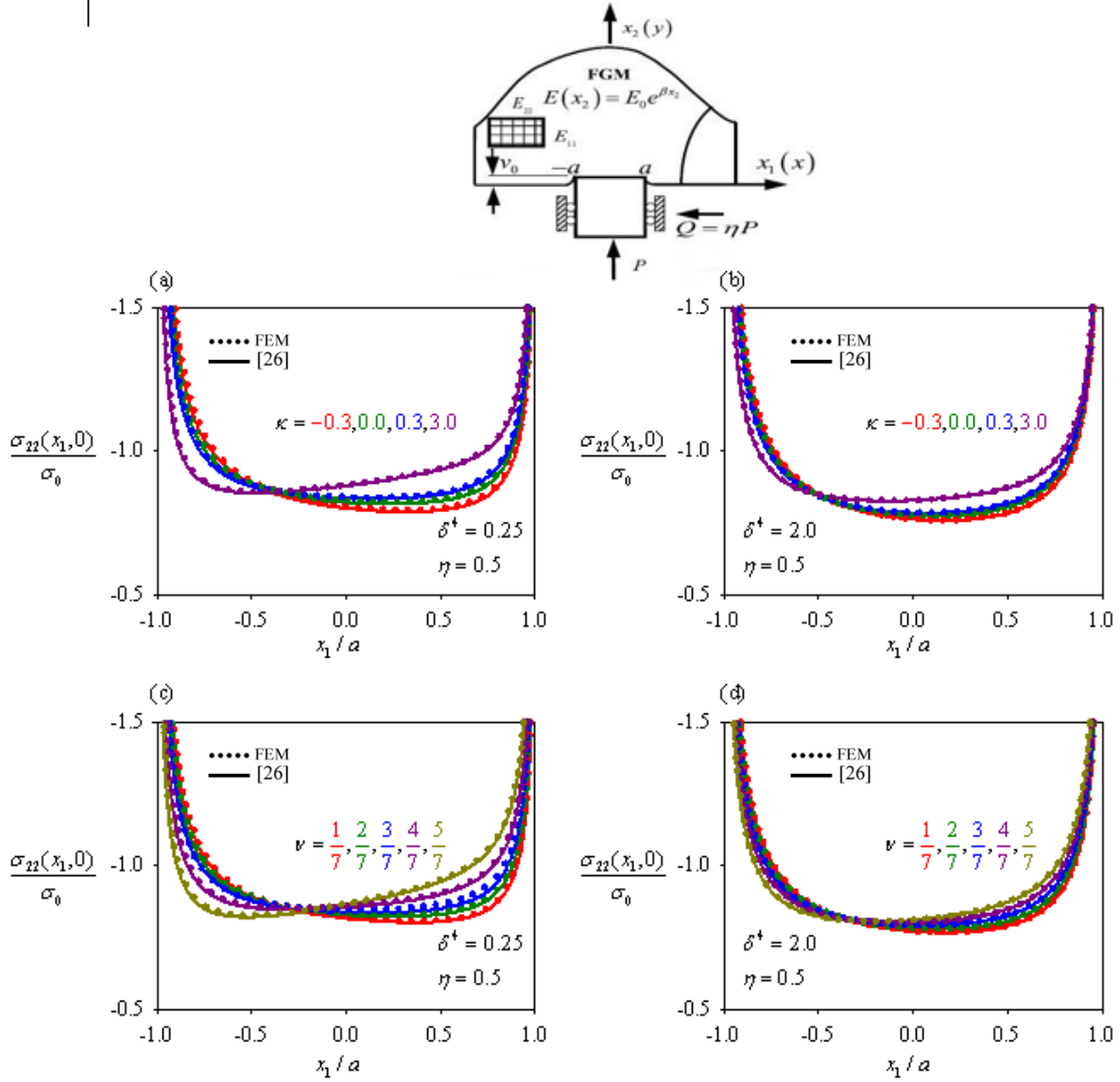


Figure 4.4. Contact stress, $\sigma_{22}(x_1, 0)$, distribution at the surface of the graded orthotropic half-plane loaded by a flat punch for various values of the shear parameter, κ , and the effective Poisson's ratio, ν , with the parameters, $\beta a = 1.0, \eta = 0.5, \sigma_0 = P/2a$, (a) $\delta^4 = 0.25, \nu = 3/7$, (b) $\delta^4 = 2.0, \nu = 3/7$, (c) $\delta^4 = 0.25, \kappa = 0.5$, (d) $\delta^4 = 2.0, \kappa = 0.5$

distribution are not symmetrical because of the value of the coefficient of friction, η , is not equal to zero.

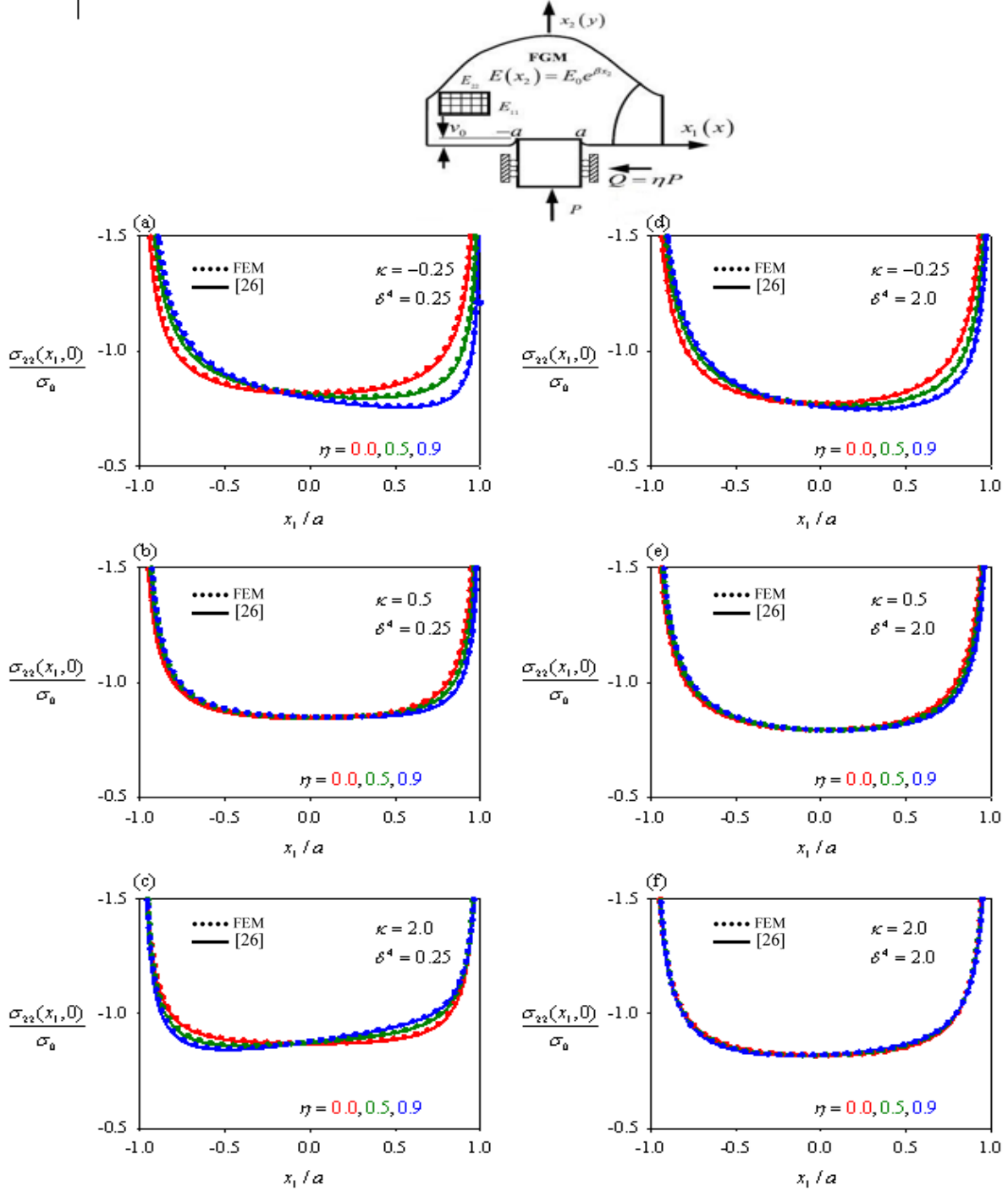


Figure 4.5. Contact stress, $\sigma_{22}(x_1, 0)$, distribution at the surface of the graded orthotropic half-plane loaded by a flat punch for various values of the coefficient of friction, η , with the parameters, $\beta a = 1.0$, $\nu = 3/7$, $\sigma_0 = P/2a$, (a) $\delta^4 = 0.25, \kappa = -0.25$, (b) $\delta^4 = 0.25, \kappa = 0.5$, (c) $\delta^4 = 0.25, \kappa = 2.0$, (d) $\delta^4 = 2.0, \kappa = -0.25$, (e) $\delta^4 = 2.0, \kappa = 0.5$, (f) $\delta^4 = 2.0, \kappa = 2.0$

4.2. Results And Discussions For The Graded Orthotropic Half-Plane Problem Loaded By A Rigid Triangular Punch

The finite element results of the contact stress distribution of the graded orthotropic half-plane problem intended by a triangular punch is given in Figures 4.6, 4.7, 4.8 respectively. The finite element solutions are compared with the analytical results in [19]. Again, the non-dimensional material properties (E , δ^4 , κ , ν), the coefficient of friction, η , and the inhomogeneity parameters, βb , are chosen between the ranges, ($-0.25 \leq \kappa \leq 3.0$, $0.2 \leq \delta^4 \leq 2.0$, $1/7 \leq \nu \leq 5/7$, $0.0 \leq \eta \leq 0.9$, $0.0 \leq \beta b \leq 2.0$) for the finite element analyses.

In the Figure 4.6, the contact stress distribution is plotted for the different values of the inhomogeneity parameter, βb , and the coefficient of friction, η . The values of the stiffness ratio δ^4 , the shear parameter, κ , and the effective Poisson's ratio, ν , are fixed at $\delta^4 = 0.2$, $\kappa = -0.1$ and $\nu = 3/7$ respectively. As it is readily apparent that, the effect of the coefficient of friction, η , on the contact stress distribution is quite large. For instance, for a constant value of the x_1/b , as the coefficient of friction, η , increases, the value of the contact stresses increases in every case in Figure 4.6. Moreover, it would be quiet wrong if the effect of the inhomogeneity parameter, βb , is ignored. The effect of the inhomogeneity parameter, βb , can easily be seen by comparing the Figures 4.6(c) and 4.6(d). As the inhomogeneity parameter, βb , increases, the resulting contact stress decreases.

In the Figure 4.7, the effect of the inhomogeneity parameter, βb , and the shear parameter, κ , on the contact stress distribution is given. An excellent match is obtained between the finite element solutions and the analytical results. For the frictionless contact, given in the Figures 4.7(a) and 4.7(c), where the $\eta = 0.0$, the inhomogeneity parameter, βb , has a major effect on the contact stresses. As the inhomogeneity parameter, βb , increases, the resulting contact stress distribution decreases. Furthermore, the effect of the inhomogeneity parameter, βb , is more evident for the larger shear parameter, κ , values. In the Figures 4.7(b) and 4.7(d), the effect of the inhomogeneity parameter is also investigated, but in under frictional contact condition, where the $\eta = 0.9$. Again, the effect of the inhomogeneity parameter, βb , is quite large on the

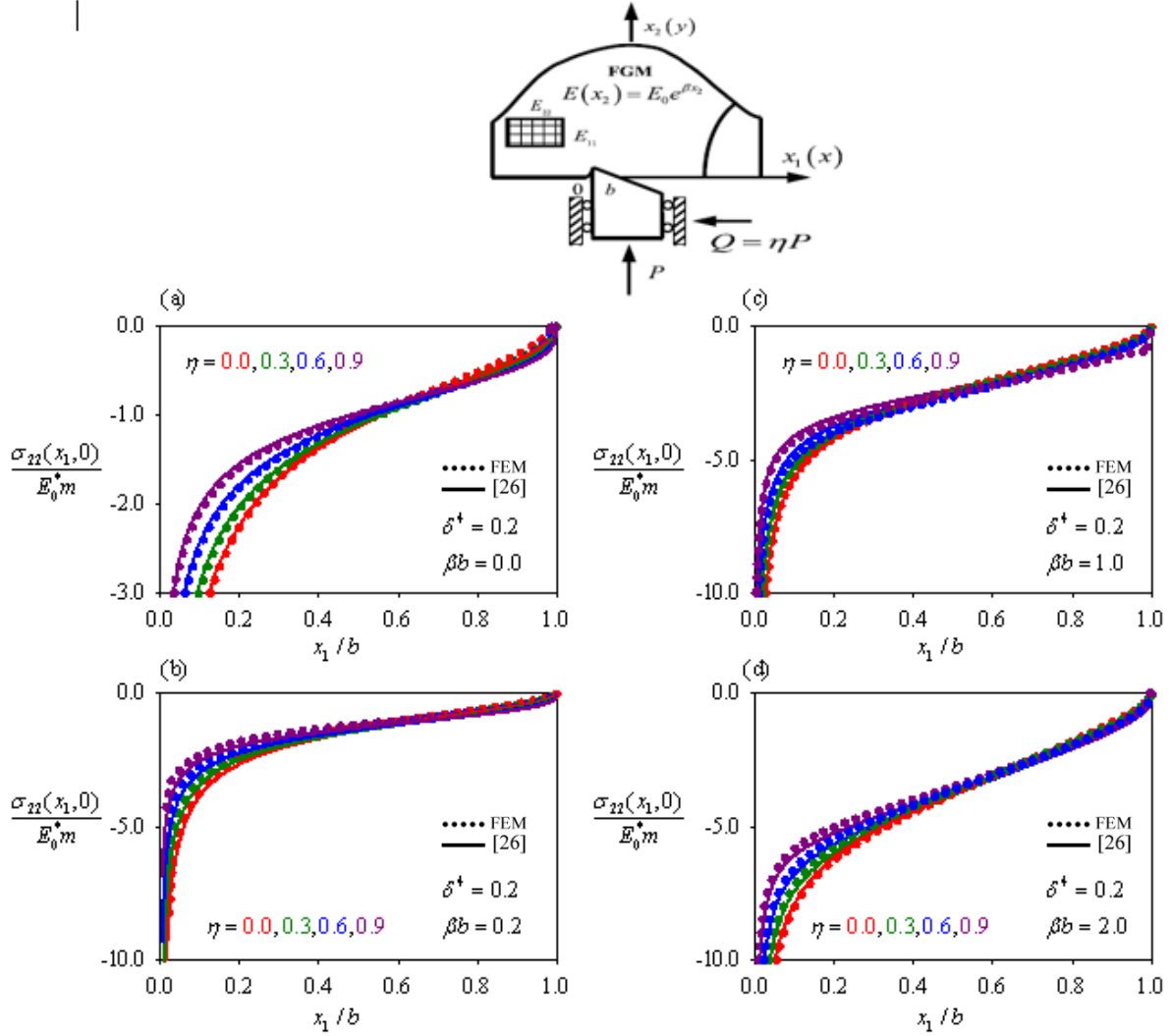


Figure 4.6. Contact stress, $\sigma_{22}(x_1, 0)$, distribution at the surface of the graded orthotropic half-plane loaded by a triangular punch for various values of the coefficient of friction, η , with the

parameters $\nu = 3/7$, $\delta^4 = 0.2$, $\kappa = -0.1$, $E_0^* = E_0$, (a) $\kappa = -0.1$, $\beta b = 0.0$ (b) $\kappa = -0.1$,

$\beta b = 0.2$, (c) $\kappa = -0.1$, $\beta b = 1.0$, (d) $\kappa = -0.1$, $\beta b = 2.0$

contact stress distribution. However, the effect of the coefficient of friction, η , is quiet smaller if a comparison is made between the Figures 4.7(a) and 4.7(b), 4.7(c) and 4.7(d).

The effect of the stiffness ratio, δ^4 , for the various values of the shear parameter, κ , and the coefficient of friction, η , is plotted in Figure 4.8. For the smaller values of the shear parameter, κ , and for both frictionless and frictional contact where $\eta = 0.0$ and $\eta = 0.9$, the curves of the

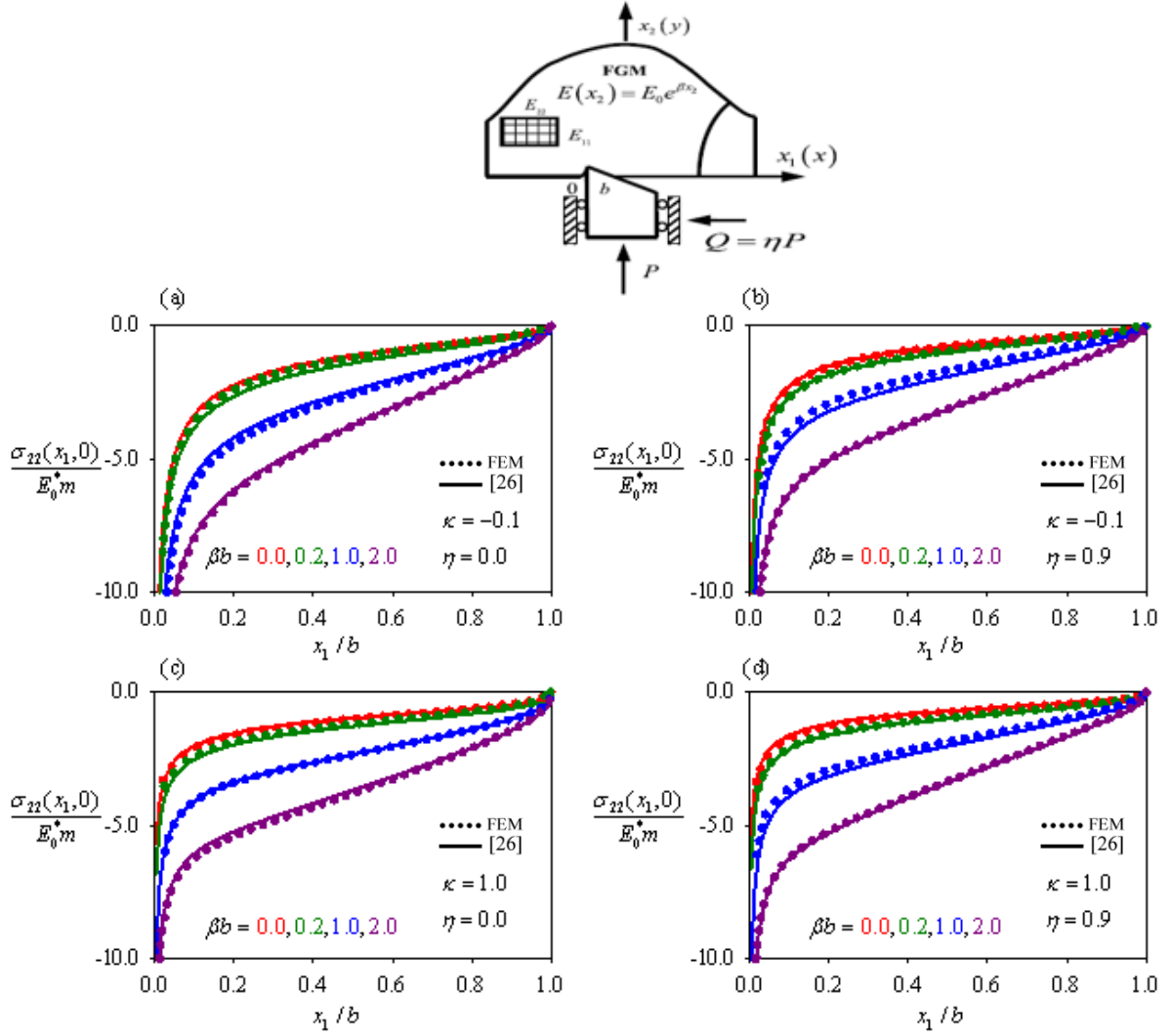


Figure 4.7. Contact stress, $\sigma_{22}(x_1, 0)$, distribution at the surface of the graded orthotropic half-plane loaded by a triangular punch for various values of the inhomogeneity parameter, βb , with the parameters $\nu = 3/7$, $\delta^4 = 0.2$, $E_0^* = E_0$, (a) $\eta = 0.0$, $\kappa = -0.1$ (b) $\eta = 0.9$, $\kappa = -0.1$, (c) $\eta = 0.0$, $\kappa = 1.0$, (d) $\eta = 0.9$, $\kappa = 1.0$

contact stress distribution is more straight for the lower values of the stiffness ratio, δ^4 , such as $\delta^4 = 0.25$. Moreover, as the stiffness ratio, δ^4 , increases, the resulting contact stresses, increases for a fixed value of x_1/b . The stiffness parameter, κ , also has an effect on the contact stress distribution. For the higher values of the stiffness parameter, κ , the resulting value of the contact stresses increases. On the other hand, again the coefficient of friction, η , has a slightly

minor effect on the contact stress distribution. The only noticeable difference from the coefficient of friction, η , can be seen in the Figures 4.8(a) and 4.8(b) respectively. However, the resulting contact stresses has higher values for the case of the frictional contact, where $\eta = 0.9$ than the case of the frictionless contact, where $\eta = 0.0$.

Note that, one of the most prominent thing for the graded orthotropic half-plane problem indented by triangular punch is that the contact stress distribution is bounded at the smooth end of the triangular punch, where x_1/b is equal to zero, and singular at the sharp end of the triangular punch, where $x_1 = b$. Consequently, this expression can be seen in a straightforward way in the Figures 4.6, 4.7 and 4.8.

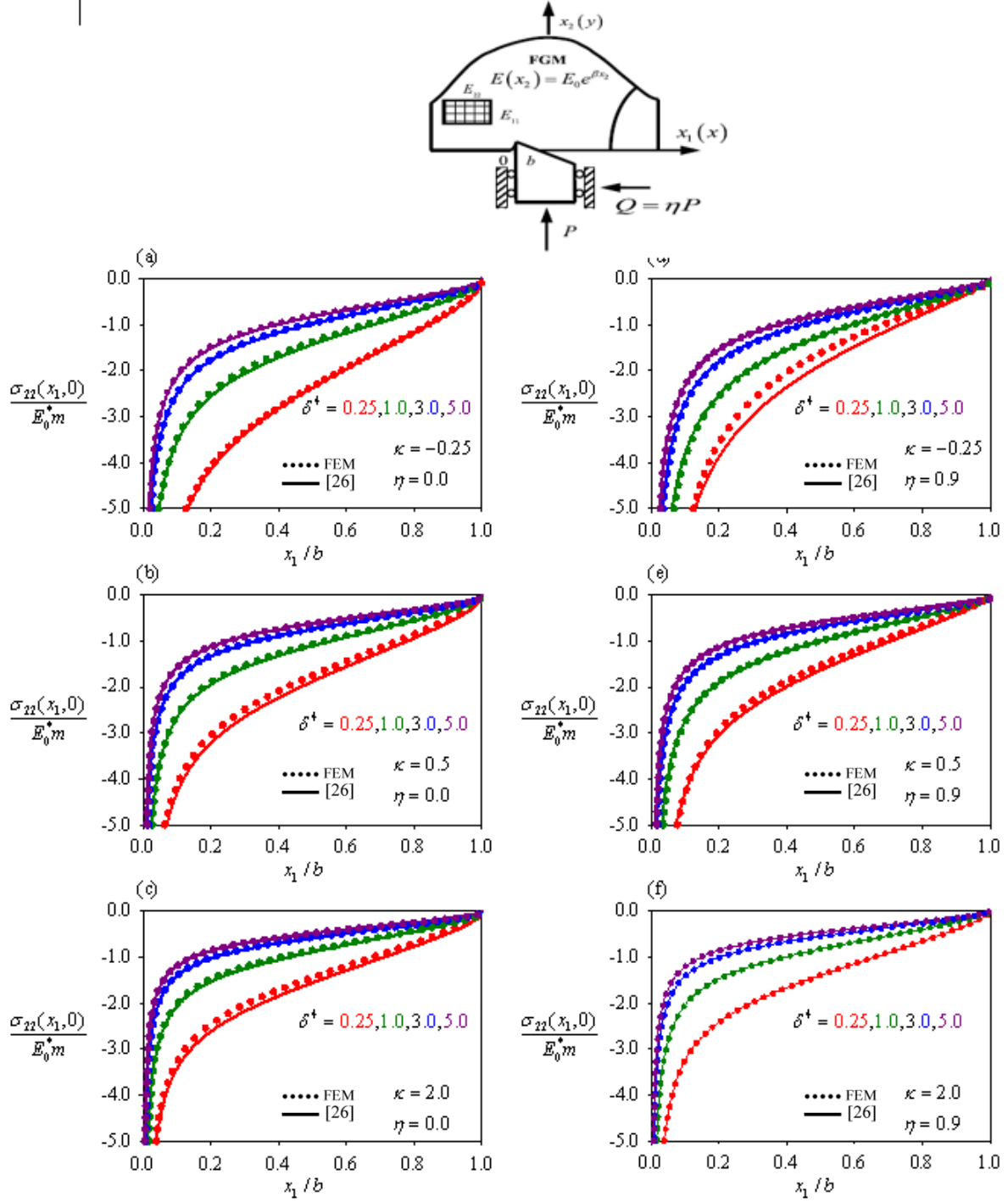


Figure 4.8. Contact stress, $\sigma_{22}(x_1, 0)$, distribution at the surface of the graded orthotropic half-plane loaded by a triangular punch for various values of the stiffness ratio, δ^4 , with the parameters $\nu = 3/7$, $\beta b = 1.0$, $E_0^* = E_0$, (a), $\eta = 0.0$, $\kappa = -0.25$ (b) $\eta = 0.0$, $\kappa = 0.5$, (c) $\eta = 0.0$, $\kappa = 2.0$, (d) $\eta = 0.5$, $\kappa = -0.25$, (e) $\eta = 0.5$, $\kappa = 0.5$, (f) $\eta = 0.5$, $\kappa = 2.0$

4.3. Results And Discussions For The Graded Orthotropic Half-Plane Problem Loaded By A Rigid Circular Punch

The influences of the variation of the inhomogeneity parameter, $\beta(a+b)$, the shear parameter, κ , the stiffness ratio, δ^4 , the effective Poisson's ratio and the coefficient of friction, η , on the contact stresses and the in-plane stresses for the graded orthotropic half-plane problem indented by a circular punch is examined in the Figures 4.9, 4.10, 4.11, 4.12 and 4.13. Moreover, the difference on the values of the dimensionless load, $P^* = P/(ER)$, that calculated from the finite element solutions and the analytical results are illustrated in the Tables 4.1, 4.2, 4.3 and 4.4, similarly the difference on the values of the contact boundaries, a/R and b/R , can be found in the Tables 4.5, 4.6, 4.7 and 4.8 respectively.

The non-dimensional material properties, $(E, \delta^4, \kappa, \nu)$, the coefficient of friction, η , and the inhomogeneity parameter, $\beta(a+b)$ are chosen between the following ranges, $(-0.1 \leq \kappa \leq 3, 0.25 \leq \delta^4 \leq 2.0, 1/7 \leq \nu \leq 5/7, 0.0 \leq \eta \leq 0.5, 0.0 \leq \beta(a+b) \leq 3.0)$ for the finite element solutions. For the Figures 4.9 to 4.12, the contact stresses and the in-plane stresses are normalized by using $P/(a+b)$.

The effect of the inhomogeneity parameter, $\beta(a+b)$, and the coefficient of friction, η , on the contact stress and the in-plane stress distribution are plotted in Figure 4.9, where the contact stress and the in-plane stress are donated by $\sigma_{22}(x_1, 0)$ and $\sigma_{11}(x_1, 0)$. An excellent match is obtained between the finite element solutions and the analytical results. The shear parameter, κ , and the stiffness ratio, δ^4 , are fixed at $\kappa=1.0$ and $\delta^4=1.0$ in order to catch the non-homogeneous orthotropic behavior. From the Figures 4.9(a) and 4.9(c), the difference in the coefficient of friction, η , affects the prone characteristic of the contact stresses. Moreover, it can easily be seen that, as the value of the coefficient of friction, η , changes from $\eta=0.0$ to $\eta=0.5$, the value of the contact stresses, for a fixed value of the $x_1/R=0.0$, is not seem to change much. Also, the effect of the inhomogeneity parameter, $\beta(a+b)$, is quite small on the value and the prone characteristic of the contact stresses. One can easily observe from the Figure 4.9(b) and 4.9(d) that, the coefficient of friction, η , has a major effect on the in-plane stress

distribution. The value that the in-plane stress get from the varying values of the inhomogeneity parameter, $\beta(a+b)$, is quiet large for the frictionless contact than the frictional contact for a fixed value of the $x_1/R=0.0$. Furthermore, the coefficient of friction, η , also has a major effect on the prone characteristic of the in-plane stress distribution that plotted in the Figure 4.9(d). Additionally, the varying inhomogeneity parameter, $\beta(a+b)$, affects the value of the in-plane stresses reasonably in the Figures 4.9(b) and 4.9(d).

Figure 4.10. presents the effect of the inhomogeneity parameter, $\beta(a+b)$, and the shear parameter, κ , on the contact stress and the in-plane stress distribution for the fixed values of the stiffness ratio, δ^4 , and the coefficient of friction, η . The effect of the inhomogeneity parameter, $\beta(a+b)$, on the contact stresses is quietly low for the smaller values of the shear parameter, κ . However, it has a major effect on the values of the in-plane stress distribution that plotted in the Figures 4.10b, 4.10d and 4.10f for all the varying shear parameters, κ , values. As the inhomogeneity parameter, $\beta(a+b)$, increases, the value of the resulting contact stresses decreases. Furthermore, the contact stress and the in-plane stress distribution is symmetrical, whereat the coefficient of friction, η , is zero in all the cases in the Figure 4.10. The increment on the shear parameter, κ , has affects the contact stress and the in-plane stress distribution in a very shallow way and it may be merely seen from the Figures 4.10(b), 4.10(d) and 4.10(f).

The contact stress and in-plane stress distribution for the various values of the inhomogeneity parameter, $\beta(a+b)$, and the shear parameter, κ , is illustrated in Figure 4.11. As the inhomogeneity parameter, $\beta(a+b)$, increases, the curves of the contact stresses begin to prone to the trailing edge of the contact. Furthermore, the effect of the shear parameter, κ , on the contact stresses can simply be understood from comparing the Figures 4.11(a), 4.11(c) and 4.11(e) respectively. As the shear parameter, κ , increases, the peak values of the contact stress distribution decreases. Moreover, the in-plane stress distribution for the frictional contact are plotted in the Figures 4.11(b), 4.11(d) and 4.11(f). Once again, the inhomogeneity parameter, $\beta(a+b)$, affects the in-plane stress distribution greatly. By the same token, as the shear parameter, κ , increases, the curves of the in-plane stress distribution between the contact zone becomes more linear.

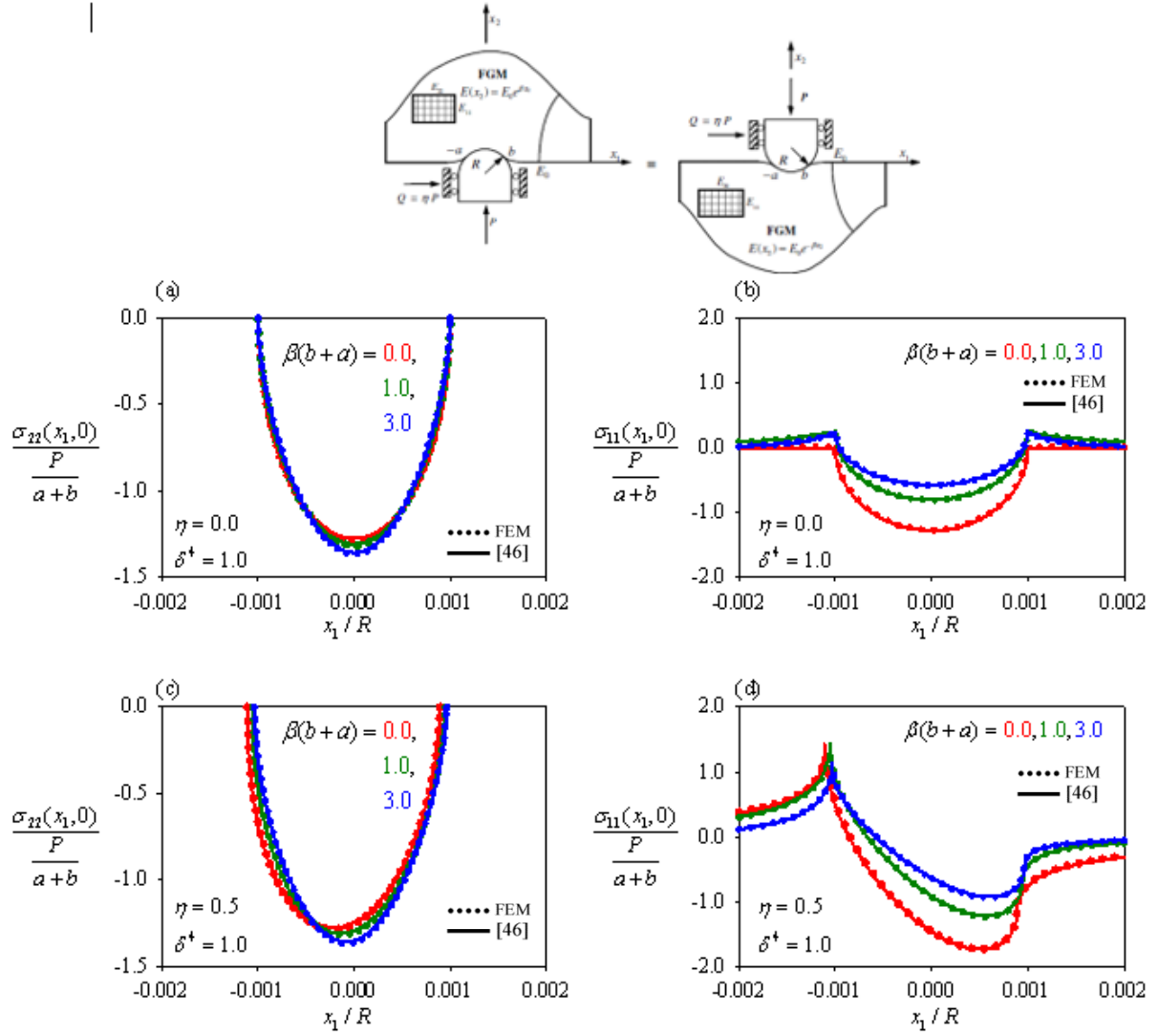


Figure 4.9. Contact stress, $\sigma_{22}(x_1, 0)$, and the in-plane stress, $\sigma_{11}(x_1, 0)$, distribution at the surface of the graded orthotropic half-plane loaded by a circular punch for various values of the inhomogeneity parameter, $\beta(a+b)$, with the parameters $\kappa=1.0$, $\delta^4=1.0$, $\nu=0.3$, $(a+b)/R=0.002$ (a) $\eta=0.0$, $\delta^4=1.0$ (b) $\eta=0.0$, $\delta^4=1.0$ (c) $\eta=0.5$, $\delta^4=1.0$ (d) $\eta=0.5$, $\delta^4=1.0$

Figure 4.12 depicts the contact stress and the in-plane stress distributions with the different combinations of the stiffness ratio, δ^4 , and the shear parameter, κ for the graded orthotropic half-plane intended by a sliding circular punch. Note that, the inhomogeneity parameter, $\beta(a+b)$, and the coefficient of friction, η , are fixed at, $\beta(a+b)=0.5$ and $\eta=0.5$. As given

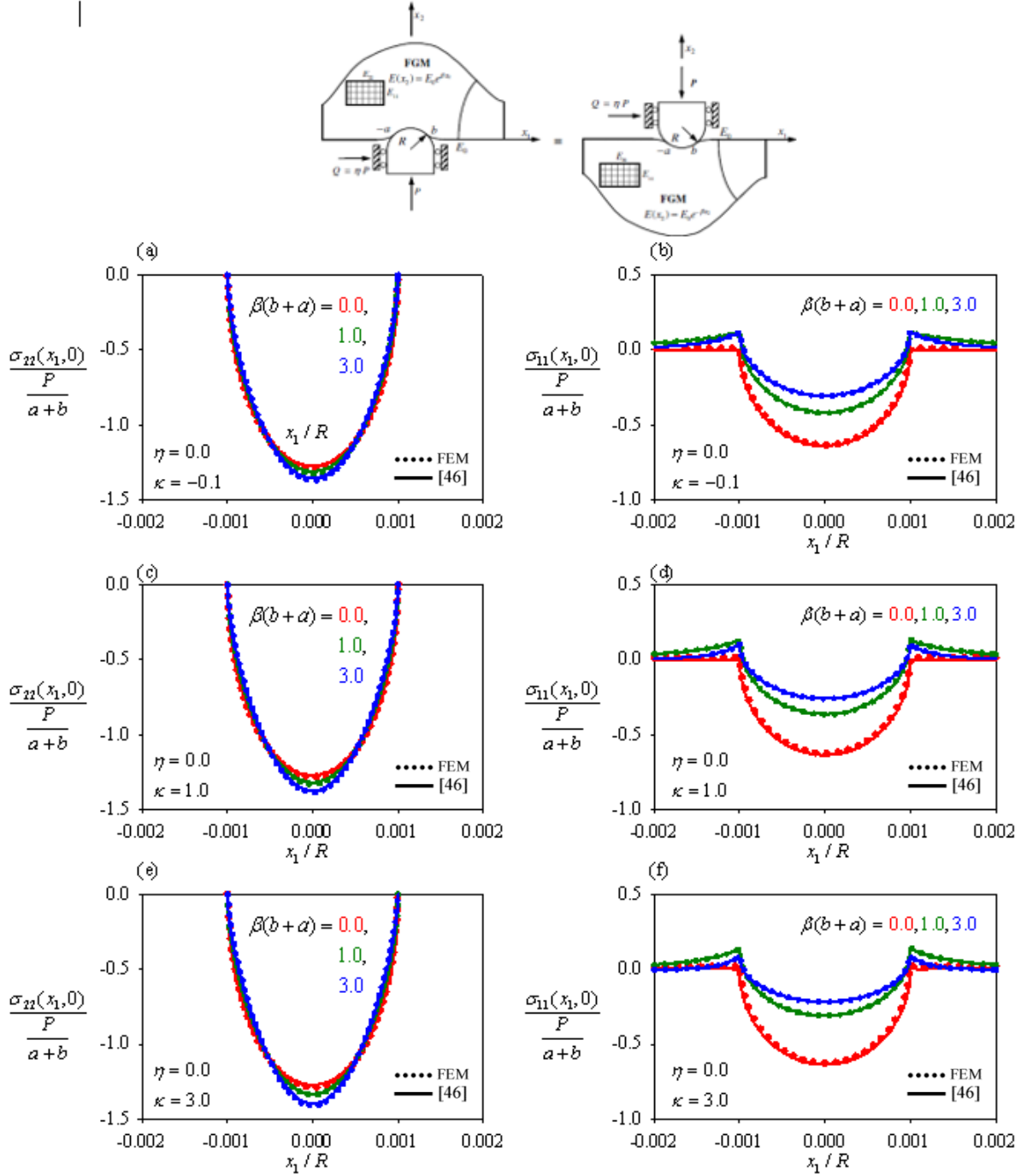


Figure 4.10. Contact stress, $\sigma_{22}(x_1, 0)$, and the in-plane stress, $\sigma_{11}(x_1, 0)$, distribution at the surface of the graded orthotropic half-plane loaded by a circular punch for various values of the inhomogeneity parameter, $\beta(a+b)$, and the shear parameter, κ , with the parameters $\delta^4 = 0.25$, $\nu = 0.3$, $(a+b)/R = 0.002$ (a) $\eta = 0.0$, $\kappa = -0.1$ (b) $\eta = 0.0$, $\kappa = -0.1$ (c) $\eta = 0.0$, $\kappa = 1.0$ (d) $\eta = 0.0$, $\kappa = 1.0$ (e) $\eta = 0.0$, $\kappa = 3.0$ (f) $\eta = 0.0$, $\kappa = 3.0$

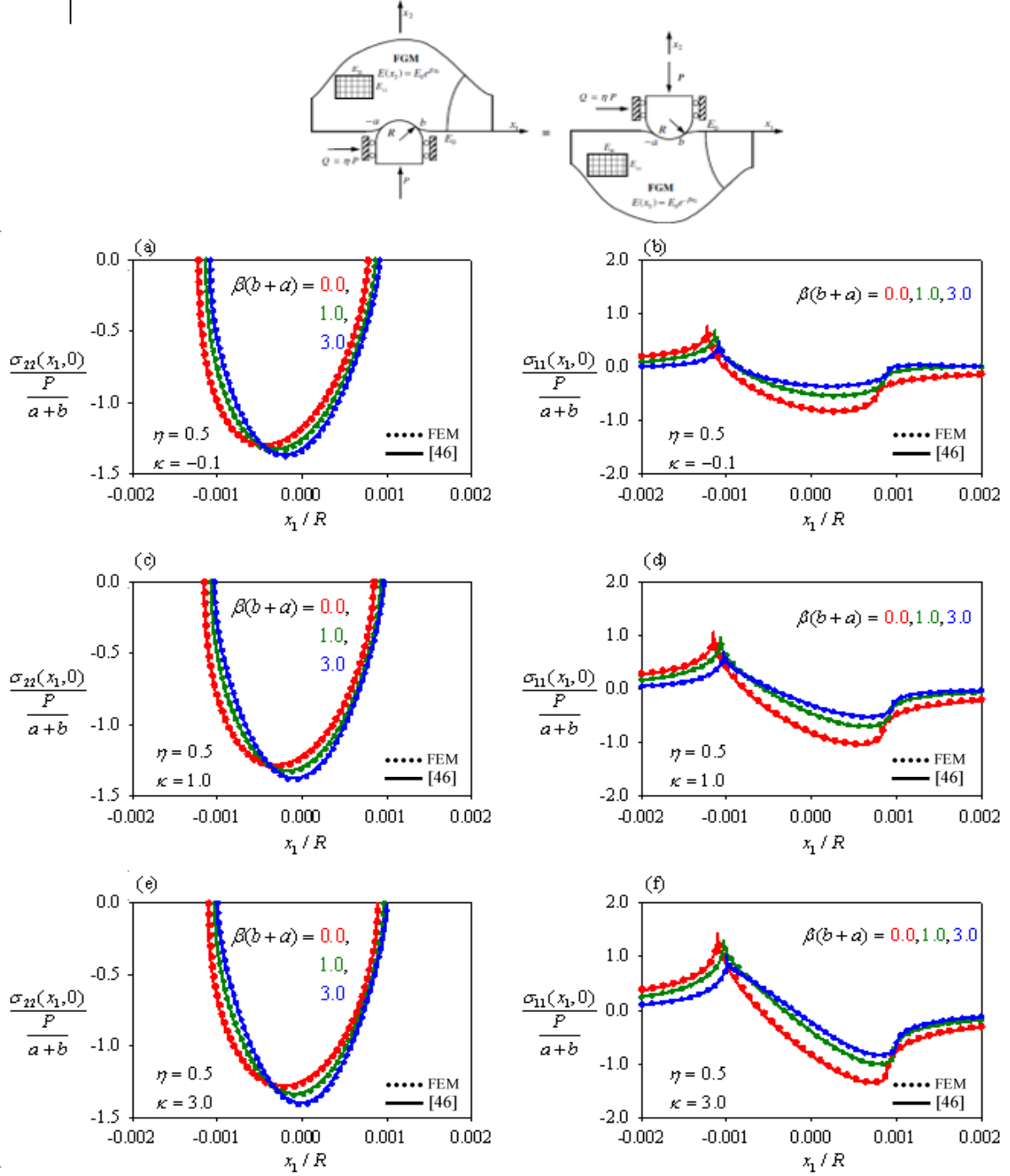


Figure 4.11. Contact stress, $\sigma_{22}(x_1, 0)$, and the in-plane stress, $\sigma_{11}(x_1, 0)$, distribution at the surface of the graded orthotropic half-plane loaded by a circular punch for various values of the inhomogeneity parameter, $\beta(a+b)$, and the shear parameter, κ , with the parameters $\delta^4 = 0.25$, $\nu = 0.3$, $(a+b)/R = 0.002$ (a) $\eta = 0.5$, $\kappa = -0.1$ (b) $\eta = 0.5$, $\kappa = -0.1$ (c) $\eta = 0.5$, $\kappa = 1.0$ (d) $\eta = 0.5$, $\kappa = 1.0$ (e) $\eta = 0.5$, $\kappa = 3.0$ (f) $\eta = 0.5$, $\kappa = 3.0$

in the Figures 4.12(a) and 4.12(c), the increment on the shear parameter, κ , makes the contact stress distribution to prone to the trailing edge of the contact. Furthermore, once again for the Figures 4.12(a) and 4.12(c), as the stiffness ratio increases, δ^4 , the effect of the shear parameter, κ , on the contact stress distribution reduces. On the other hand, as it can be seen from the Figures 4.12(b) and 4.12(d), the shear parameter, κ , has a major effect on the in-plane stress distribution. As the shear parameter, κ , the value of the in-plane stress, $\sigma_{11}(x_1, 0)$, on the trailing edge of the contact increases for the both values of the stiffness ratio, δ^4 . Furthermore, the value of the in-plane stress at the center of the contact, where the $x_1 / R = 0.0$, increases as the shear parameter, κ , increases. Moreover, because of the coefficient of friction, η , is equal to $\eta = 0.5$, both of the contact stress and the in-plane stress distributions are unsymmetrical.

Figure 4.13. illustrates the influence of the effective Poisson's ratio, ν , and the stiffness ratio, δ^4 , on the distribution of the contact stresses and the in-plane stresses with a fixed value of the inhomogeneity parameter, $\beta(a+b)$ and the shear parameter, κ , at $\beta(a+b) = 0.5$ and $\kappa = 0.5$. From the Figures 4.13(a) and 4.13(c), as the effective Poisson's ratio, ν , increases, the resulting contact stress distribution begins to prone to the trailing edge of the contact. However, this effect is quite minor in the Figure 4.13(c), because of the higher value of the stiffness ratio, δ^4 . Moreover, for the Figures 4.13(b) and 4.13(d), the difference on the resulting values of the in-plane stresses from the effect of the effective Poisson's ratio is quiet low, ν . Additionally, the stiffness ratio, δ^4 , has a quite bigger effect on the distribution of the in-plane stresses rather than the distribution of the contact stresses. Also, the contact stress and the in-plane stress distributions are not symmetrical, because of the effect of the coefficient of friction, η . Note that, it is important to delineate that, the distribution of the contact stresses is bounded at the leading and the trailing edges of the contact and are equal to zero. Furthermore, as expected, the distribution of the in-plane stresses has a peak at the leading edge of the contact.

In Table 4.1, the comparison of the dimensionless load, $P^* = P / (ER)$, for the graded orthotropic half-plane intended by a sliding circular punch is given. For the frictionless and the frictional contact, where $\eta = 0.0$, $\eta = 0.5$ respectively, as the inhomogeneity parameter, $\beta(a+b)$, increases, the dimensionless load, $P^* = P / (ER)$, increases. Moreover, the change on

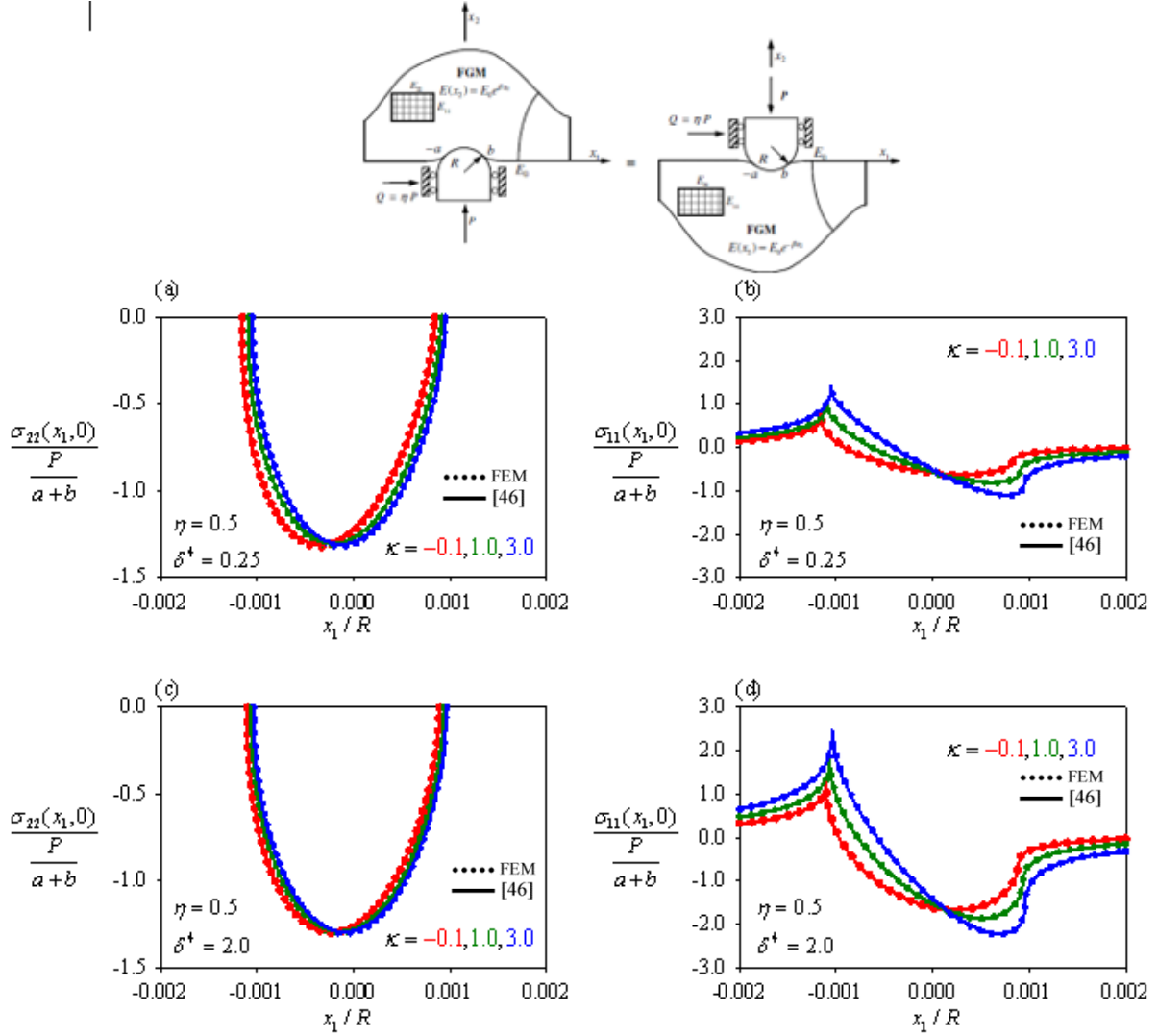


Figure 4.12. Contact stress, $\sigma_{22}(x_1, 0)$, and the in-plane stress, $\sigma_{11}(x_1, 0)$, distribution at the surface of the graded orthotropic half-plane loaded by a circular punch for various values of the shear parameter, κ , and the stiffness ratio, δ^4 , with the parameters, $\nu = 0.3$, $\beta(a+b) = 0.5$, $(a+b)/R = 0.002$ (a) $\eta = 0.5$, $\delta^4 = 0.25$ (b) $\eta = 0.5$, $\delta^4 = 0.25$ (c) $\eta = 0.5$, $\delta^4 = 2.0$ (d) $\eta = 0.5$, $\delta^4 = 2.0$

the coefficient of friction from $\eta = 0.0$ to $\eta = 0.5$ has a minor effect on the values of the dimensionless load, $P^* = P/(ER)$. Furthermore, an excellent match is obtained between the finite element solutions and the analytical results.

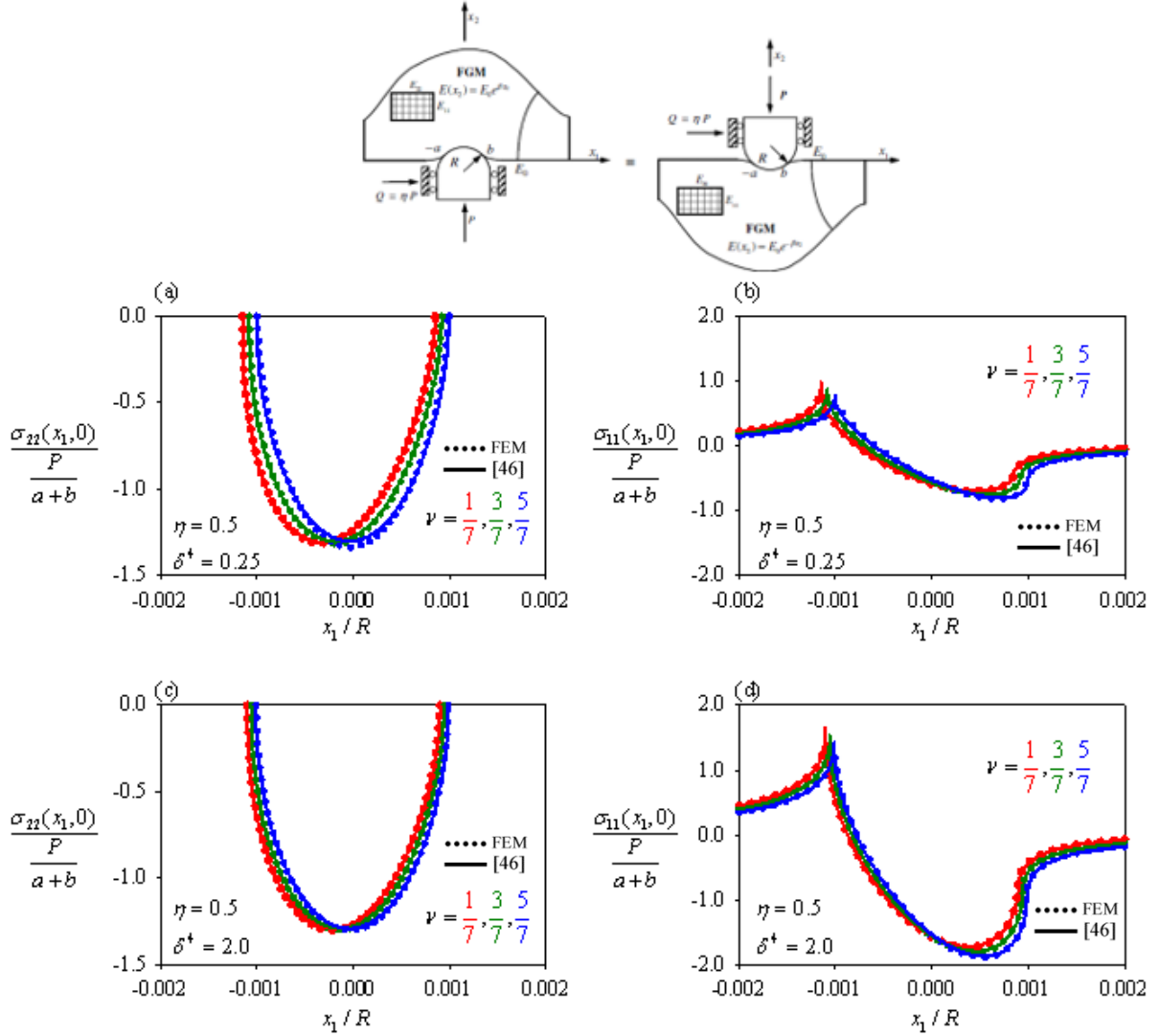


Figure 4.13. Contact stress, $\sigma_{22}(x_1, 0)$, and the in-plane stress, $\sigma_{11}(x_1, 0)$, distribution at the surface of the graded orthotropic half-plane loaded by a circular punch for various values of the effective Poisson's ratio, ν , and the stiffness ratio, δ^4 , with the parameters, $\kappa = 0.5$, $\beta(a+b) = 0.5$, $(a+b)/R = 0.002$ (a) $\eta = 0.5$, $\delta^4 = 0.25$ (b) $\eta = 0.5$, $\delta^4 = 0.25$ (c) $\eta = 0.5$, $\delta^4 = 2.0$ (d) $\eta = 0.5$, $\delta^4 = 2.0$

Table 4.2 shows the contact boundaries, a/R and b/R , for the graded orthotropic half-plane intended by a sliding circular punch. For the frictionless contact, where $\eta = 0.0$, the contact boundaries are, a/R and b/R , are equal and the effect of the inhomogeneity parameter, $\beta(a+b)$, on the contact boundaries, a/R and b/R , is quite small. Moreover, as the coefficient

Table 4.1. Comparison of the true percentage error (%) of the dimensionless load, $P^* = P/(ER)$, for a graded orthotropic half-plane loaded by a circular punch for various values of the inhomogeneity parameter, $\beta(a+b)$, with the parameters $\nu = 0.3$, $\delta^4 = 1.0$, $\kappa = 1.0$, $(a+b)/R = 0.002$

$\delta^4 = 1.0$		$P/(ER)$	$P/(ER)$	Error
		Analytical [46]	FEM	(%)
$\eta = 0.0$	$\beta(a+b) = 0.0$	0.7873E-06	0.7806E-06	0.8
	$\beta(a+b) = 1.0$	1.1325E-06	1.1236E-06	0.7
	$\beta(a+b) = 3.0$	1.8160E-06	1.8018E-06	0.7
$\eta = 0.5$	$\beta(a+b) = 0.0$	0.7777E-06	0.7716E-06	0.7
	$\beta(a+b) = 1.0$	1.1201E-06	1.1025E-06	1.5
	$\beta(a+b) = 3.0$	1.8015E-06	1.7825E-06	1.0

of friction, η , on the contact area is not equal to zero, the contact boundaries, a/R and b/R , are not symmetrical. Furthermore, for the frictional contact, where $\eta = 0.5$, as the inhomogeneity parameter increases, the resulting value of the contact boundary, a/R increases.

Table 4.3 displays the comparison of the dimensionless load, $P^* = P/(ER)$, for the various values of the coefficient of friction, η , shear parameter, κ , and the inhomogeneity parameter, $\beta(a+b)$. As it can be seen from the Table 4.3, an impeccable match is obtained between the finite element solutions and the analytical results. The increment on the shear parameter, κ , decreases the value of the dimensionless load, $P^* = P/(ER)$, for the both frictionless and frictional contact, where $\eta = 0.0$ and $\eta = 0.5$ respectively. Moreover, as the inhomogeneity parameter increases, the dimensionless load, $P^* = P/(ER)$, decreases. Additionally, the difference in the coefficient of friction, η , has a quiet minor effect on the dimensionless load, $P^* = P/(ER)$ for the varying values of the shear parameter, κ , and the inhomogeneity parameter, $\beta(a+b)$.

Table 4.2. Comparison of the contact boundaries, a/R and b/R , for a graded orthotropic half-plane loaded by a circular stamp for various punch of the inhomogeneity parameter,

$\beta(a+b)$, with the parameters $\nu = 0.3$, $\delta^4 = 1.0$, $\kappa = 1.0$, $(a+b)/R = 0.002$

$\delta^4 = 1.0$		a/R	a/R FEM	b/R	b/R FEM
		Analytical [46]		Analytical	
$\eta = 0.0$	$\beta(a+b) = 0.0$		1.008E-03		1.008E-03
	$\beta(a+b) = 1.0$	1.000E-03	1.007E-03	1.000E-03	1.007E-03
	$\beta(a+b) = 3.0$		1.007E-03		1.007E-03
$\eta = 0.5$	$\beta(a+b) = 0.0$	1.110E-03	1.115E-03	0.889E-03	0.900E-03
	$\beta(a+b) = 1.0$	1.058E-03	1.070E-03	0.941E-03	0.961E-03
	$\beta(a+b) = 3.0$	1.058E-03	1.048E-03	0.941E-03	0.972E-03

The effect of the coefficient of friction, η , shear parameter, κ , and the inhomogeneity parameter, $\beta(a+b)$, on the contact boundaries, a/R and b/R , are illustrated in the Table 4.4. Note that, the stiffness ratio, δ^4 , is fixed at $\delta^4 = 0.25$. As the coefficient of friction, η , is equal to zero, the contact boundaries, a/R and b/R , are symmetrical. Moreover, under the presence of the friction, the increment of the shear parameter, κ , decreases the value of the contact boundary, a/R for the various values of the inhomogeneity parameter, $\beta(a+b)$. Additionally, the increment of the inhomogeneity parameter, $\beta(a+b)$, increases the value of the contact boundary, b/R , for the increasing value of the shear parameter, κ .

Comparison of the dimensionless load, $P^* = P/(ER)$, for the various values of the stiffness ratio, δ^4 , and the shear parameter, κ , is given in the Table 4.5. Note that, the coefficient of friction, η , and the inhomogeneity parameter, $\beta(a+b)$, are fixed at $\eta = 0.5$ and $\beta(a+b) = 0.5$. It can be easily seen that, as the stiffness ratio, δ^4 , increases, the resulting value of the dimensionless load, $P^* = P/(ER)$, decreases for the different values of shear parameter, κ . Furthermore, the shear parameter, κ , has a major effect on the dimensionless load, $P^* = P/(ER)$. As the shear parameter, κ , increases, the resulting value of the dimensionless

Table 4.3. Comparison of the dimensionless load, $P^* = P/(ER)$, for a graded orthotropic half-plane loaded by a circular punch for various values of the inhomogeneity parameter,

$\beta(a+b)$, and the shear parameter, κ , with the parameters $\nu = 0.3$, $\delta^4 = 0.25$,

$$(a+b)/R = 0.002$$

$\delta^4 = 0.25$		$P/(ER)$		Error (%)	
		Analytical [46]	FEM		
$\eta = 0.0$	$\kappa = -0.1$	$\beta(a+b) = 0.0$	1.6558E-06	1.6287E-06	1.6
		$\beta(a+b) = 1.0$	2.4149E-06	2.3641E-06	2.1
		$\beta(a+b) = 3.0$	3.8353E-06	3.7588E-06	1.9
	$\kappa = 1.0$	$\beta(a+b) = 0.0$	1.1135E-06	1.1038E-06	0.8
		$\beta(a+b) = 1.0$	1.7997E-06	1.7730E-06	1.4
		$\beta(a+b) = 3.0$	3.1824E-06	3.1370E-06	1.4
	$\kappa = 3.0$	$\beta(a+b) = 0.0$	0.7854E-06	0.7761E-06	1.1
		$\beta(a+b) = 1.0$	1.4402E-06	1.4291E-06	0.7
		$\beta(a+b) = 3.0$	2.7967E-06	2.7533E-06	1.5
$\eta = 0.5$	$\kappa = -0.1$	$\beta(a+b) = 0.0$	1.5720E-06	1.5504E-06	1.3
		$\beta(a+b) = 1.0$	2.2956E-06	2.2564E-06	1.7
		$\beta(a+b) = 3.0$	3.6765E-06	3.6145E-06	1.6
	$\kappa = 1.0$	$\beta(a+b) = 0.0$	1.0868E-06	1.0753E-06	1.0
		$\beta(a+b) = 1.0$	1.7633E-06	1.7442E-06	1.0
		$\beta(a+b) = 3.0$	3.1412E-06	3.1026E-06	1.2
	$\kappa = 3.0$	$\beta(a+b) = 0.0$	0.7758E-06	0.7689E-06	0.8
		$\beta(a+b) = 1.0$	1.4279E-06	1.4116E-06	1.1
		$\beta(a+b) = 3.0$	2.7873E-06	2.7461E-06	1.4

load, $P^* = P/(ER)$ decreases. Additionally, as illustrated in the Table 4.6, the resulting value of the contact boundary, a/R decreases, and b/R increases, as the stiffness ratio, δ^4 , increases. Besides, one can get a lower value for the a/R , because of the increment on the

Table 4.4. Comparison of the contact boundaries, a/R and b/R , for a graded orthotropic half-plane loaded by a circular punch for various values of the inhomogeneity parameter,

$\beta(a+b)$, and the shear parameter, κ , with the parameters $\nu = 0.3$, $\delta^4 = 0.25$,

$$(a+b)/R = 0.002$$

$\delta^4 = 0.25$		a/R	a/R	b/R	b/R
		Analytical [46]	FEM	Analytical	FEM
$\eta = 0.0$	$\beta(a+b) = 0.0$		1.017E-03		1.017E-03
	$\kappa = -0.1$ $\beta(a+b) = 1.0$		1.022E-03		1.022E-03
	$\beta(a+b) = 3.0$		1.020E-03		1.020E-03
	$\beta(a+b) = 0.0$		1.009E-03		1.009E-03
	$\kappa = 1.0$ $\beta(a+b) = 1.0$	1.000E-03	1.015E-03	1.000E-03	1.015E-03
	$\beta(a+b) = 3.0$		1.015E-03		1.015E-03
	$\beta(a+b) = 0.0$		1.012E-03		1.012E-03
	$\kappa = 3.0$ $\beta(a+b) = 1.0$		1.008E-03		1.008E-03
	$\beta(a+b) = 3.0$		1.016E-03		1.016E-03
$\eta = 0.5$	$\beta(a+b) = 0.0$	1.000E-03	1.240E-03	1.000E-03	0.788E-03
	$\kappa = -0.1$ $\beta(a+b) = 1.0$	1.000E-03	1.151E-03	1.000E-03	0.884E-03
	$\beta(a+b) = 3.0$	1.087E-03	1.102E-03	0.912E-03	0.932E-03
	$\beta(a+b) = 0.0$	1.155E-03	1.165E-03	0.845E-03	0.856E-03
	$\kappa = 1.0$ $\beta(a+b) = 1.0$	1.071E-03	1.081E-03	0.928E-03	0.940E-03
	$\beta(a+b) = 3.0$	1.036E-03	1.048E-03	0.964E-03	0.976E-03
	$\beta(a+b) = 0.0$	1.110E-03	1.110E-03	0.889E-03	0.907E-03
	$\kappa = 3.0$ $\beta(a+b) = 1.0$	1.033E-03	1.044E-03	0.966E-03	0.979E-03
	$\beta(a+b) = 3.0$	0.997E-04	1.015E-03	1.002E-03	1.015E-03

shear parameter, κ . Note that, the contact boundaries, a/R and b/R , are unsymmetrical because of the presence of the friction on the contact zone.

Table 4.7 and 4.8 shows the variation of the values of the dimensionless load, $P^* = P/(ER)$, and the contact boundaries, a/R and b/R , for the various values of the effective Poisson's

Table 4.5. Comparison of the dimensionless load, $P^* = P / (ER)$, for a graded orthotropic half-plane loaded by a circular punch for various values of the shear parameter, κ , and the stiffness ratio, δ^4 , with the parameters $\nu = 0.3$, $\beta(a+b) = 0.5$, $(a+b)/R = 0.002$

$\beta(a+b)=0.5$		$P/(ER)$	$P/(ER)$	Error (%)	
		Analytical [46]	FEM		
$\eta=0.5$	$\delta^4=0.25$	$\kappa=-0.1$	1.9452E-06	1.9150E-06	1.5
		$\kappa=1.0$	1.4289E-06	1.4116E-06	1.2
		$\kappa=3.0$	1.1023E-06	1.0897E-06	1.1
	$\delta^4=2.0$	$\kappa=-0.1$	1.1058E-06	1.0974E-06	0.7
		$\kappa=1.0$	0.7805E-06	0.7756E-06	0.6
		$\kappa=3.0$	0.5819E-06	0.5773E-06	0.7

Table 4.6. Comparison of the contact boundaries, a/R and b/R , for a graded orthotropic half-plane loaded by a circular punch for various values of the shear parameter, κ , and the stiffness ratio, δ^4 , with the parameters $\nu = 0.3$, $\beta(a+b) = 0.5$, $(a+b)/R = 0.002$

$\beta(a+b)=0.5$		a/R	a/R FEM	b/R	b/R FEM	
		Analytical [46]		Analytical [46]		
$\eta=0.5$	$\delta^4=0.25$	$\kappa=-0.1$	1.159E-03	1.174E-03	0.840E-03	0.857E-03
		$\kappa=1.0$	1.092E-03	1.102E-03	0.907E-03	0.922E-03
		$\kappa=3.0$	1.053E-03	1.062E-03	0.946E-03	0.960E-03
	$\delta^4=2.0$	$\kappa=-0.1$	1.106E-03	1.114E-03	0.893E-03	0.900E-03
		$\kappa=1.0$	1.064E-03	1.068E-03	0.935E-03	0.943E-03
		$\kappa=3.0$	1.039E-03	1.046E-03	0.960E-03	0.969E-03

ratio, ν , and the stiffness ratio, κ . For instance, the value of the dimensionless load, $P^* = P / (ER)$, increases, as the value of the effective Poisson's ratio, ν , increases and decreases, as the value of the stiffness ratio, δ^4 , decreases. Moreover, as the effective Poisson's ratio, ν , increases, the value of contact boundary, a/R , decreases for the values $\delta^4 = 0.25$ and

Table 4.7. Comparison of the dimensionless load, $P^* = P / (ER)$, for a graded orthotropic half-plane loaded by a circular punch for various values of the effective Poisson's ratio, ν , and the stiffness ratio, δ^4 , with the parameters $\kappa = 0.5$, $\beta(a+b) = 0.5$, $(a+b) / R = 0.002$

$\beta(a+b) = 0.5$			$P / (ER)$ Analytical [46]	$P / (ER)$ FEM	Error (%)
$\eta = 0.5$	$\delta^4 = 0.25$	$\nu = 1/7$	1.5379E-06	1.5341E-06	0.2
		$\nu = 3/7$	1.6380E-06	1.6159E-06	1.3
		$\nu = 5/7$	1.7282E-06	1.6596E-06	3.9
	$\delta^4 = 2.0$	$\nu = 1/7$	0.8661E-06	0.8585E-06	0.8
		$\nu = 3/7$	0.8953E-06	0.8894E-06	0.6
		$\nu = 5/7$	0.9223E-06	0.9144E-06	0.8

Table 4.8. Comparison of the contact boundaries, a / R and b / R , for a graded orthotropic half-plane loaded by a circular punch for various values of the effective Poisson's ratio, ν , and the stiffness ratio, δ^4 , with the parameters $\kappa = 0.5$, $\beta(a+b) = 0.5$, $(a+b) / R = 0.002$

$\beta(a+b) = 0.5$			a / R Analytical [46]	a / R FEM	b / R Analytical [46]	b / R FEM
$\eta = 0.5$	$\delta^4 = 0.25$	$\nu = 1/7$	1.149E-03	1.151E-03	0.851E-03	0.854E-03
		$\nu = 3/7$	1.083E-03	1.095E-03	0.917E-03	0.932E-03
		$\nu = 5/7$	1.005E-03	1.042E-03	0.994E-03	1.041E-03
	$\delta^4 = 2.0$	$\nu = 1/7$	1.100E-03	1.108E-03	0.900E-03	0.909E-03
		$\nu = 3/7$	1.058E-03	1.065E-03	0.941E-03	0.948E-03
		$\nu = 5/7$	1.013E-03	1.022E-03	0.987E-03	0.994E-03

$\delta^4 = 2.0$. In addition to, the value of contact boundary, b / R , increases, as the stiffness ratio, δ^4 , increases for the assorted values of the effective Poisson's ratio, ν . Also, it can easily be seen that, the finite element solutions are quite good converged to the analytical results.

4.4. Results And Discussions For The Graded Orthotropic Half-Plane Problem Loaded By A Rigid Semi-Circular Punch

The influences of the non-dimensional material properties (E , δ^4 , κ , ν), the coefficient of friction, η , and the inhomogeneity parameters, $\beta(a+b)$, on the contact pressure distribution for the graded orthotropic half-plane problem indented by a semi-circular punch is given in the Figures 4.14, 4.15, 4.16, 4.17 and 4.18 respectively. Additionally, the difference on the values of the dimensionless load, $P^* = P/(ER)$, are illustrated in the Tables 4.9, 4.10, 4.11 and 4.12, similarly the difference on the values of the contact boundaries, a/R and b/R , can be found in the Tables 4.13, 4.14, 4.15 and 4.16 respectively. Identical non-dimensional material properties, (E , δ^4 , κ , ν), the coefficient of friction, η , and the inhomogeneity parameter, $\beta(b)$ are chosen for the finite element solutions. Furthermore, the contact stresses and the in-plane stresses are normalized with using $P/(a+b)$.

The effect of the inhomogeneity parameter, $\beta(b)$, and the coefficient of friction, η , on the distribution of the contact stresses and the in-plane stresses is given in the Figure 4.14. An excellent match is obtained between the finite element solutions and the analytical results. Note that, the graded half-plane is isotropic, because of the shear parameter, κ and the stiffness ratio, δ^4 , are equal to $\kappa = 1.0$ and $\delta^4 = 1.0$. For the Figure 4.14(a), as the inhomogeneity parameter, $\beta(b)$, increases, the resulting value of the contact stresses decreases. On the other hand, the increment on the inhomogeneity parameter, $\beta(b)$, increases the resulting value of the in-plane stresses.. Moreover, for the frictional contact, that illustrated in the Figures 4.14(c) and 4.14(d), as the inhomogeneity parameter, $\beta(b)$, increases, the value of the contact stresses decreases and the value of the in-plane stresses increases. Furthermore, the variation of the coefficient of friction, η , has a quiet minor effect on the contact stress and the in-plane stress distributions for the Figures 4.14(a) and 4.14(c), however, it has a major effect on the in-plane stress distribution that given in the Figures 4.14(b) and 4.14(d).

In the Figures 4.15 and 4.16, the variation of the contact stress and the in-plane stress distribution for the different values of the inhomogeneity parameter, $\beta(b)$, the shear parameter, κ , and the coefficient of friction, η , is investigated. As it can be seen from the Figures 4.15(a),

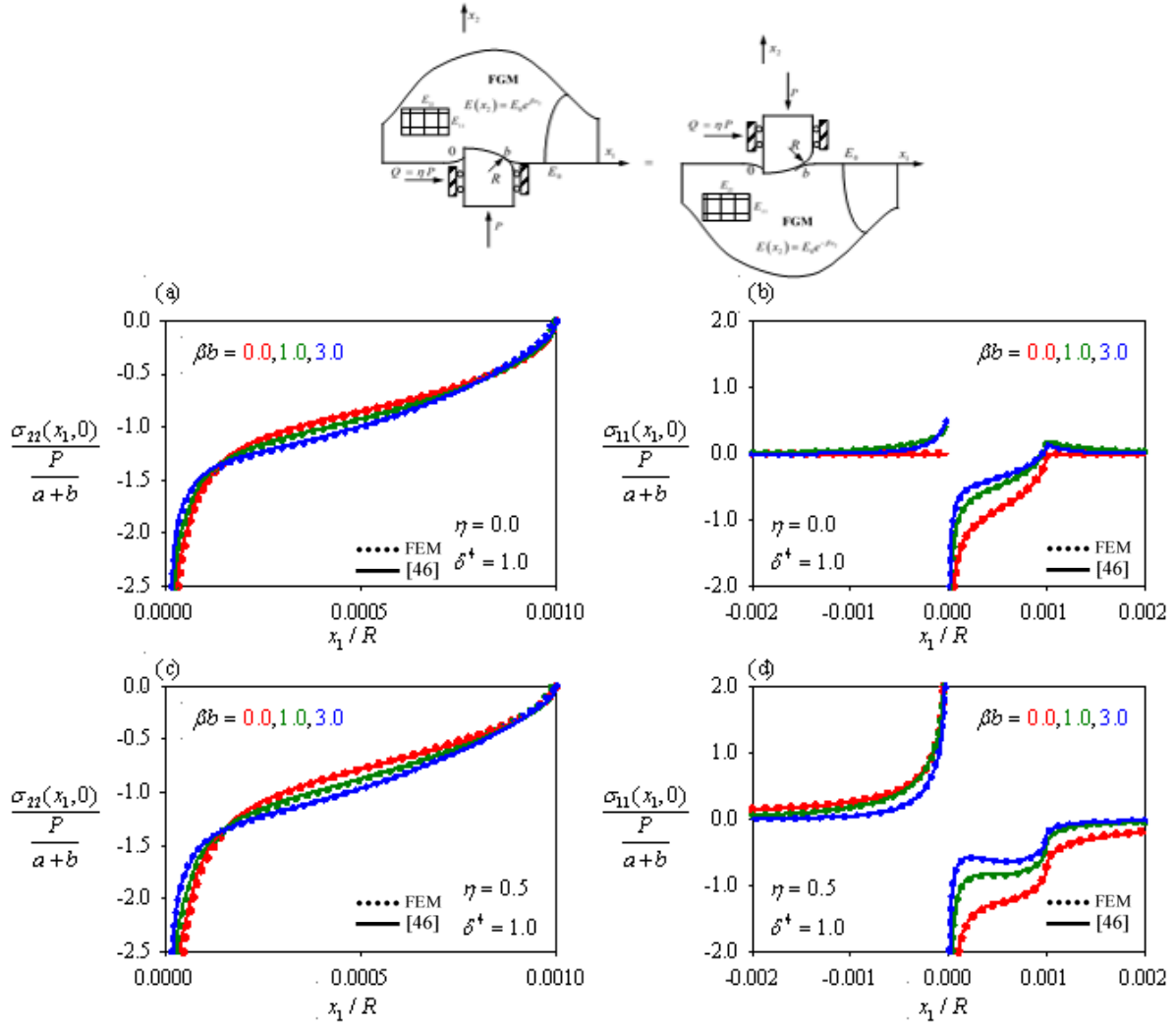


Figure 4.14. Contact stress, $\sigma_{22}(x_1, 0)$, and the in-plane stress, $\sigma_{11}(x_1, 0)$, distribution at the surface of the graded orthotropic half-plane loaded by a semi-circular punch for various values of the inhomogeneity parameter, $\beta(b)$, with the parameters $\kappa = 1.0$, $\delta^4 = 1.0$, $\nu = 0.3$, $b/R = 0.001$ (a) $\eta = 0.0$, $\delta^4 = 1.0$ (b) $\eta = 0.0$, $\delta^4 = 1.0$ (c) $\eta = 0.5$, $\delta^4 = 1.0$ (d) $\eta = 0.5$, $\delta^4 = 1.0$

4.15(c), 4.15(e) and 4.16(a), 4.16(c), 4.16(e), the variation of the shear parameter, κ , has a slightly minor effect on the variation of the contact stresses for the both frictionless and the frictional contact, where the value of the coefficient of friction is $\eta = 0.0$ and $\eta = 0.5$ respectively. Moreover, as illustrated in the Figures 4.15 and 4.16, as the inhomogeneity parameter, $\beta(b)$, increases, the resulting contact stress distribution decreases. In addition to,

the variation of the contact stress distribution illustrated in the Figure 4.16 is more evident than in the Figure 4.15, because as the coefficient of friction, η , increases, the effect of the inhomogeneity parameter, for the frictionless and frictional contact are illustrated in the Figures 4.15(b), 4.15(d), 4.15(f) and 4.16(b), 4.16(d), 4.16(f) respectively. It can easily be seen from the Figure 4.15, the shear parameter, κ , has a quiet small effect on the in-plane distribution, wherefore the contact is frictionless. However, for the Figure 4.16, on account of the coefficient of friction, η , is equal to $\eta = 0.5$, the shear parameter, κ , has a significant effect on the in-plane stress distribution. Also, the increment on the value of the inhomogeneity parameter, $\beta(b)$, increases the in-plane stress distribution for the both values of the $\eta = 0.0$ and $\eta = 0.5$.

The Figure 4.17 presents the effect of the shear parameter, κ , for the various values of the stiffness ratio, δ^4 . The coefficient of friction, η , and the inhomogeneity parameter, $\beta(b)$, are fixed at $\eta = 0.5$ and $\beta(b) = 0.5$ respectively. As it can be seen from the Figures 4.17(a) and 4.17(c), the effect of the shear parameter, κ , is quite low on the contact stress distribution. However, for the in-plane stress distribution depicted in the Figures 4.17(b) and 4.17(d), the variation of the shear parameter, κ , greatly affects the in-plane stress distribution.

In the Figure 4.18, the contact stress and the in-plane stress distribution for the various values of the effective Poisson's ratio, ν , and the stiffness ratio, δ^4 , is illustrated. As expected, the curves of the contact stress and the in-plane stress distribution are not affected widely by the change of the effective Poisson's ratio, ν .

Table 4.9, shows the comparison of the dimensionless load, $P^* = P/(ER)$, for a non-homogeneous orthotropic half plane loaded by a semi-circular punch for the various values of the inhomogeneity parameter, $\beta(b)$. The stiffness ratio, δ^4 , and the shear parameter, κ , are fixed at, $\delta^4 = 1.0$ and $\kappa = 1.0$. An excellent match is obtained between the finite element solutions and the analytical results. As it can be readily seen from the Table 4.9 that, as the inhomogeneity parameter, $\beta(b)$, increases, the resulting value of the dimensionless load, $P^* = P/(ER)$, significantly increases for the both frictionless and frictional contact. On the other hand, the change in the value of the coefficient of friction, η , also changes the value of

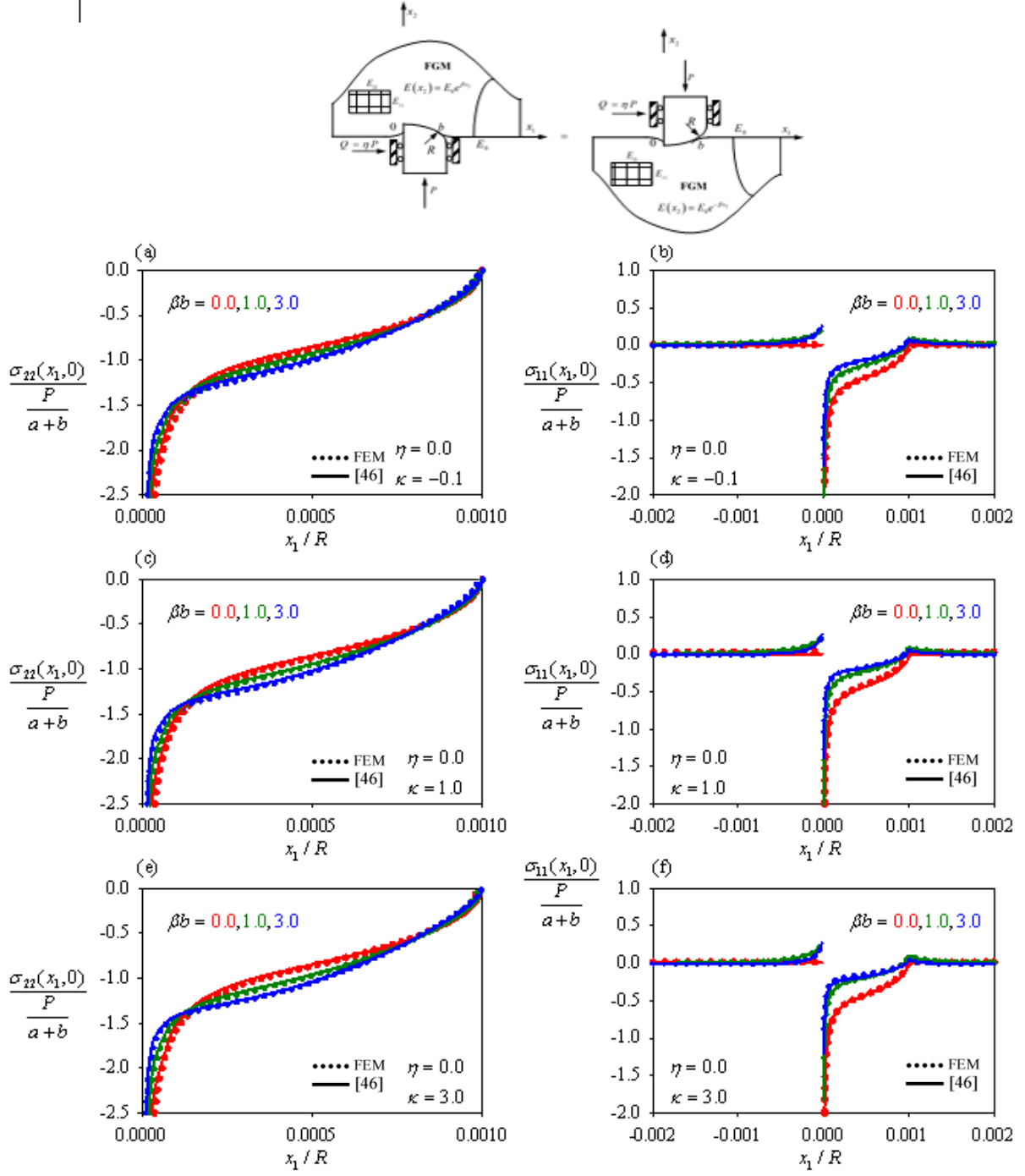


Figure 4.15. Contact stress, $\sigma_{22}(x_1, 0)$, and the in-plane stress, $\sigma_{11}(x_1, 0)$, distribution at the surface of the graded orthotropic half-plane loaded by a semi-circular punch for various values of the inhomogeneity parameter, $\beta(b)$, and the shear parameter, κ , with the parameters $\delta^4 = 0.25$, $\nu = 0.3$, $b/R = 0.001$ (a) $\eta = 0.0$, $\kappa = -0.1$ (b) $\eta = 0.0$, $\kappa = -0.1$ (c) $\eta = 0.0$, $\kappa = 1.0$ (d) $\eta = 0.0$, $\kappa = 1.0$ (e) $\eta = 0.0$, $\kappa = 3.0$ (f) $\eta = 0.0$, $\kappa = 3.0$

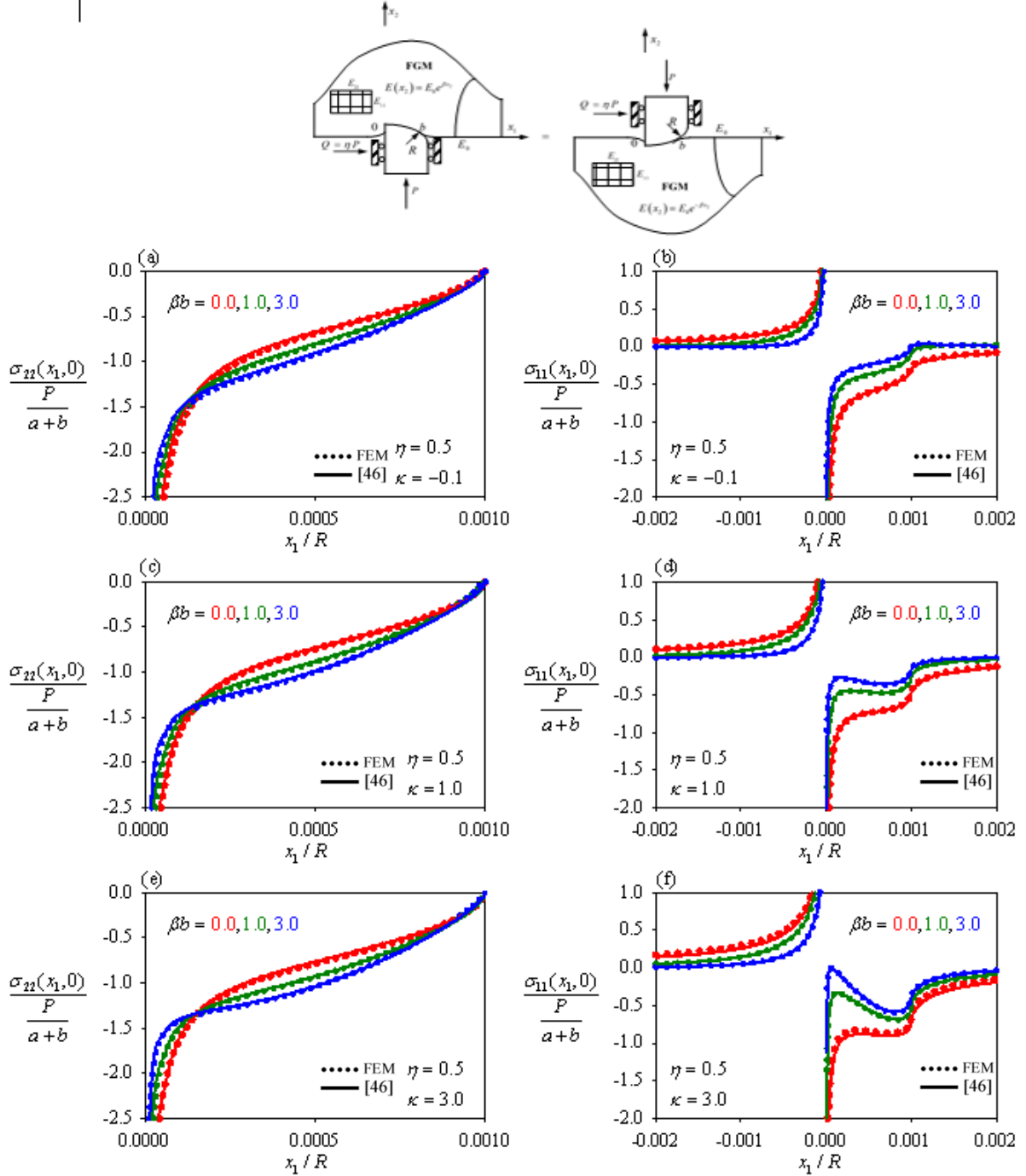


Figure 4.16. Contact stress, $\sigma_{22}(x_1, 0)$, and the in-plane stress, $\sigma_{11}(x_1, 0)$, distribution at the surface of the graded orthotropic half-plane loaded by a semi-circular punch for various values of the inhomogeneity parameter, $\beta(b)$, and the shear parameter, κ , with the parameters $\delta^4 = 0.25$, $\nu = 0.3$, $b/R = 0.001$ (a) $\eta = 0.5$, $\kappa = -0.1$ (b) $\eta = 0.5$, $\kappa = -0.1$ (c) $\eta = 0.5$, $\kappa = 1.0$ (d) $\eta = 0.5$, $\kappa = 1.0$ (e) $\eta = 0.5$, $\kappa = 3.0$ (f) $\eta = 0.5$, $\kappa = 3.0$

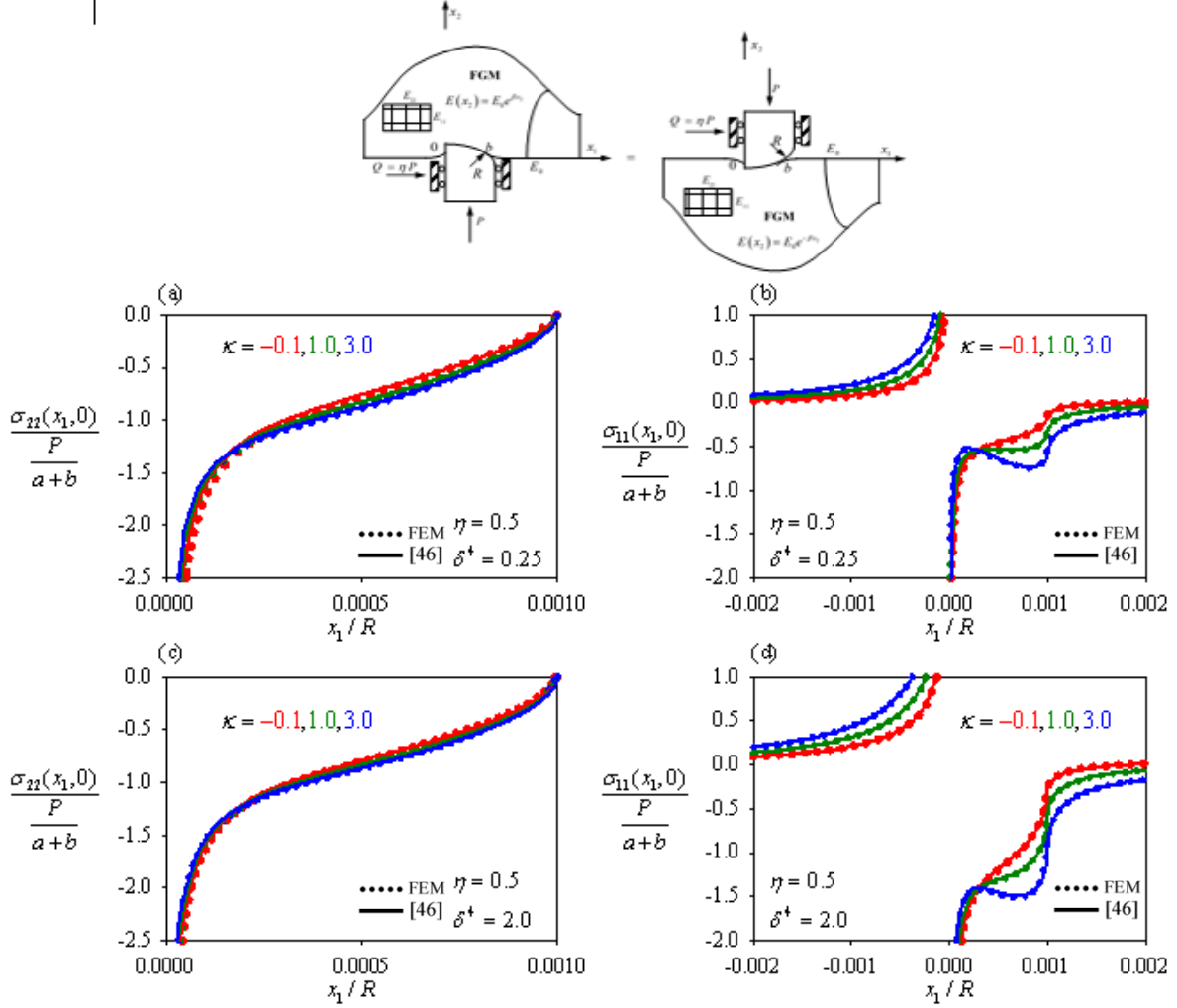


Figure 4.17. Contact stress, $\sigma_{22}(x_1, 0)$, and the in-plane stress, $\sigma_{11}(x_1, 0)$, distribution at the surface of the graded orthotropic half-plane loaded by a semi-circular punch for various values of the shear parameter, κ , and the stiffness ratio, δ^4 , with the parameters, $\nu = 0.3$, $\beta(b) = 0.5$, $b/R = 0.001$ (a) $\eta = 0.5$, $\delta^4 = 0.25$ (b) $\eta = 0.5$, $\delta^4 = 0.25$ (c) $\eta = 0.5$, $\delta^4 = 2.0$ (d) $\eta = 0.5$, $\delta^4 = 2.0$

the dimensionless load, $P^* = P/(ER)$.

In Table 4.10, the calculated values of the contact boundary, b/R , from the finite element solutions are given. The effect of the inhomogeneity parameter, $\beta(b)$, and the coefficient of friction, η , on the contact boundary, b/R , is slightly low.

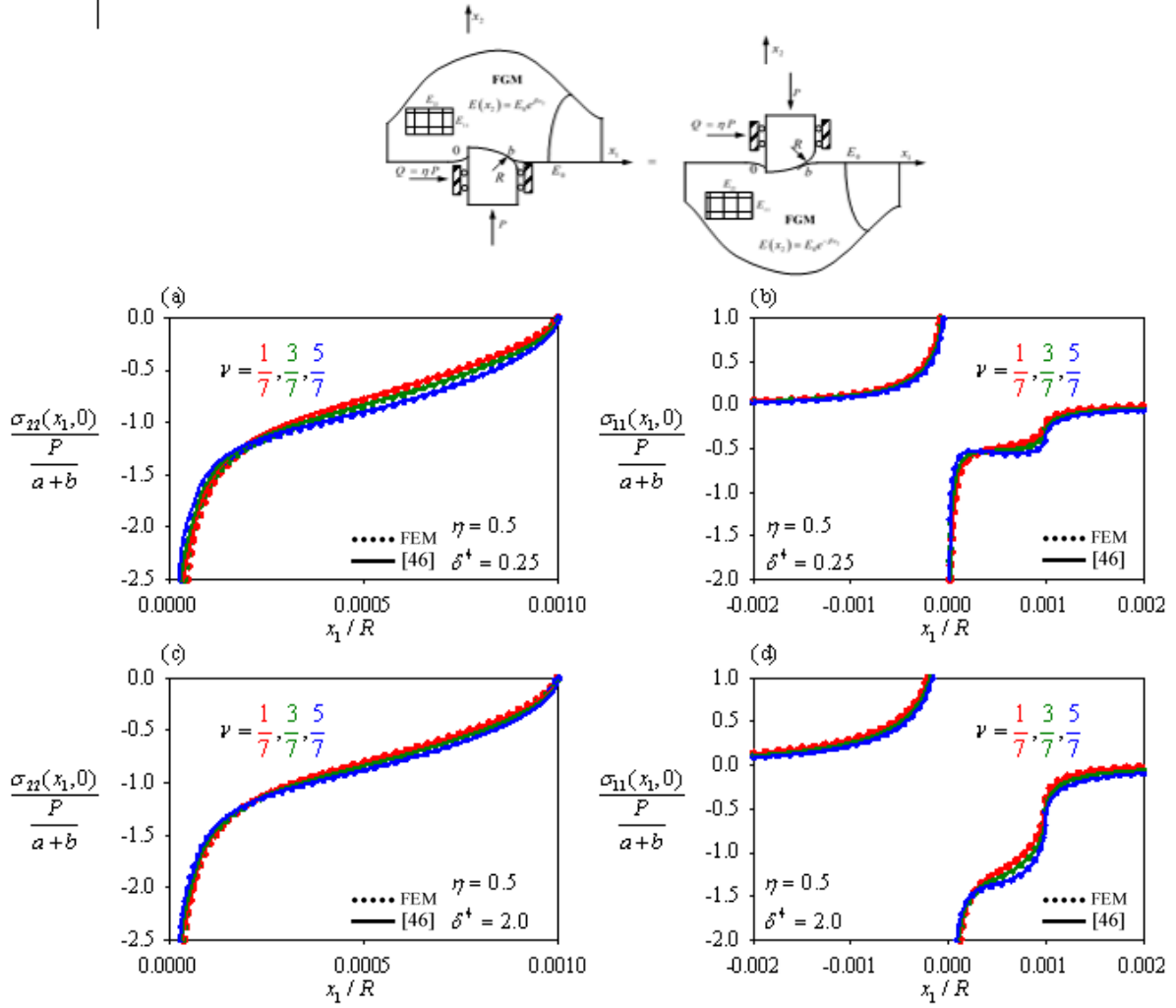


Figure 4.18. Contact stress, $\sigma_{22}(x_1, 0)$, and the in-plane stress, $\sigma_{11}(x_1, 0)$, distribution at the surface of the graded orthotropic half-plane loaded by a semi-circular punch for various values of the effective Poisson's ratio, ν , and the stiffness ratio, δ^4 , with the parameters, $\kappa = 0.5$, $\beta(b) = 0.5$, $b/R = 0.001$ (a) $\eta = 0.5$, $\delta^4 = 0.25$ (b) $\eta = 0.5$, $\delta^4 = 0.25$ (c) $\eta = 0.5$, $\delta^4 = 2.0$ (d) $\eta = 0.5$, $\delta^4 = 2.0$

The effect of the inhomogeneity parameter, $\beta(b)$, the shear parameter, κ , and the coefficient of friction, η , on the variation of the dimensionless load, $P^* = P/(ER)$, is given in the Table 4.11. Again the finite element solutions are quiet good converged to the analytical results. For the frictionless contact, where, $\eta = 0.0$, the change on the value of the inhomogeneity parameter, $\beta(b)$, affects the value of the dimensionless load greatly. For instance, as the value

Table 4.9. Comparison of the dimensionless load, $P^* = P / (ER)$, for a non-homogeneous orthotropic half-plane loaded by a semi-circular punch for various values of the inhomogeneity parameter, $\beta(b)$, with the parameters $\nu = 0.3$, $\delta^4 = 1.0$, $\kappa = 1.0$, $b / R = 0.001$

$\delta^4 = 1.0$		$P / (ER)$ Analytical [46]	$P / (ER)$ FEM	Error (%)
$\eta = 0.0$	$\beta(b) = 0.0$	0.5905E-06	0.5848E-06	0.9
	$\beta(b) = 1.0$	0.9338E-06	0.9208E-06	1.3
	$\beta(b) = 3.0$	1.6221E-06	1.5924E-06	1.8
$\eta = 0.5$	$\beta(b) = 0.0$	0.6799E-06	0.6693E-06	1.5
	$\beta(b) = 1.0$	0.9877E-06	0.9652E-06	2.2
	$\beta(b) = 3.0$	1.6612E-06	1.6420E-06	1.1

Table 4.10. Comparison of the contact boundary, b / R , for a non-homogeneous orthotropic half-plane loaded by a semi-circular punch for various values of the inhomogeneity parameter, $\beta(b)$, with the parameters $\nu = 0.3$, $\delta^4 = 1.0$, $\kappa = 1.0$, $b / R = 0.001$

$\delta^4 = 1.0$		b / R Analytical [46]	b / R FEM
$\eta = 0.0$	$\beta(b) = 0.0$		1.010E-03
	$\beta(b) = 1.0$	1.000E-03	1.014E-03
	$\beta(b) = 3.0$		1.019E-03
$\eta = 0.5$	$\beta(b) = 0.0$		1.016E-03
	$\beta(b) = 1.0$	1.000E-03	1.023E-03
	$\beta(b) = 3.0$		1.012E-03

of the inhomogeneity parameter, $\beta(b)$, increases, the resulting value of the dimensionless load, $P^* = P / (ER)$, also increases for the both values of the shear parameter, κ . Furthermore, the calculated value of the dimensionless load, $P^* = P / (ER)$, for the frictional contact, is bigger

Table 4.11. Comparison of the dimensionless load, $P^* = P / (ER)$, for a non-homogeneous orthotropic half-plane loaded by a semi-circular punch for various values of the inhomogeneity parameter, $\beta(b)$, and the shear parameter, κ , with the parameters $\nu = 0.3$,

$$\delta^4 = 0.25, b / R = 0.001$$

$\delta^4 = 0.25$		$P / (ER)$		Error (%)	
		Analytical [46]	FEM		
$\eta = 0.0$	$\kappa = -0.1$	$\beta(b) = 0.0$	1.2418E-06	1.2077E-06	2.7
		$\beta(b) = 1.0$	1.9916E-06	1.9268E-06	3.2
		$\beta(b) = 3.0$	3.4071E-06	3.2982E-06	3.1
	$\kappa = 1.0$	$\beta(b) = 0.0$	0.8351E-06	0.8186E-06	1.9
		$\beta(b) = 1.0$	1.5191E-06	1.4881E-06	2.0
		$\beta(b) = 3.0$	2.9147E-06	2.8249E-06	3.0
	$\kappa = 3.0$	$\beta(b) = 0.0$	0.5890E-06	0.5757E-06	2.2
		$\beta(b) = 1.0$	1.2447E-06	1.2143E-06	2.4
		$\beta(b) = 3.0$	2.6171E-06	2.5641E-06	2.0
$\eta = 0.5$	$\kappa = -0.1$	$\beta(b) = 0.0$	1.6353E-06	1.5949E-06	2.4
		$\beta(b) = 1.0$	2.2321E-06	2.1771E-06	2.4
		$\beta(b) = 3.0$	3.5936E-06	3.4819E-06	3.1
	$\kappa = 1.0$	$\beta(b) = 0.0$	1.0142E-06	0.9965E-06	1.7
		$\beta(b) = 1.0$	1.6106E-06	1.5674E-06	2.6
		$\beta(b) = 3.0$	2.9710E-06	2.9070E-06	2.1
	$\kappa = 3.0$	$\beta(b) = 0.0$	0.6780E-06	0.6613E-06	2.4
		$\beta(b) = 1.0$	1.2666E-06	1.2500E-06	1.3
		$\beta(b) = 3.0$	2.5473E-06	2.4913E-06	2.2

than the calculated value of the dimensionless load, $P^* = P / (ER)$, for the frictionless contact. This can be easily seen by fixing the values of the shear parameter, κ , and the inhomogeneity

parameter, $\beta(b)$, while changing the value of the coefficient friction, η , from $\eta = 0.0$ to $\eta = 0.5$.

The Table 4.12, depicts the variation of the contact boundary, b/R , under the same conditions in the Table 4.11. It may be safely said that, the variation of the shear parameter, κ , inhomogeneity parameter, $\beta(b)$, and the coefficient of friction, η , pretty much has no effect on the variation of the values of the contact boundary, b/R .

The dimensionless load, $P^* = P/(ER)$, obtained by the finite element solutions for the various values of the stiffness ratio, δ^4 , and the shear parameter, κ , is compared with the analytical results in the Table 4.13. The matching between the finite elements solutions and the analytical results are quite decent. It can be prominently said that, as the shear parameter, κ , increases, the computed value of the dimensionless load, $P^* = P/(ER)$, decreases. Furthermore, the enhancement on the value of the stiffness ratio, δ^4 , also decreases the resulting value of the dimensionless load, $P^* = P/(ER)$. Moreover, in the Table 4.14, the demonstration of the values of the contact boundary, b/R , for the finite element solutions and the analytical results are given. Although, the shear parameter, κ , and the stiffness ratio, δ^4 , varies from $\kappa = -0.1$ to $\kappa = 3.0$ and $\delta^4 = 0.25$ to $\delta^4 = 2.0$, the alteration on the values of contact boundary, b/R , is nebulously low.

In the Table 4.15, the efficacy of the effective Poisson's ratio, ν , on the graded orthotropic half-plane intended by a semi-circular punch, for the various values of the stiffness ratio, δ^4 , is illustrated. Although, the matching between the finite element solutions and the analytical results are impeccably good, the variation of the effective Poisson's ratio, ν , has no major impact on the values of the dimensionless load, $P^* = P/(ER)$. On the other hand, the stiffness ratio, δ^4 , has a substantial impact on the variation of the dimensionless load, $P^* = P/(ER)$. To illustrate with, as the stiffness ratio, δ^4 , increases, the resulting value of the dimensionless load, $P^* = P/(ER)$, nearly decreases by half for the fixed values of the effective Poisson's ratio, ν . Moreover, there is almost no variety on the values of the contact boundary, b/R , as the effective Poisson's ratio, ν , and the stiffness ratio, δ^4 , has nearly zero. Furthermore, the

Table 4.12. Comparison of the contact boundary, b/R , for a non-homogeneous orthotropic half-plane loaded by a semi-circular punch for various values of the inhomogeneity parameter, $\beta(b)$, and the shear parameter, κ , with the parameters $\nu = 0.3$, $\delta^4 = 0.25$, $b/R = 0.001$

$\delta^4 = 0.25$		b/R Analytical [46]	b/R FEM
$\eta = 0.0$	$\kappa = -0.1$	$\beta(b) = 0.0$	1.028E-03
		$\beta(b) = 1.0$	1.034E-03
		$\beta(b) = 3.0$	1.033E-03
	$\kappa = 1.0$	$\beta(b) = 0.0$	1.020E-03
		$\beta(b) = 1.0$	1.021E-03
		$\beta(b) = 3.0$	1.032E-03
	$\kappa = 3.0$	$\beta(b) = 0.0$	1.023E-03
		$\beta(b) = 1.0$	1.025E-03
		$\beta(b) = 3.0$	1.021E-03
$\eta = 0.5$	$\kappa = -0.1$	$\beta(b) = 0.0$	1.025E-03
		$\beta(b) = 1.0$	1.025E-03
		$\beta(b) = 3.0$	1.032E-03
	$\kappa = 1.0$	$\beta(b) = 0.0$	1.018E-03
		$\beta(b) = 1.0$	1.028E-03
		$\beta(b) = 3.0$	1.022E-03
	$\kappa = 3.0$	$\beta(b) = 0.0$	1.025E-03
		$\beta(b) = 1.0$	1.013E-03
		$\beta(b) = 3.0$	1.023E-03

resulting value of the contact boundary, b/R , is almost constant for the various values of effective Poisson's ratio, ν , and the stiffness ratio, δ^4 .

Table 4.13. Comparison of the dimensionless load, $P^* = P / (ER)$, for a non-homogeneous orthotropic half-plane loaded by a semi-circular punch for various values of the shear parameter, κ , and the stiffness ratio, δ^4 , with the parameters $\nu = 0.3$, $\beta(b) = 0.5$, $b / R = 0.001$

$\beta(b)=0.5$		$P/(ER)$	$P/(ER)$	Error (%)	
		Analytical [46]	FEM		
$\eta=0.5$	$\delta^4=0.25$	$\kappa=-0.1$	1.8970E-06	1.8315E-06	3.4
		$\kappa=1.0$	1.2889E-06	1.2602E-06	2.2
		$\kappa=3.0$	0.9591E-06	0.9434E-06	1.6
	$\delta^4=2.0$	$\kappa=-0.1$	0.9897E-06	0.9746E-06	1.5
		$\kappa=1.0$	0.6656E-06	0.6589E-06	1.0
		$\kappa=3.0$	0.4864E-06	0.4807E-06	1.1

Table 4.14. Comparison of the contact contact boundary, b / R , for a non-homogeneous orthotropic half-plane loaded by a semi-circular punch for various values of the shear parameter, κ , and the stiffness ratio, δ^4 , with the parameters $\nu = 0.3$, $\beta(b) = 0.5$, $b / R = 0.001$

$\beta(b) = 0.5$		b / R Analytical	b / R FEM
		[46]	
$\eta = 0.5$	$\delta^4 = 0.25$	$\kappa = -0.1$	1.000E-03
		$\kappa = 1.0$	1.023E-03
		$\kappa = 3.0$	1.017E-03
	$\delta^4 = 2.0$	$\kappa = -0.1$	1.000E-03
		$\kappa = 1.0$	1.010E-03
		$\kappa = 3.0$	1.012E-03

Table 4.15. Comparison of the dimensionless load, $P^* = P / (ER)$, for a non-homogeneous orthotropic half-plane loaded by a semi-circular punch for various values of the effective Poisson's ratio, ν , and the stiffness ratio, δ^4 , with the parameters $\kappa = 0.5$, $\beta(b) = 0.5$, $b / R = 0.001$

$\beta(a+b) = 0.5$		$P / (ER)$		Error (%)
		Analytical [46]	FEM	
$\eta = 0.5$	$\delta^4 = 0.25$	$\nu = 1/7$	1.4837E-06	1.4493E-06
		$\nu = 3/7$	1.4520E-06	1.4184E-06
		$\nu = 5/7$	1.3859E-06	1.3578E-06
	$\delta^4 = 2.0$	$\nu = 1/7$	0.7701E-06	0.7530E-06
		$\nu = 3/7$	0.7558E-06	0.7474E-06
		$\nu = 5/7$	0.7354E-06	0.7267E-06

Table 4.16. Comparison of the contact contact boundary, b / R , for a non-homogeneous orthotropic half-plane loaded by a semi-circular punch for various values of the effective Poisson's ratio, ν , and the stiffness ratio, δ^4 , with the parameters $\kappa = 0.5$, $\beta(b) = 0.5$, $b / R = 0.001$

$\beta(a+b) = 0.5$		b / R Analytical [46]	b / R FEM
$\eta = 0.5$	$\delta^4 = 0.25$	$\nu = 1/7$	1.000E-03
		$\nu = 3/7$	1.024E-03
		$\nu = 5/7$	1.021E-03
	$\delta^4 = 2.0$	$\nu = 1/7$	1.000E-03
		$\nu = 3/7$	1.023E-03
		$\nu = 5/7$	1.011E-03

4.5. Von-Mises Stress Distributions For The Graded Orthotropic Half-Plane Loaded By a Rigid Circular Punch

The Figure 4.19 and the Figure 4.20 shows the impact of the inhomogeneity parameter, $\beta(a+b)$, and the shear parameter, κ , on the Von-Mises stress distribution for the graded orthotropic half-plane loaded by a circular punch. The calculated Von-Mises stress values are normalized by using the maximum Hertzian contact pressure, $P_0 = 2P/(\pi R)$, where P and R donates the value of the force acting on the punch circular punch and the radius of the circular punch respectively. The Von-Mises stress distribution is divided into eight uniform contours to illustrate the stress distribution in a better way. As there is no friction on the contact zone, the Von-Mises stress distribution is symmetrical 4.19. It can be easily seen from the Figure 4.19 that, the inhomogeneity parameter, $\beta(a+b)$, and the shear parameter, κ , has a significant effect on the Von-Mises stress distribution. For the higher values of the shear parameter, the resulting Von-Mises stress distribution shrinks. Furthermore, the red blended contour distribution becomes more evident as the shear parameter, κ , increases, and one can easily see that by comparing the Figures 4.19(a) and 4.19(g), 4.19(b) and 4.19(h), 4.19(c) and 4.19(i). Moreover, as the inhomogeneity parameter, $\beta(a+b)$, increases, the red blended contour distribution also increases. Additionally, as the shear parameter, κ , increases, the resulting Von-Mises stress distribution becomes more vertical. On the other hand, because of the presence of the friction on the contact zone, the Von-Mises stress distribution is unsymmetrical in the Figure 4.20. Note that, the Von-Mises stress distributions prone to the trailing edge of contact, as the coefficient of friction, η , has a positive value. The effect of the coefficient of friction, η , is more evident on the contour distribution for the smaller values of the shear parameter, κ , and the higher values of the inhomogeneity parameter, $\beta(a+b)$. Over and above that, the Von-Mises stress distribution becomes more vertical as the shear parameter, κ , increases.

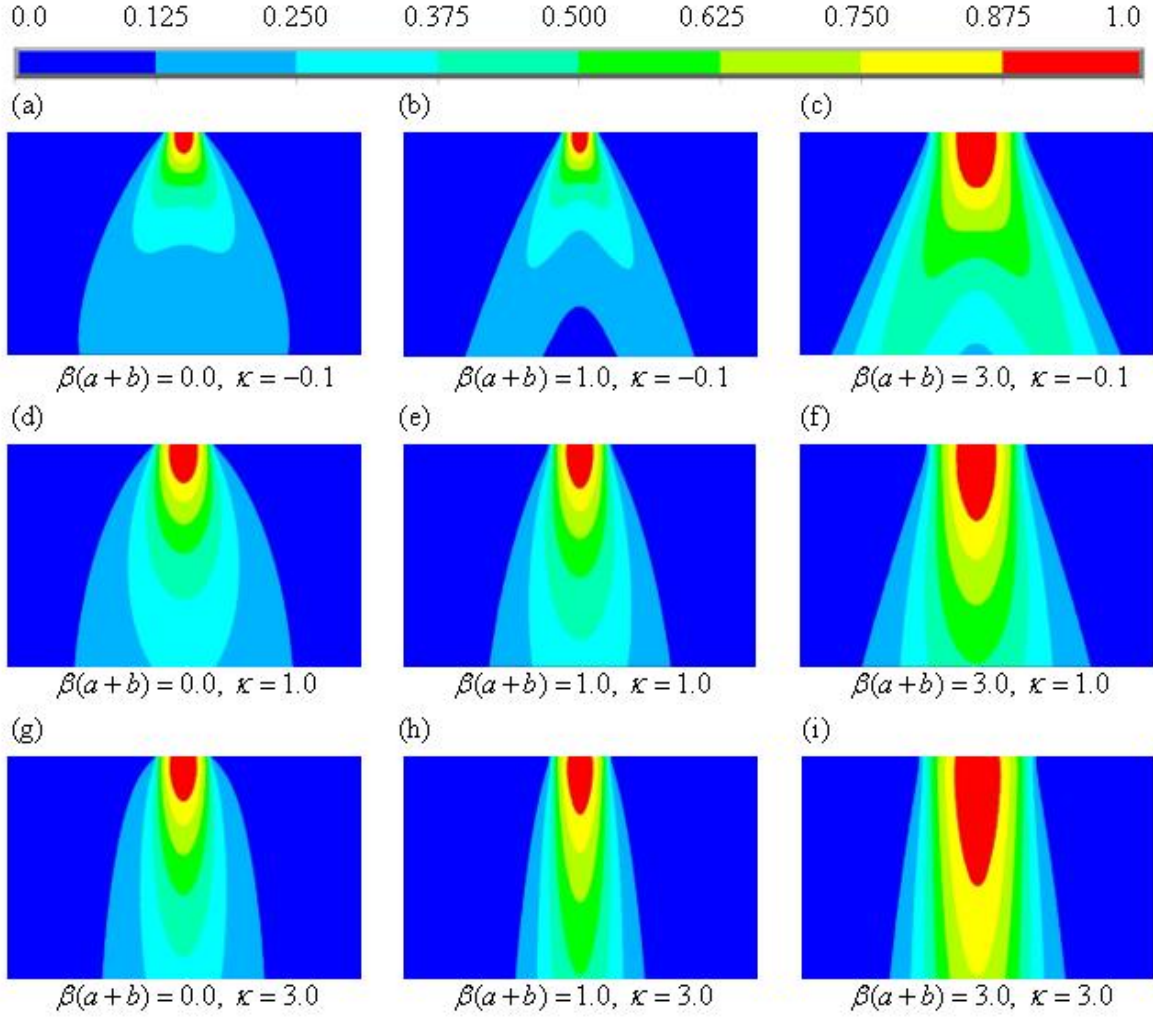


Figure. 4.19. Von-Mises stress distribution at the surface of graded orthotropic half-plane, normalized by the maximum Hertzian contact pressure, $P_0 = 2P / (\pi R)$ loaded by a circular punch for various values of the inhomogeneity parameter, $\beta(a+b)$, and the shear parameter, κ , with the parameters, $\nu = 0.3$, $\delta^4 = 0.25$, $\eta = 0.0$, $(a+b)/R = 0.002$

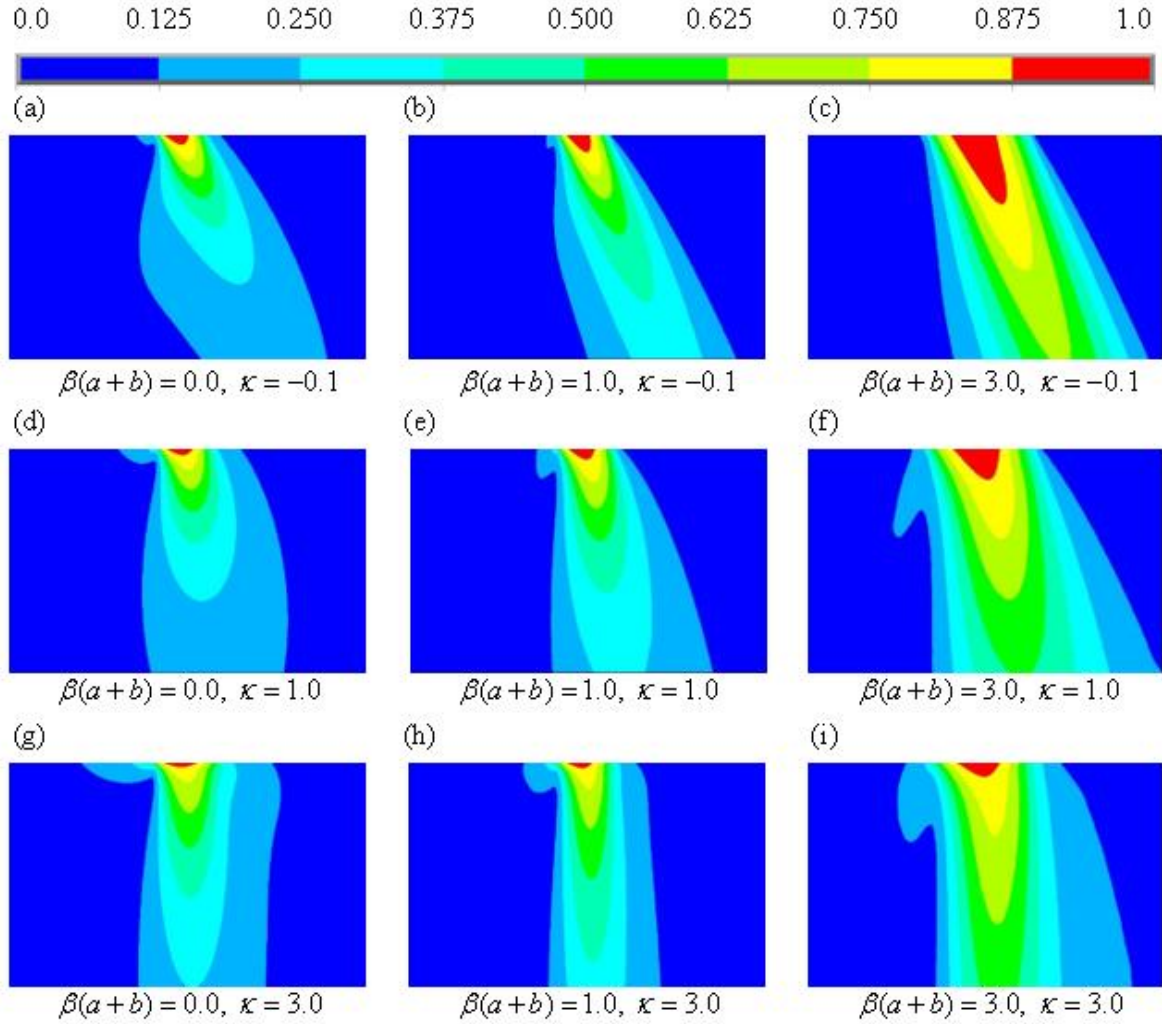


Figure. 4.20. Von-Mises stress distribution at the surface of graded orthotropic half-plane, normalized by the maximum Hertzian contact pressure, $P_0 = 2P / (\pi R)$ loaded by a circular punch for various values of the inhomogeneity parameter, $\beta(a+b)$, and the shear parameter,

κ , with the parameters, $\nu = 0.3$, $\delta^4 = 0.25$, $\eta = 0.5$, $(a+b)/R = 0.002$

4.6. Von-Mises Stress Distributions For The Graded Orthotropic Half-Plane Loaded By a Rigid Semi-Circular Punch

In the Figure 4.21, the effect of the inhomogeneity parameter, $\beta(b)$, and the shear parameter, κ , on the Von-Mises stress distribution is investigated. The stiffness ratio, δ^4 , and the effective Poisson's ratio, ν , are fixed at, $\delta^4 = 0.25$ and $\nu = 0.3$. Although there is no friction on the contact area, the resulting Von-Mises stress distribution is not symmetrical because of the geometry of the semi-circular punch profile. Furthermore, it can be easily seen from the Figure 4.21 that, because of the singularity on the leading edge of the semi-circular punch, the Von-Mises stress distribution are almost equal to zero on the left side of the Figures 4.21(a) to 4.21(i). The increment on the value of the shear parameter, κ , shrinks the resulting Von-Mises stress distribution. For the lower values of the shear parameter, κ , the effect of the inhomogeneity parameter, $\beta(b)$, on the Von-Mises stress distribution is slightly low. As the inhomogeneity parameter, $\beta(b)$, increases, the resulting distribution of the Von-Mises stresses rapidly changes for the higher values of the shear parameter, κ .

The Von-Mises stress distribution for the graded orthotropic half-plane intended by a semi-circular punch is illustrated in the Figure 4.22. The inhomogeneity parameter and the shear parameter are chosen between the ranges of, $0.0 < \beta(b) < 3.0$ and $-0.1 < \kappa < 3.0$. For the homogeneous orthotropic half-planes that illustrated in the Figures 4.22(a), 4.22(d) and 4.22(g), the increment on the shear parameter, κ , changes the orientation of the Von-Mises stress distribution from the trailing edge to the singular edge. Moreover, the change on the Von-Mises stress distribution is noteworthy, as the inhomogeneity parameter, $\beta(b)$, increases. The red blended contour is excessively small, as the leading edge of the contact is singular at a point.

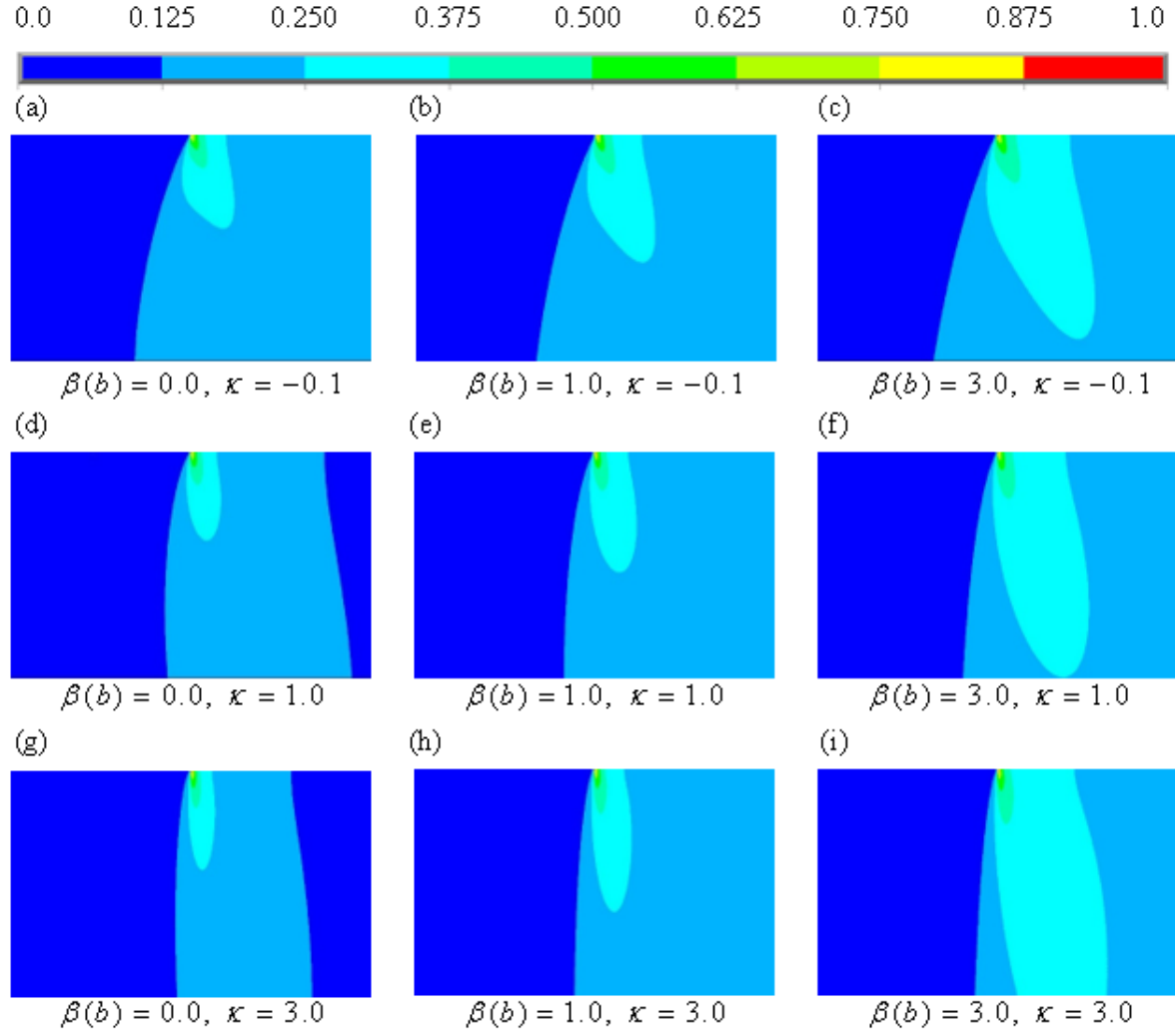


Figure. 4.21. Von-Mises stress distribution at the surface of graded orthotropic half-plane, normalized by the maximum Hertzian contact pressure, $P_0 = 2P/(\pi R)$ loaded by a semi-circular punch for various values of the inhomogeneity parameter, $\beta(b)$, and the shear parameter, κ , with the parameters, $\nu = 0.3$, $\delta^4 = 0.25$, $\eta = 0.0$, $b/R = 0.001$

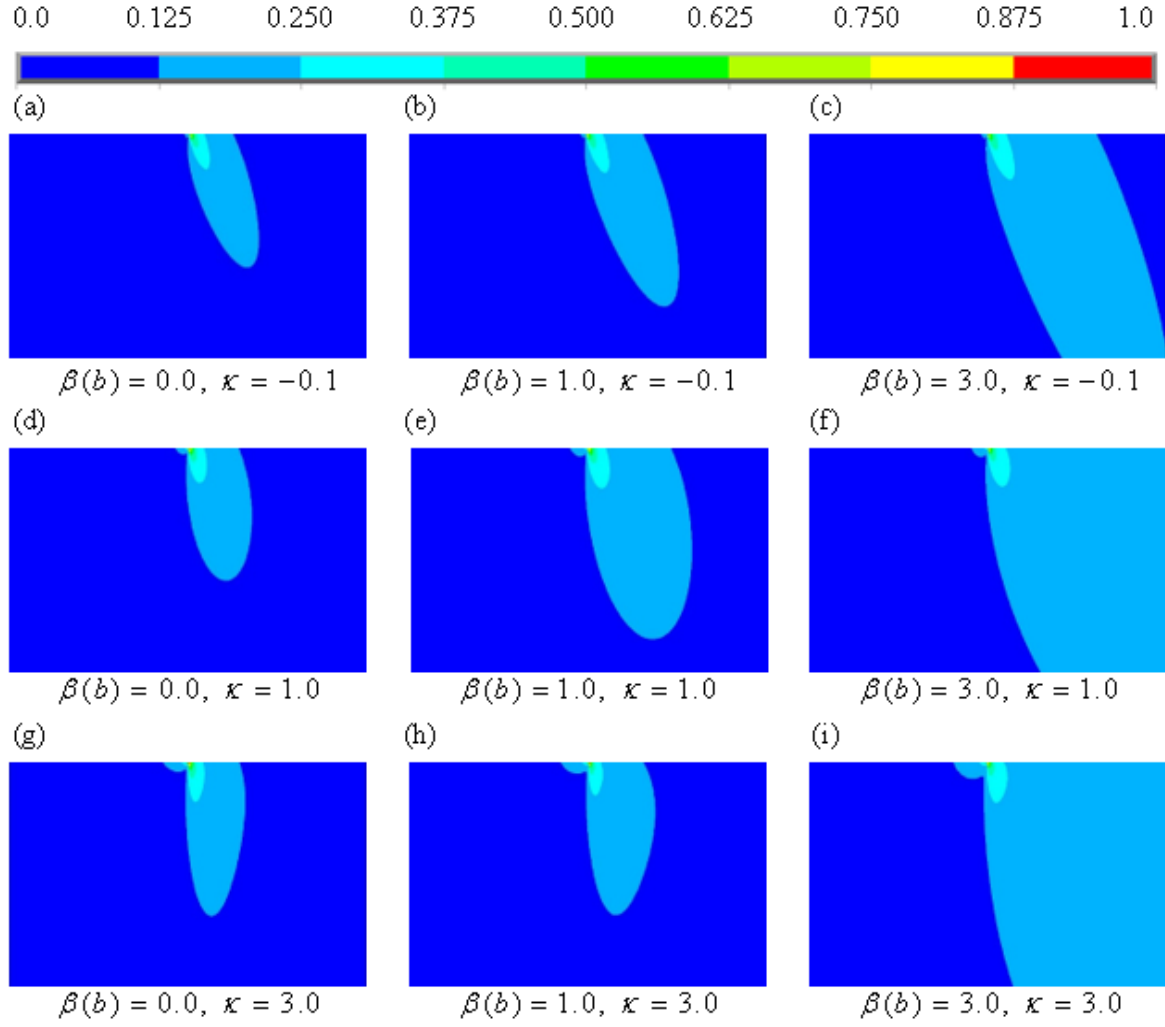


Figure. 4.22. Von-Mises stress distribution at the surface of graded orthotropic half-plane, normalized by the maximum Hertzian contact pressure, $P_0 = 2P/(\pi R)$ loaded by a semi-circular punch for various values of the inhomogeneity parameter, $\beta(b)$, and the shear parameter, κ , with the parameters, $\nu = 0.3$, $\delta^4 = 0.25$, $\eta = 0.5$, $b/R = 0.001$

5. CONCLUDING REMARKS

In this master thesis, the comparison of the finite element solutions and the analytical results are obtained for the graded orthotropic half-plane that is loaded by a rigid punch with an arbitrary profile. As illustrated in the Section 3, four different punch profiles (flat, triangular, circular and semi-circular punch profiles) are used in order to conduct and compare the finite element solutions with the analytical results. The major findings and the most prominent points can be summarized below.

- The contact stress distributions that calculated with using the finite element solutions for the graded orthotropic half-plane that is loaded by a flat punch are in an excellent agreement with the analytical results that given in the [19].
- As expected, the contact stress distribution is singular at the sharp edges $(-a, a)$ of the contact zone for the graded graded orthotropic half-plane that is loaded by a flat punch.
- The changing values of the shear parameter and the coefficient of friction has a significant effect on the contact stress distribution. As the coefficient of friction increases, the curves of the contact stresses, begin to prone to the leading edge of the contact for the values of the shear parameter less than one.
- The effective Poisson's ratio has only an immaterial efficacy on the contact stress distribution.
- The calculated contact stress distributions by using the finite element method is under a unique concordance with the analytical results for the triangular punch profile [19].
- The contact stress distribution is bounded at the smooth edge, $x_1 = b$, of the contact and singular at the sharp edge, $x_1 = 0$, of the contact.
- An excellent match is obtained between the finite element solutions and the analytical results for the graded orthotropic half-plane problem intended by a circular punch [46].
- For the frictional cases, the contact stress distribution prone to the trailing edge of the contact, as the value of the coefficient of friction is positive for all the cases.
- The inhomogeneity parameter significantly affects the contact stress distribution and the in-plane stress distribution.

- The increment on the value of the effective Poisson's ratio is quiet inefficacious on the contact stress and the in-plane stress distribution.
- The increment on the value of the inhomogeneity parameter, increases the calculated value of the dimensionless load for the both frictionless and the frictional contact cases.
- In all cases, as the shear parameter increases, the resulting value of the dimensionless load decreases.
- For the frictional contact, the increment on the stiffness ratio, increases the resulting values of the dimensionless load for the various values of the effective Poisson's ratio.
- For all the cases, the change of the non-dimensional material properties, inhomogeneity parameter and the coefficient of friction has a slightly low effect on the resulting values of the contact boundaries.
- The Von-Mises stress distribution for the graded orthotropic half-plane punch loaded by a circular punch is symmetrical for the frictionless case.
- As the shear parameter and the inhomogeneity parameter increases, the Von-Mises stress distribution becomes more vertical.
- Under the presence of friction, the Von-Mises stress distribution prones to the trailing edge of the contact.
- The finite element solutions are quiet well converged to the analytical results for the semi-circular punch profile [46].
- The effective Poisson's ratio has not a substantial effect on the distribution of the contact stresses and the in-plane stresses.
- As the inhomogeneity parameter increases, the resulting contact stress distribution decreases for the various values of the shear parameter, coefficient of friction.
- The contact stress and the in-plane stress distribution is singular at the leading edge, $x_1 / R = 0.0$, of the contact.
- The shear parameter has a trivial effect on the contact stress distribution for the various values of the stiffness ratio.
- As the inhomogeneity parameter increases, the dimensionless load increases and as the shear parameter increases, the resulting dimensionless load decreases.

- The Von-Mises stress distribution becomes more evident, as the stiffness ratio and the inhomogeneity parameter increases for the both frictionless and the frictional contact.
- For the smaller values of the shear parameter, the Von-Mises stress distribution prones to the trailing edge of the contact, as the coefficient of friction is positive.



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Kucuksucu, A., Guler, M.A., Yilmaz, K.B., Yildirim, B., Contact mechanics of a rigid parabolic stamp sliding over a graded orthotropic half-plane, (in preparation), **2016** has submitted to **ISFGM^s 2016 Symposium**, <https://www.isfgms2016.org/>.



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Tarih: 22/06/2016

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Sliding Over A Graded Orthotropic Medium

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